JOPE DIME / PRIVATE MAINTAINER EDITION

Evidence reference

Volume 2 of ASM — Maintainer and User Manual

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Contents

A All thirty datasets and all 522 feature records 3

A.1 All thirty datasets at a glance 5
A.2 Regression datasets 8
A.3 Binary classification datasets 15
A.4 Multiclass classification datasets 32
A.5 Explicitly omitted or unknown statistics 46

B Historical training, checkpoint lineage and original loss trajectories 48

B.1 Dated lineage and evidence coverage 49
B.2 Legacy Python training: gen31 and round 1 50
B.3 Native training: the 108-block audit and the A100 recovery lineage 56
B.4 Current publication regeneration and provenance 64

C Historical GP targets, rankings, Elo and attribution limits 65

C.1 Did GP targets get better? 65
C.2 Existing historical rankings and Elo 69

D Dated data estate, recovery evidence and binary formats 77

D.1 Configured product stores and local roots 78
D.2 Additional archive, replica and custody metadata 80
D.3 Additional host data roots 81
D.4 Recorded copy and retention gaps 82
D.5 Recovery evidence: what is preserved, and what was lost 83
D.6 Retained corpus sizes 84
D.7 Binary format: the exact packed teaching record 85

E Full task catalogue, source/schema index, receipts and glossary 89

E.1 Primary acceptance items by stage 91
E.2 Coordinated future backlog: 24 proposals 113
E.3 Planned execution and review ownership 118
E.4 Source index 120
E.5 Machine-readable companions 124
E.6 Receipts 127
E.7 Reproduction boundaries 128
E.8 Names and glossary 129

APPENDIX A

All thirty datasets and all 522 feature records

IN THIS APPENDIX

A.1 All thirty datasets at a glance 5
- **A.1.1** Per-dataset summary 5
- **A.1.2** Frozen dataset coverage 7

A.2 Regression datasets 8
- **A.2.1** 197_cpu_act 8
- **A.2.2** 225_puma8NH 9
- **A.2.3** 230_machine_cpu 9
- **A.2.4** 522_pm10 10
- **A.2.5** 529_pollen 10
- **A.2.6** 579_fri_c0_250_5 11
- **A.2.7** auto_insurance_price 12
  - **A.2.7.1** Summary and feature statistics 12
  - **A.2.7.2** Category frequencies 13
- **A.2.8** 690_visualizing_galaxy 13
- **A.2.9** 712_chscase_geyser1 14
- **A.2.10** 503_wind 14

A.3 Binary classification datasets 15
- **A.3.1** breast_cancer_wisconsin_diagnostic 15
- **A.3.2** credit_approval_germany 17
  - **A.3.2.1** Summary and feature statistics 17
  - **A.3.2.2** Category frequencies 18
- **A.3.3** congressional_voting_records 19
  - **A.3.3.1** Summary and feature statistics 19
  - **A.3.3.2** Category frequencies 20
- **A.3.4** haberman 21
- **A.3.5** ionosphere 21
- **A.3.6** sonar 23
  - **A.3.6.1** Summary, identities and class counts 23
  - **A.3.6.2** Feature statistics 24
- **A.3.7** spambase 26
  - **A.3.7.1** Summary, identities and class counts 26
  - **A.3.7.2** Feature statistics 26
- **A.3.8** phoneme 28
- **A.3.9** tic_tac_toe 29
  - **A.3.9.1** Summary and feature statistics 29
  - **A.3.9.2** Category frequencies 30
- **A.3.10** horse_colic_surgery 30
  - **A.3.10.1** Summary and feature statistics 30
  - **A.3.10.2** Category frequencies 31

A.4 Multiclass classification datasets 32
- **A.4.1** iris 32
- **A.4.2** wine_recognition 33
- **A.4.3** car_evaluation 33
  - **A.4.3.1** Summary and feature statistics 33
  - **A.4.3.2** Category frequencies 34
- **A.4.4** balance_scale 34
  - **A.4.4.1** Summary and feature statistics 34
  - **A.4.4.2** Category frequencies 35
- **A.4.5** dermatology 35
  - **A.4.5.1** Summary, identities and class counts 35
  - **A.4.5.2** Feature statistics 36
  - **A.4.5.3** Category frequencies 37
- **A.4.6** segmentation 39
- **A.4.7** vehicle 40
- **A.4.8** optdigits 42
  - **A.4.8.1** Summary, identities and class counts 42

A.4.8.2 Feature statistics 42
A.4.8.3 Category frequencies 44
A.4.9 penguins 45
A.4.10 tae 46
A.5 Explicitly omitted or unknown statistics 46

All statistics below use development rows only. Numeric values use finite observations; missing cells are counted separately. Standard deviation is the population statistic. Medians and the additional 1%, 5%, 25%, 75%, 95%, 99% quantiles in JSON linearly interpolate sorted values at `(n−1)p`. Category counts retain exact stored strings. Original class ordering and fold class counts are preserved. Upstream metadata, provenance URLs, source byte/hash pins, group-size histograms and all file identities remain in JSON.

A.1 All thirty datasets at a glance

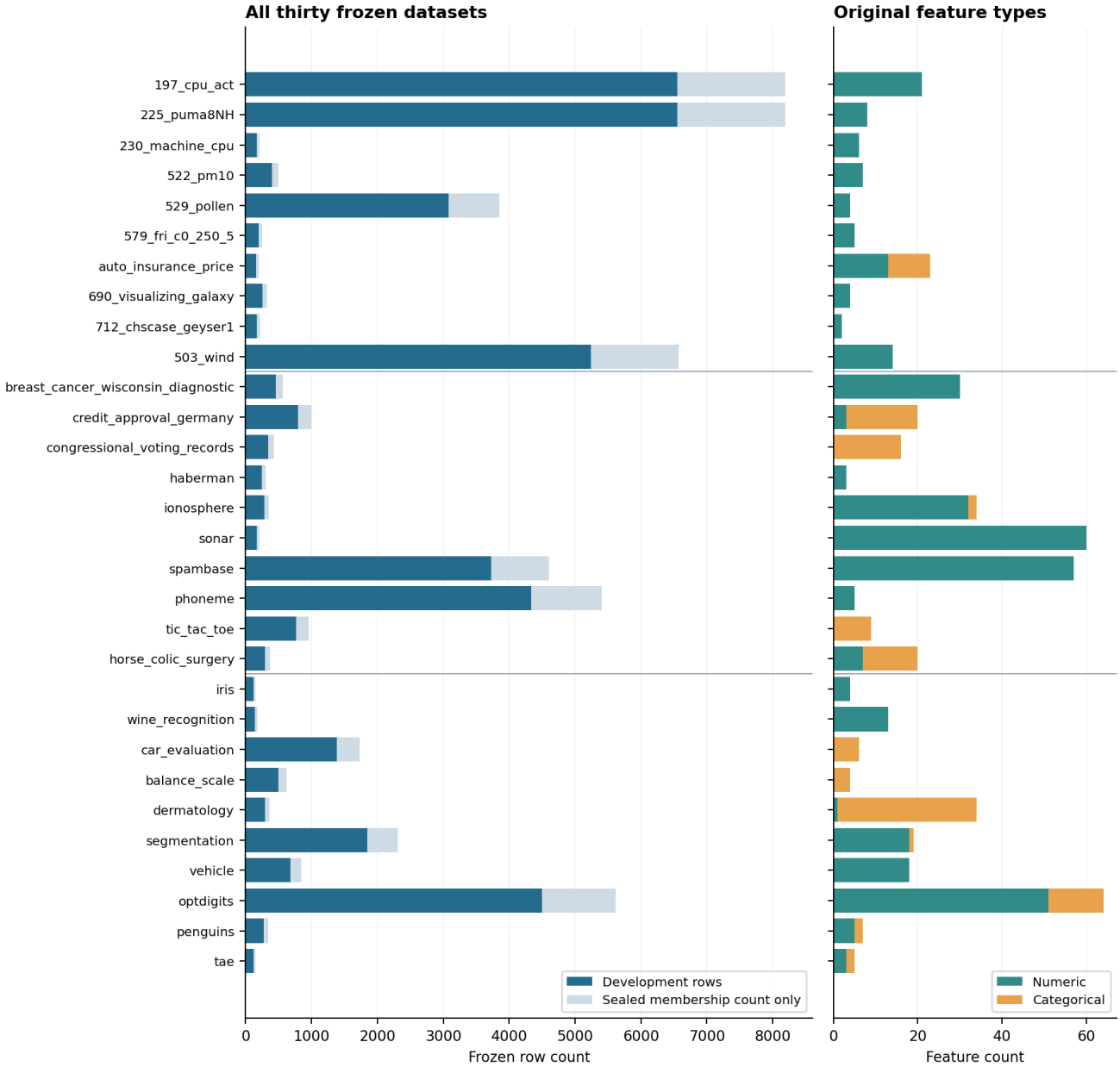
A.1.1 Per-dataset summary

Dataset	Task	Total rows	Development	Sealed metadata	Numeric features	Categorical features	Missing development cells	Fold sizes
197_cpu_act	regression	8192	6553	1639	21	0	0	1311,1311,1311,1310,1310
225_puma8NH	regression	8192	6553	1639	8	0	0	1311,1311,1311,1310,1310
230_machine_cpu	regression	209	169	40	6	0	0	34,32,37,33,33
522_pm10	regression	500	402	98	7	0	0	81,81,91,67,82
529_pollen	regression	3848	3078	770	4	0	0	616,616,616,615,615
579_fri_c0_250_5	regression	250	200	50	5	0	0	40,40,40,40,40
auto_insurance_price	regression	201	162	39	13	10	10	35,32,31,31,33
690_visualizing_galaxy	regression	323	258	65	4	0	0	52,52,52,51,51
712_chscase_geyser1	regression	222	172	50	2	0	0	32,34,38,35,33
503_wind	regression	6574	5238	1336	14	0	0	1066,1074,1030,1032,1036
breast_cancer_wisconsin_diagnostic	binary	569	455	114	30	0	0	91,91,91,91,91

Dataset	Task	Total rows	Development	Sealed metadata	Numeric features	Categorical features	Missing development cells	Fold sizes
credit_approval_germany	binary	1000	800	200	3	17	0	160,160,160,160,160
congressional_voting_records	binary	435	347	88	0	16	0	77,57,64,84,65
haberman	binary	306	245	61	3	0	0	49,49,46,50,51
ionosphere	binary	351	281	70	32	2	0	55,57,56,57,56
sonar	binary	208	166	42	60	0	0	34,33,33,33,33
spambase	binary	4601	3723	878	57	0	0	696,781,710,789,747
phoneme	binary	5404	4330	1074	5	0	0	866,867,866,870,861
tic_tac_toe	binary	958	766	192	0	9	0	154,153,153,153,153
horse_colic_surgery	binary	368	292	76	7	13	1583	58,61,58,58,57
iris	multiclass	150	122	28	4	0	0	25,23,25,25,24
wine_recognition	multiclass	178	142	36	13	0	0	29,29,28,28,28
car_evaluation	multiclass	1728	1382	346	0	6	0	277,277,276,276,276
balance_scale	multiclass	625	500	125	0	4	0	100,100,100,100,100
dermatology	multiclass	366	292	74	1	33	0	59,59,58,58,58
segmentation	multiclass	2310	1848	462	18	1	0	364,372,369,373,370
vehicle	multiclass	846	677	169	18	0	0	135,136,135,136,135
optdigits	multiclass	5620	4496	1124	51	13	0	900,899,899,899,899
penguins	multiclass	344	275	69	5	2	19	55,55,55,55,55
tae	multiclass	151	124	27	3	2	0	20,25,27,30,22

A.1.2 Frozen dataset coverage

The following chart covers all thirty immutable datasets in task order: regression, binary, multiclass. It preserves the original feature-type metadata and row assignments. Sealed counts come from membership metadata; sealed target bodies were not opened. The per-dataset sections under §A.2 Regression datasets, §A.3 Binary classification datasets and §A.4 Multiclass classification datasets give readable numeric tables, class/category frequencies and every feature record.



Regression / binary / multiclass: ten each. Sealed target bodies were not opened.
Source: pinned plot-dataset-sizes.csv and dataset-inventory.json; 23 September 2026.

Figure A.1 All thirty datasets: frozen rows and feature types

Totals are 55,029 rows, 44,048 development rows, 10,981 sealed membership records and 522 summed feature records. Feature counts are not a union of semantic variables across datasets. The chart and the detailed tables share the same pinned inventory.

A.2 Regression datasets

A.2.1 197_cpu_act

Task `regression`; metric `rmse`; 6553 development rows; 1639 sealed rows counted from assignments; 21 numeric and 0 categorical features. Development groups: 6553; all groups: 8192; largest group: 1. Original fold sizes:[1311, 1311, 1311, 1310, 1310]; other-four-fold fit sizes:[5242, 5242, 5242, 5243, 5243]. Declared grouping columns: none.

Development file SHA-256: `c987b431681784b87f12f3c98e660fd0aa76f08d01fe005695f41470b3750019`. Assignment SHA-256: `624d44ee0a5e2214e8edf256f8d9c028ca3f3765237401e3c84cf9c232a60bc7`. Original source SHA-256: `d00fd6bac2eda0821a04ff10277663d211127745a19a474c0feee41a16914fdc`. Original license: unknown; repository license: MIT.

Development target: min 0, median 89, max 99, mean 84.01938, population SD 18.290468; missing 0.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
lread	numeric	0	207	0	7	1845	19.349306	53.733421
lwrite	numeric	0	177	0	1	575	12.922173	29.496482
scall	numeric	0	3696	109	2053	12493	2307.3473	1631.4764
sread	numeric	0	752	6	166	5318	209.85686	199.84585
swrite	numeric	0	601	7	117	5456	149.81505	162.43571
fork	numeric	0	207	0	0.80000001	20.120001	1.8901007	2.4752517
exec	numeric	0	349	0	1.2	59.560001	2.7978773	5.2634111
rchar	numeric	0	6414	278	122879	2526649	195586.22	238487.78
wchar	numeric	0	6397	1498	46082	1801623	95334.943	140638.4
pgout	numeric	0	362	0	0	81.440002	2.2342881	5.2216254
ppgout	numeric	0	670	0	0	168.06	5.8141904	14.712169
pgfree	numeric	0	927	0	0	523	11.634935	31.594236
pgscan	numeric	0	1004	0	0	1237	21.142667	69.957532
atch	numeric	0	219	0	0	211.58	1.0821486	5.4985106
pgin	numeric	0	745	0	2.8	126.45	8.2493835	13.75562
ppgin	numeric	0	955	0	3.8099999	292.60999	12.334296	22.180491
pflt	numeric	0	2626	0	64.199997	899.79999	110.20706	114.35173
vflt	numeric	0	3310	0.2	121.56	1365	185.84375	190.4443
runqsz	numeric	0	253	1	2	2823	19.619182	128.21611
freemem	numeric	0	2757	55	579	12027	1768.3189	2497.6881
freeswap	numeric	0	6154	2	1137126	2042280	1328586	421591.38

A.2.2 225_puma8NH

Task `regression`; metric `rmse`; 6553 development rows; 1639 sealed rows counted from assignments; 8 numeric and 0 categorical features. Development groups: 6553; all groups: 8192; largest group: 1. Original fold sizes:[1311, 1311, 1311, 1310, 1310]; other-four-fold fit sizes:[5242, 5242, 5242, 5243, 5243]. Declared grouping columns: none.

Development file SHA-256: `31a4ce74a20613e192cab948518ab391feb0b70b0863fbe263d288d4a5df9c6e`. Assignment SHA-256: `f901eda7d22c08a8e7d6b8eb8455f21de4fc58eb4cb497657d2a56bd035cc8e1`. Original source SHA-256: `0be6203e167cc5e7b038368dbfe0a7790d5dc423c9d7e42887907c5c03f81c27`. Original license: unknown; repository license: MIT.

Development target: min-12.415322, median 1.054127, max 11.87619, mean 1.1382282, population SD 5.6448802; missing 0.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
theta1	numeric	0	6547	-1.884729	0.0060899998	1.884735	-0.00220025 74	1.0823712
theta2	numeric	0	6543	-1.8834831	0.016564	1.883306	0.010064784	1.0937572
theta3	numeric	0	6548	-1.884449	0.023234	1.884046	0.0076206044	1.0832023
thetad1	numeric	0	6548	-1.884632	-0.013532	1.884923	-0.007727921	1.0913692
thetad2	numeric	0	6549	-1.8847851	-0.030689999	1.884077	-0.013800457	1.09539
thetad3	numeric	0	6546	-1.884558	0.020455999	1.884746	0.0078690332	1.0864734
tau1	numeric	0	6534	-0.5999950 2	0.0047479998	0.599011	0.00079091185	0.34369033
tau2	numeric	0	6533	-0.5998719 9	0.0048799999	0.59991199	0.0012599463	0.34712905

A.2.3 230_machine_cpu

Task `regression`; metric `rmse`; 169 development rows; 40 sealed rows counted from assignments; 6 numeric and 0 categorical features. Development groups: 152; all groups: 190; largest group: 5. Original fold sizes:[34, 32, 37, 33, 33]; other-four-fold fit sizes:[135,137,132,136,136]. Declared grouping columns: none.

Development file SHA-256: `07c9b448d18c0479cb817235efff5ec242fd3723329dcd69582d39043d150c87`. Assignment SHA-256: `68d86fbc518c72ee20dc5feefac8ad2e5a2f7c36ea9613b2755c7177f6f1bf82`. Original source SHA-256: `937b0a23076f948343b8b7ac005bc70ea7080c391218fb4896462699b8cb59c4`. Original license: unknown; repository license: MIT.

Development target: min 6, median 50, max 1150, mean 111.53254, population SD 169.51452; missing 0.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
MYCT	numeric	0	56	17	110	1500	196.59172	248.93309
MMIN	numeric	0	20	64	2000	32000	2901.7751	3928.1762
MMAX	numeric	0	21	64	8000	64000	12481.811	12389.434
CACH	numeric	0	22	0	8	256	24.633136	38.093256
CHMIN	numeric	0	15	0	3	52	4.9940828	7.2483396
CHMAX	numeric	0	28	0	8	176	19.100592	27.884089

A.2.4 522_pm10

Task `regression`; metric `rmse`; 402 development rows; 98 sealed rows counted from assignments; 7 numeric and 0 categorical features. Development groups: 218; all groups: 273; largest group: 6. Original fold sizes:[81, 81, 91, 67, 82]; other-four-fold fit sizes:[321,321,311,335,320]. Declared grouping columns:['day'].

Development file SHA-256: `a33090c6e33323c48afd74338f72b1caa12376b6af4af00f7a48f6c03168df4c`. Assignment SHA-256: `13a00e1a058422110f6ff4caabdac7f648e5da976e10ed0a27be077534e469dd`. Original source SHA-256: `ecbabaa5a989a8c11f5a57e874c135f386da5379d6689ae6d5d4264a652fa5e0`. Original license: unknown; repository license: MIT.

Development target: min 0.69314998, median 3.3322001, max 5.3845, mean 3.3171902, population SD 0.85366052; missing 0.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
cars_per_hour	numeric	0	374	3.8066599	7.4974751	8.3520803	7.0094372	1.0713765
temperature_at_2m	numeric	0	197	-19	0.80000001	19.799999	0.51965173	6.3551238
wind_speed	numeric	0	71	0.40000001	2.8	9.3999996	3.1445274	1.7968553
temperature_diff_2m_25m	numeric	0	56	-5	0	4	0.11666667	0.97697158
wind_direction	numeric	0	288	5	103.2	358	140.67065	84.378122
hour_of_day	numeric	0	24	1	12	24	12.266169	6.8120686
day	numeric	0	218	32	206.5	605	303.22139	197.99917

A.2.5 529_pollen

Task `regression`; metric `rmse`; 3078 development rows; 770 sealed rows counted from assignments; 4 numeric and 0 categorical features. Development groups: 3078; all groups: 3848; largest group: 1. Original fold sizes:[616,616,616,615,615]; other-four-fold fit sizes:[2462, 2462, 2462, 2463, 2463]. Declared grouping columns: none.

Development file SHA-256: `4afdc668b7c7bdde004b651e639ee62542518fa9629a4b9a9d18f42d2bfe5cf5`. Assignment SHA-256: `225ab47f4b84b8b29f808751a567dc06e49af57f77d918402dc98929494b9aad`. Original source SHA-256: `2e9414f0216e75296ba4211c567dd8a60d269273b38f8e00bd8fea3a37d2e654`. Original license: unknown; repository license: MIT.

Development target: min-12.0391, median-0.0282, max 10.4934, mean 0.013061306, population SD 3.1652511; missing 0.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
RIDGE	numeric	0	3052	-23.283899	-0.1736	21.406601	-0.040160139	6.4118802
NUB	numeric	0	3047	-16.393499	-0.26875	17.258301	-0.017482649	5.1239751
CRACK	numeric	0	3057	-26.4652	-0.12075	25.6434	-0.07717969	7.9651552
WEIGHT	numeric	0	3067	-34.035198	-0.1534	35.802799	0.062779366	10.099992

A.2.6 579_fri_c0_250_5

Task `regression`; metric `rmse`; 200 development rows; 50 sealed rows counted from assignments; 5 numeric and 0 categorical features. Development groups: 200; all groups: 250; largest group: 1. Original fold sizes:[40, 40, 40, 40, 40]; other-four-fold fit sizes:[160,160,160,160,160]. Declared grouping columns: none.

Development file SHA-256: `9c2608cc2a14cb52e26f174518606fbc4bb7ed48b04aa4c62d840a7dc6cbd365`. Assignment SHA-256: `fa08e1237360b16d98c581d6c62695cb6c8631cca9940ba1671a6b8678e60790`. Original source SHA-256: `d32597bb1043df296ac3be1fff926646b26f268f89185310ea898e59b5fff896`. Original license: unknown; repository license: MIT.

Development target: min-2.2208376, median-0.050367098, max 2.1689749, mean-0.045378203, population SD 1.0368873; missing 0.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
oz1	numeric	0	200	-1.7117687	-0.040313881	1.7466414	-0.0097722925	0.99221231
oz2	numeric	0	200	-1.7675385	0.11001987	1.7166964	0.017360429	1.0225496
oz3	numeric	0	200	-1.798439	-0.013159709	1.6682318	0.024478181	0.97661234
oz4	numeric	0	200	-1.7556303	0.029955601	1.6519692	-0.027132471	0.97989126
oz5	numeric	0	200	-1.8355519	0.022033866	1.7011706	-0.017172821	1.0096145

A.2.7 auto_insurance_price

A.2.7.1 Summary and feature statistics

Task `regression`; metric `rmse`; 162 development rows; 39 sealed rows counted from assignments; 13 numeric and 10 categorical features. Development groups: 154; all groups: 193; largest group: 2. Original fold sizes:[35, 32, 31, 31, 33]; other-four-fold fit sizes:[127,130,131,131,129]. Declared grouping columns: none.

Development file SHA-256: `6c0719a5e4433e653e8997fc1f19842fd3aae044531e7639484d24d879d7f971`. Assignment SHA-256: `9769b5f1510b1048afe81125f8f7dbb1272ab81825305af5414c8da6c52c7b1f`. Original source SHA-256: `244c05ca33384bcdd519ff88fad036ae7140a5438b76a8248401cdc065582ac7`. Original license: CC BY 4.0; repository license: MIT.

Development target: min 5118, median 10096.5, max 45400, mean 13127.222, population SD 8034.0417; missing 0.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
make	categorical	0	21	n/a	n/a	n/a	n/a	n/a
fuel_type	categorical	0	2	n/a	n/a	n/a	n/a	n/a
aspiration	categorical	0	2	n/a	n/a	n/a	n/a	n/a
number_of_doors	categorical	0	2	n/a	n/a	n/a	n/a	n/a
body_style	categorical	0	5	n/a	n/a	n/a	n/a	n/a
drive_wheels	categorical	0	3	n/a	n/a	n/a	n/a	n/a
engine_location	categorical	0	2	n/a	n/a	n/a	n/a	n/a
wheel_base	numeric	0	49	86.6	96.9	120.9	98.664815	6.1873474
length	numeric	0	67	141.1	172.5	208.1	174.10123	12.321366
width	numeric	0	40	60.3	65.4	72	65.856173	2.1333027
height	numeric	0	46	47.8	54.1	59.8	53.730247	2.5110641
curb_weight	numeric	0	140	1488	2407.5	4066	2548.8457	518.36907
engine_type	categorical	0	6	n/a	n/a	n/a	n/a	n/a
number_of_cylinders	categorical	0	7	n/a	n/a	n/a	n/a	n/a
engine_size	numeric	0	40	61	110	326	126.39506	42.981816
fuel_system	categorical	0	7	n/a	n/a	n/a	n/a	n/a
bore	numeric	4	35	2.54	3.31	3.94	3.3218987	0.26890813
stroke	numeric	4	33	2.07	3.29	4.17	3.2515823	0.31914744
compression_ratio	numeric	0	28	7	9	23	10.003148	3.7645327
horsepower	numeric	1	52	48	95	262	103.68323	38.08872

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
peak_rpm	numeric	1	21	4150	5200	6600	5129.8137	480.66268
city_mpg	numeric	0	26	13	24.5	49	25.024691	6.2370365
highway_mpg	numeric	0	29	16	30	54	30.518519	6.6640942

A.2.7.2 Category frequencies

Exact development category frequencies:

- `make`: `{"0": 3, "1": 6, "11": 11, "12": 16, "13": 9, "14": 5, "15": 4, "16": 1, "17": 3, "18": 12, "19": 25, "2": 5, "20": 11, "21": 8, "3": 2, "4": 6, "5": 11, "6": 2, "7": 3, "8": 13, "9": 6}`. Categories absent from the other four folds: `{"0": ["16"], "1": [], "2": [], "3": [], "4": []}`.
- `fuel_type`: `{"0": 14, "1": 148}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `aspiration`: `{"0": 134, "1": 28}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `number_of_doors`: `{"0": 93, "1": 69}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `body_style`: `{"0": 7, "1": 22, "2": 75, "3": 54, "4": 4}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `drive_wheels`: `{"0": 8, "1": 96, "2": 58}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `engine_location`: `{"0": 159, "1": 3}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `engine_type`: `{"0": 11, "1": 10, "2": 112, "3": 15, "4": 10, "5": 4}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `number_of_cylinders`: `{"0": 4, "1": 1, "2": 124, "3": 9, "4": 20, "5": 3, "6": 1}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": ["6"], "3": ["1"], "4": []}`.
- `fuel_system`: `{"0": 9, "1": 55, "2": 3, "3": 14, "5": 72, "6": 8, "7": 1}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": ["7"], "4": []}`.

A.2.8 690_visualizing_galaxy

Task `regression`; metric `rmse`; 258 development rows; 65 sealed rows counted from assignments; 4 numeric and 0 categorical features. Development groups: 258; all groups: 323; largest group: 1. Original fold sizes:[52, 52, 52, 51, 51]; other-four-fold fit sizes:[206,206,206,207,207]. Declared grouping columns: none.

Development file SHA-256: `5c060a8e2459df5e8a533003febab98755a531f3bce0426ac1b617edf2445659`. Assignment SHA-256: `d3f63cf22ce07441cdb3e01d77160d3c36a6630c99c5e7533f9b5563e06db892`. Original source SHA-256: `5fe909e08a622224135a2dcfd409bbe10abab4919f917a07b771c3fb00580852`. Original license: unknown; repository license: MIT.

Development target: min 1409, median 1581, max 1769, mean 1588.6085, population SD 91.295658; missing 0.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
eastwest	numeric	0	258	-29.666929	-0.42766218	29.48414	-0.60256838	12.015002
northsouth	numeric	0	258	-49.107689	1.8613807	49.888725	2.4837183	22.042518
angle	numeric	0	7	12.5	102.5	133	81.486434	38.60098
radialposition	numeric	0	245	-51.099998	-1.8	55.700001	-1.5406977	25.187122

A.2.9 712_chscase_geyser1

Task `regression`; metric `rmse`; 172 development rows; 50 sealed rows counted from assignments; 2 numeric and 0 categorical features. Development groups: 135; all groups: 169; largest group: 4. Original fold sizes:[32, 34, 38, 35, 33]; other-four-fold fit sizes:[140,138,134,137,139]. Declared grouping columns: none.

Development file SHA-256: `a00344615df0ba8b22227753d6931e72c039bed451500c74c794a91038f63459`. Assignment SHA-256: `161ae799380a7919870d5ae493d2c026c6f36fda9fc2dae192ad5f9009fb205b`. Original source SHA-256: `9231e5c6bb08d6a7fab8509e4a8330063a5e3da77b75ff3b9dc412557ed3e334`. Original license: unknown; repository license: MIT.

Development target: min 43, median 75, max 95, mean 71.098837, population SD 12.668703; missing 0.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
col_1	numeric	0	16	1	16	23	12.209302	7.6502261
col_2	numeric	0	33	1.7	4.05	5.1999998	3.5883721	1.0771861

A.2.10 503_wind

Task `regression`; metric `rmse`; 5238 development rows; 1336 sealed rows counted from assignments; 14 numeric and 0 categorical features. Development groups: 172; all groups: 216; largest group: 31. Original fold sizes:[1066, 1074, 1030, 1032, 1036]; other-four-fold fit sizes:[4172, 4164, 4208, 4206, 4202]. Declared grouping columns:['year', 'month'].

Development file SHA-256: `3bb9b2400ba47561c0df144b7a0d2b8583ad589425d3ada4d0c6313c331d4fd2`. Assignment SHA-256: `1cc0ecb4f63ebddab23f309899171f350164ff1b91b81c144e32ad57640cf281`. Original source SHA-256: `2cc1e040bdc20737674a00039cd80e16758c511e697e524ed17e4335ec93e9a6`. Original license: unknown; repository license: MIT.

Development target: min 1.75, median 14.88, max 42.540001, mean 15.517566, population SD 6.6603293; missing 0.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
year	numeric	0	18	61	70	78	69.660939	5.171068
month	numeric	0	12	1	6	12	6.358152	3.3938973

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
day	numeric	0	31	1	16	31	15.737686	8.8047809
RPT	numeric	0	644	0.67000002	11.63	35.799999	12.284956	5.5789555
VAL	numeric	0	594	0.37	10.08	33.369999	10.579704	5.2260336
ROS	numeric	0	589	1.5	10.83	33.84	11.576588	4.9764516
KIL	numeric	0	439	0	5.71	28.459999	6.2592974	3.5713878
SHA	numeric	0	575	0.13	9.9200001	33.630001	10.369643	4.8910479
BIR	numeric	0	446	0	6.79	26.040001	7.0677301	3.9515225
DUB	numeric	0	565	0	9.0799999	30.370001	9.6849523	4.9379271
CLA	numeric	0	520	0	8.04	31.08	8.4148511	4.4694644
MUL	numeric	0	490	0	8.1199999	25.799999	8.4560596	4.1545162
CLO	numeric	0	520	0.039999999	8.21	28.209999	8.6498683	4.5013668
BEL	numeric	0	664	0.57999998	12.42	42.380001	13.044351	5.8033753

A.3 Binary classification datasets

A.3.1 breast_cancer_wisconsin_diagnostic

Task `binary`; metric `auroc`; 455 development rows; 114 sealed rows counted from assignments; 30 numeric and 0 categorical features. Development groups: 455; all groups: 569; largest group: 1. Original fold sizes:[91, 91, 91, 91, 91]; other-four-fold fit sizes:[364,364,364,364,364]. Declared grouping columns: none.

Development file SHA-256: `cc8bdc81b49bef92ae2aee74a4e919ea5d2515465adef12b7ebae53733e33fc6`. Assignment SHA-256: `41c454f412740e2996d3b7894af2e6d7c75042bdae9db222d14427e1ca7fb64c`. Original source SHA-256: `b3845ddf3afff808cd6449daa546caf4d8b18f109b7ebb2344990e4c37fea291`. Original license: CC BY 4.0; repository license: MIT.

Frozen class order: `["0", "1"]`. Development counts: `{"0": 284, "1": 171}`. Counts by original fold: `{"0": {"0": 55, "1": 36}, "1": {"0": 58, "1": 33}, "2": {"0": 61, "1": 30}, "3": {"0": 57, "1": 34}, "4": {"0": 53, "1": 38}}`.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
radius_mean	numeric	0	379	6.981	13.37	28.11	14.128822	3.5074831
texture_mean	numeric	0	395	9.71	18.83	39.28	19.337451	4.3409678
perimeter_mean	numeric	0	423	43.79	86.18	188.5	91.964747	24.188136
area_mean	numeric	0	437	143.5	551.1	2501	654.93604	347.93468

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
smoothness_mean	numeric	0	386	0.05263	0.09578	0.1634	0.096036352	0.014378444
compactness_mean	numeric	0	433	0.01938	0.09263	0.3454	0.10405275	0.053448375
concavity_mean	numeric	0	432	0	0.06154	0.4268	0.088762678	0.080580341
n_concave_mean	numeric	0	432	0	0.03326	0.2012	0.048554477	0.038664424
symmetry_mean	numeric	0	358	0.106	0.1784	0.304	0.18096022	0.028034982
fractal_dimension_mean	numeric	0	412	0.04996	0.06161	0.09744	0.062755077	0.00713061
radius_se	numeric	0	438	0.1144	0.3186	2.873	0.40416242	0.28494741
texture_se	numeric	0	421	0.3602	1.127	4.885	1.2224974	0.56657768
perimeter_se	numeric	0	432	0.757	2.244	21.98	2.8611211	2.0984043
area_se	numeric	0	428	6.802	24.25	542.2	40.441947	47.366591
smoothness_se	numeric	0	442	0.001713	0.006471	0.03113	0.0070618176	0.0030840353
compactness_se	numeric	0	438	0.002252	0.02048	0.1354	0.025522848	0.017892893
concavity_se	numeric	0	431	0	0.02586	0.396	0.03220903	0.031603666
n_concave_se	numeric	0	409	0	0.01083	0.05279	0.011810613	0.0063615603
symmetry_se	numeric	0	407	0.007882	0.01868	0.07895	0.020614919	0.008722101 8
fractal_dimension_se	numeric	0	437	0.0008948	0.003224	0.02984	0.0037881501	0.0024496254
radius_worst	numeric	0	387	7.93	14.98	36.04	16.256743	4.7838454
texture_worst	numeric	0	420	12.02	25.4	49.54	25.688879	6.1068704
perimeter_worst	numeric	0	419	50.41	97.96	251.2	107.09422	33.240634
area_worst	numeric	0	440	185.2	688.6	4254	878.64	556.25614
smoothness_worst	numeric	0	352	0.07117	0.1312	0.2226	0.13195897	0.023257748
compactness_worst	numeric	0	429	0.02729	0.2113	1.058	0.25405004	0.15864571
concavity_worst	numeric	0	432	0	0.2267	1.252	0.27054573	0.20677617

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
n_concave_worst	numeric	0	396	0	0.09861	0.291	0.11400506	0.065796181
symmetry_worst	numeric	0	408	0.1565	0.281	0.6638	0.2903422	0.064265468
fractal_dimension_worst	numeric	0	430	0.05504	0.07999	0.2075	0.084133473	0.018414541

A.3.2 credit_approval_germany

A.3.2.1 Summary and feature statistics

Task `binary`; metric `auroc`; 800 development rows; 200 sealed rows counted from assignments; 3 numeric and 17 categorical features. Development groups: 800; all groups: 1000; largest group: 1. Original fold sizes:[160,160,160,160,160]; other-four-fold fit sizes:[640,640,640,640,640]. Declared grouping columns: none.

Development file SHA-256: `34ffaac6a51ea0ac2e153ba828ce6b2437f685f6d181bb175356a1285c36ef7d`. Assignment SHA-256: `42532d431f177698ca65270388a542d6a687c9634648d523d601ea7763e5a7c4`. Original source SHA-256: `93e819f4b2d26df1cd5eb9805bd2fe15767941c6c8d66c58eba86d9249cb8ce1`. Original license: CC BY 4.0; repository license: MIT.

Frozen class order: `["0", "1"]`. Development counts: `{"0": 552, "1": 248}`. Counts by original fold: `{"0": {"0": 109, "1": 51}, "1": {"0": 107, "1": 53}, "2": {"0": 118, "1": 42}, "3": {"0": 115, "1": 45}, "4": {"0": 103, "1": 57}}`.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
credit_height	numeric	0	752	276	2330	18424	3317.4475	2867.9399
credit_duration	numeric	0	30	4	18	72	20.98125	12.204237
purpose	categorical	0	10	n/a	n/a	n/a	n/a	n/a
credit_guarantee	categorical	0	3	n/a	n/a	n/a	n/a	n/a
rate_fraction_income	categorical	0	4	n/a	n/a	n/a	n/a	n/a
credit_history	categorical	0	5	n/a	n/a	n/a	n/a	n/a
existing_credits_at_bank	categorical	0	4	n/a	n/a	n/a	n/a	n/a
other_credits	categorical	0	3	n/a	n/a	n/a	n/a	n/a
age	numeric	0	53	19	34	75	35.77375	11.346919
personal_status	categorical	0	4	n/a	n/a	n/a	n/a	n/a

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
dependents	categorical	0	2	n/a	n/a	n/a	n/a	n/a
employment_type	categorical	0	4	n/a	n/a	n/a	n/a	n/a
employment_length	categorical	0	5	n/a	n/a	n/a	n/a	n/a
foreign_worker	categorical	0	2	n/a	n/a	n/a	n/a	n/a
housing_status	categorical	0	3	n/a	n/a	n/a	n/a	n/a
residence_duration	categorical	0	4	n/a	n/a	n/a	n/a	n/a
checking_account_status	categorical	0	4	n/a	n/a	n/a	n/a	n/a
savings_account_status	categorical	0	5	n/a	n/a	n/a	n/a	n/a
assets	categorical	0	4	n/a	n/a	n/a	n/a	n/a
telephone	categorical	0	2	n/a	n/a	n/a	n/a	n/a

A.3.2.2 Category frequencies

Exact development category frequencies:

- `purpose`: `{"0": 181, "1": 78, "2": 151, "3": 226, "4": 9, "5": 15, "6": 41, "7": 8, "8": 81, "9": 10}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `credit_guarantee`: `{"0": 726, "1": 37, "2": 37}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `rate_fraction_income`: `{"0": 108, "1": 177, "2": 131, "3": 384}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `credit_history`: `{"0": 31, "1": 38, "2": 424, "3": 72, "4": 235}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `existing_credits_at_bank`: `{"0": 499, "1": 274, "2": 23, "3": 4}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `other_credits`: `{"0": 112, "1": 38, "2": 650}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `personal_status`: `{"0": 41, "1": 240, "2": 447, "3": 72}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `dependents`: `{"0": 671, "1": 129}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `employment_type`: `{"0": 16, "1": 166, "2": 492, "3": 126}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `employment_length`: `{"0": 51, "1": 131, "2": 261, "3": 145, "4": 212}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.

- `foreign_worker`: `{"0": 768, "1": 32}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `housing_status`: `{"0": 144, "1": 571, "2": 85}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `residence_duration`: `{"0": 104, "1": 243, "2": 113, "3": 340}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `checking_account_status`: `{"0": 212, "1": 211, "2": 58, "3": 319}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `savings_account_status`: `{"0": 485, "1": 84, "2": 54, "3": 33, "4": 144}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `assets`: `{"0": 226, "1": 187, "2": 264, "3": 123}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `telephone`: `{"0": 477, "1": 323}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.

A.3.3 congressional_voting_records

A.3.3.1 Summary and feature statistics

Task `binary`; metric `auroc`; 347 development rows; 88 sealed rows counted from assignments; 0 numeric and 16 categorical features. Development groups: 274; all groups: 342; largest group: 8. Original fold sizes:[77, 57, 64, 84, 65]; other-four-fold fit sizes:[270,290,283,263,282]. Declared grouping columns: none.

Development file SHA-256: `77382af6acc208516bcf44d9ff2911848b90c9893eca5499353e79dd7aa98c0c`. Assignment SHA-256: `3a38db7f711af084a16a69625d5ac782b2591fd2aa0682f0074afaa39bdc45c1`. Original source SHA-256: `fbee2fea90f1e1ce3251869a84b87723e37ab6a512c515a4fb6b9f9d72e5ee0c`. Original license: CC BY 4.0; repository license: MIT.

Frozen class order: `["0", "1"]`. Development counts: `{"0": 209, "1": 138}`. Counts by original fold: `{"0": {"0": 58, "1": 19}, "1": {"0": 33, "1": 24}, "2": {"0": 36, "1": 28}, "3": {"0": 44, "1": 40}, "4": {"0": 38, "1": 27}}`.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
handicapped_infants	categorical	0	3	n/a	n/a	n/a	n/a	n/a
water_project_cost_sharing	categorical	0	3	n/a	n/a	n/a	n/a	n/a
adoption_of_the_budget_resolution	categorical	0	3	n/a	n/a	n/a	n/a	n/a
physician_fee_freeze	categorical	0	3	n/a	n/a	n/a	n/a	n/a
el_salvador_aid	categorical	0	3	n/a	n/a	n/a	n/a	n/a
religious_groups_in_schools	categorical	0	3	n/a	n/a	n/a	n/a	n/a

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
anti_satellite_test_ban	categorical	0	3	n/a	n/a	n/a	n/a	n/a
aid_to_nicaraguan_contras	categorical	0	3	n/a	n/a	n/a	n/a	n/a
mx_missile	categorical	0	3	n/a	n/a	n/a	n/a	n/a
immigration	categorical	0	3	n/a	n/a	n/a	n/a	n/a
synfuels_corporation_cutback	categorical	0	3	n/a	n/a	n/a	n/a	n/a
education_spending	categorical	0	3	n/a	n/a	n/a	n/a	n/a
superfund_right_to_sue	categorical	0	3	n/a	n/a	n/a	n/a	n/a
crime	categorical	0	3	n/a	n/a	n/a	n/a	n/a
duty_free_exports	categorical	0	3	n/a	n/a	n/a	n/a	n/a
export_administation_act_south_africa	categorical	0	3	n/a	n/a	n/a	n/a	n/a

A.3.3.2 Category frequencies

Exact development category frequencies:

- `handicapped_infants`: `{"0": 142, "1": 194, "2": 11}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `water_project_cost_sharing`: `{"0": 157, "1": 160, "2": 30}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `adoption_of_the_budget_resolution`: `{"0": 200, "1": 138, "2": 9}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `physician_fee_freeze`: `{"0": 147, "1": 191, "2": 9}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `el_salvador_aid`: `{"0": 177, "1": 157, "2": 13}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `religious_groups_in_schools`: `{"0": 224, "1": 113, "2": 10}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `anti_satellite_test_ban`: `{"0": 185, "1": 151, "2": 11}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `aid_to_nicaraguan_contras`: `{"0": 192, "1": 144, "2": 11}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `mx_missile`: `{"0": 164, "1": 168, "2": 15}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.

- `immigration`: `{"0": 176, "1": 166, "2": 5}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `synfuels_corporation_cutback`: `{"0": 123, "1": 211, "2": 13}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `education_spending`: `{"0": 142, "1": 182, "2": 23}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `superfund_right_to_sue`: `{"0": 169, "1": 159, "2": 19}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `crime`: `{"0": 207, "1": 127, "2": 13}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `duty_free_exports`: `{"0": 132, "1": 195, "2": 20}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `export_administation_act_south_africa`: `{"0": 219, "1": 53, "2": 75}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.

A.3.4 haberman

Task `binary`; metric `auroc`; 245 development rows; 61 sealed rows counted from assignments; 3 numeric and 0 categorical features. Development groups: 226; all groups: 283; largest group: 3. Original fold sizes:[49, 49, 46, 50, 51]; other-four-fold fit sizes:[196,196,199,195,194]. Declared grouping columns: none.

Development file SHA-256: `6261779f619ecbdf9184c49da9fae7bb4f8bbcf1ffadffd51637db2673407b3a`. Assignment SHA-256: `705d871416d7b284d96331ead05b1bf138ba0669923cbe7e28668cbc63c6d56e`. Original source SHA-256: `2f44ac12e09d2fd09d9ac70231cecbfbc333356339dec0509ad1ed96e36e94c0`. Original license: unknown; repository license: MIT.

Frozen class order: `["1", "2"]`. Development counts: `{"1": 181, "2": 64}`. Counts by original fold: `{"0": {"1": 36, "2": 13}, "1": {"1": 36, "2": 13}, "2": {"1": 33, "2": 13}, "3": {"1": 38, "2": 12}, "4": {"1": 38, "2": 13}}`.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
Age_of_patient_at_time_of_operation	numeric	0	47	30	52	83	52.665306	10.685973
Patients_year_of_operation	numeric	0	12	58	63	69	62.857143	3.3036776
Number_of_positive_axillary_nodes_detected	numeric	0	28	0	1	46	3.9346939	6.6807537

A.3.5 ionosphere

Task `binary`; metric `auroc`; 281 development rows; 70 sealed rows counted from assignments; 32 numeric and 2 categorical features. Development groups: 280; all groups: 350; largest group: 2. Original fold sizes:[55, 57, 56, 57, 56]; other-four-fold fit sizes:[226,224,225,224,225]. Declared grouping columns: none.

Development file SHA-256: `9dffeb4dac21ce8104c8bb98f1f96968b1d6262c0c47bd2676293a615ef4bf16`. Assignment SHA-256: `73798005d06153b5109154cc728c53e95fe027f65fd548079e364750c38e7f50`. Original source SHA-256: `adb6fc877404f2b55f6d59fc7322265c733f5cf0ae6d7c2e52c3468733630023`. Original license: unknown; repository license: MIT.

Frozen class order: `["0", "1"]`. Development counts: `{"0": 102, "1": 179}`. Counts by original fold: `{"0": {"0": 21, "1": 34}, "1": {"0": 26, "1": 31}, "2": {"0": 18, "1": 38}, "3": {"0": 19, "1": 38}, "4": {"0": 18, "1": 38}}`.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
0	categorical	0	2	n/a	n/a	n/a	n/a	n/a
1	categorical	0	1	n/a	n/a	n/a	n/a	n/a
2	numeric	0	174	-1	0.88103	1	0.66155744	0.47010906
3	numeric	0	215	-1	0.00892	1	0.028568683	0.44112987
4	numeric	0	160	-1	0.82683	1	0.60080947	0.53074655
5	numeric	0	206	-1	0.0127	1	0.089482954	0.46367215
6	numeric	0	180	-1	0.76001	1	0.55640886	0.49881365
7	numeric	0	204	-1	0.01128	1	0.099825658	0.52851044
8	numeric	0	194	-1	0.71875	1	0.54520381	0.48753108
9	numeric	0	211	-1	0.01681	1	0.17432249	0.47797457
10	numeric	0	197	-1	0.68627	1	0.49457996	0.55173223
11	numeric	0	212	-1	0.02825	1	0.16022107	0.49491547
12	numeric	0	193	-1	0.67028	1	0.4104532	0.61450019
13	numeric	0	215	-1	0.02105	1	0.080409858	0.50169559
14	numeric	0	188	-1	0.62936	1	0.35983819	0.64793776
15	numeric	0	214	-1	0	1	0.077333772	0.47171614
16	numeric	0	203	-1	0.62121	1	0.39399758	0.61580948
17	numeric	0	226	-1	0	1	0.0025851957	0.49900626
18	numeric	0	203	-1	0.61996	1	0.39353537	0.61746332
19	numeric	0	213	-1	0	1	-0.020650036	0.52258519
20	numeric	0	198	-1	0.52494	1	0.3334358	0.61650624
21	numeric	0	211	-1	0	1	0.014242954	0.51554817
22	numeric	0	200	-1	0.53246	1	0.35669872	0.61030843

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
23	numeric	0	209	-1	0	1	-0.036835658	0.53328823
24	numeric	0	206	-1	0.55389	1	0.39690103	0.57383195
25	numeric	0	218	-1	-0.01664	1	-0.065681388	0.50686043
26	numeric	0	206	-1	0.69811	1	0.53114655	0.51992115
27	numeric	0	227	-1	-0.02294	1	-0.051712669	0.5482146
28	numeric	0	196	-1	0.48348	1	0.36189975	0.58595804
29	numeric	0	212	-1	0	1	-0.03398274	0.51497855
30	numeric	0	199	-1	0.45114	1	0.34863491	0.56760128
31	numeric	0	209	-1	0	1	-0.014466975	0.51454688
32	numeric	0	198	-1	0.40445	1	0.33919872	0.52562623
33	numeric	0	210	-1	0	1	0.0074272242	0.4767589

Exact development category frequencies:

- `0`: `{"0": 30, "1": 251}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `1`: `{"0": 281}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.

A.3.6 sonar

A.3.6.1 Summary, identities and class counts

Task `binary`; metric `auroc`; 166 development rows; 42 sealed rows counted from assignments; 60 numeric and 0 categorical features. Development groups: 166; all groups: 208; largest group: 1. Original fold sizes:[34, 33, 33, 33, 33]; other-four-fold fit sizes:[132,133,133,133,133]. Declared grouping columns: none.

Development file SHA-256: `413956c1d626d6671e4b2e99f06ef3c9dbeab52cb947aa2dda52c5eb4d590ec4`. Assignment SHA-256: `dce5e4eb3541c056cab75d72b0be5212d4788bfec249a54a6ffe6160e82853f9`. Original source SHA-256: `64378cabaecdb50e2e1e644c1d6bc359f0627ebe313870ad09e1e41ed972529b`. Original license: unknown; repository license: MIT.

Frozen class order: `["0", "1"]`. Development counts: `{"0": 88, "1": 78}`. Counts by original fold: `{"0": {"0": 19, "1": 15}, "1": {"0": 20, "1": 13}, "2": {"0": 16, "1": 17}, "3": {"0": 15, "1": 18}, "4": {"0": 18, "1": 15}}`.

A.3.6.2 Feature statistics

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
A1	numeric	0	141	0.0015	0.0217	0.1371	0.028841566	0.023686115
A2	numeric	0	151	0.0006	0.0286	0.2339	0.037833735	0.034101163
A3	numeric	0	157	0.0015	0.0321	0.3059	0.042960843	0.037819594
A4	numeric	0	152	0.0058	0.04405	0.4264	0.053654217	0.046923156
A5	numeric	0	157	0.0067	0.0625	0.401	0.075328916	0.055578143
A6	numeric	0	159	0.0102	0.0898	0.3823	0.10617169	0.061505341
A7	numeric	0	159	0.0033	0.11245	0.3729	0.12627711	0.064775691
A8	numeric	0	162	0.0055	0.1117	0.459	0.13576145	0.088328103
A9	numeric	0	164	0.0075	0.15375	0.6828	0.18029096	0.12123601
A10	numeric	0	166	0.0113	0.1809	0.7106	0.2100741	0.138032
A11	numeric	0	162	0.0289	0.2273	0.7342	0.23626867	0.13285597
A12	numeric	0	165	0.0259	0.24905	0.706	0.24793976	0.13719085
A13	numeric	0	161	0.0184	0.25735	0.7131	0.27112349	0.13802798
A14	numeric	0	165	0.0273	0.2811	0.997	0.29665482	0.16533271
A15	numeric	0	163	0.0031	0.27805	0.9137	0.32257771	0.20514931
A16	numeric	0	162	0.0162	0.31505	0.9751	0.38494157	0.23299201
A17	numeric	0	161	0.0349	0.3242	1	0.42208735	0.26767643
A18	numeric	0	163	0.0375	0.3712	1	0.45387711	0.26423326
A19	numeric	0	164	0.0746	0.42215	1	0.49631566	0.25898032
A20	numeric	0	162	0.0656	0.4977	1	0.54843916	0.26497105
A21	numeric	0	162	0.0512	0.58605	1	0.59400783	0.25705453
A22	numeric	0	162	0.0219	0.65335	1	0.61085241	0.2582013
A23	numeric	0	157	0.0563	0.66805	1	0.63666627	0.24875836
A24	numeric	0	159	0.0239	0.68315	1	0.6634747	0.23756303
A25	numeric	0	159	0.024	0.7211	1	0.6719988	0.24365831
A26	numeric	0	155	0.0921	0.7545	1	0.69583916	0.23788158
A27	numeric	0	155	0.0481	0.7211	1	0.7011488	0.24248268
A28	numeric	0	155	0.0284	0.7337	1	0.69982289	0.2339089

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
A29	numeric	0	158	0.0144	0.6808	1	0.64515843	0.23642261
A30	numeric	0	162	0.1354	0.61945	1	0.58621928	0.21518403
A31	numeric	0	166	0.0482	0.5037	0.9657	0.51252169	0.21079284
A32	numeric	0	164	0.0404	0.44535	0.9306	0.44586867	0.20698438
A33	numeric	0	163	0.0477	0.39375	0.9708	0.42063614	0.19916545
A34	numeric	0	165	0.0212	0.3682	0.9647	0.40769639	0.23003905
A35	numeric	0	163	0.0223	0.3369	1	0.40306205	0.2611019
A36	numeric	0	164	0.0185	0.32115	1	0.3937	0.27059015
A37	numeric	0	166	0.0379	0.3103	0.9497	0.37263133	0.24345705
A38	numeric	0	164	0.0383	0.3153	1	0.34736988	0.21649059
A39	numeric	0	163	0.0407	0.28345	0.9857	0.33274458	0.20047974
A40	numeric	0	164	0.0202	0.28225	0.9297	0.32078976	0.17907208
A41	numeric	0	164	0.0438	0.26305	0.8995	0.29696867	0.17064681
A42	numeric	0	166	0.0056	0.2534	0.8246	0.27926265	0.17015944
A43	numeric	0	163	0.0159	0.2162	0.7733	0.24302651	0.14246534
A44	numeric	0	157	0.0255	0.1751	0.7762	0.21003795	0.13362077
A45	numeric	0	165	0.0046	0.148	0.7034	0.19582169	0.1514589
A46	numeric	0	161	0.008	0.1179	0.7292	0.16208976	0.13726397
A47	numeric	0	163	0.0073	0.10385	0.5522	0.12454398	0.089007856
A48	numeric	0	162	0.0052	0.0803	0.3339	0.09448494	0.063893903
A49	numeric	0	154	0.0073	0.0451	0.1981	0.053189157	0.035846172
A50	numeric	0	132	0.0006	0.0179	0.0825	0.020369277	0.013156943
A51	numeric	0	134	0.0009	0.014	0.1004	0.015896386	0.01129344
A52	numeric	0	127	0.0008	0.01125	0.0709	0.013522289	0.009936481 3
A53	numeric	0	117	0.0005	0.00955	0.039	0.010813855	0.007042289 7
A54	numeric	0	114	0.0011	0.0094	0.0352	0.010825904	0.007082550 5
A55	numeric	0	113	0.0006	0.0076	0.0447	0.009561445 8	0.007155254 7
A56	numeric	0	108	0.0004	0.00705	0.0394	0.008278915 7	0.005786719 4

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
A57	numeric	0	105	0.0003	0.00575	0.0355	0.007623494	0.0057258021
A58	numeric	0	115	0.0003	0.0059	0.044	0.0080210843	0.0065779644
A59	numeric	0	103	0.0001	0.0061	0.0332	0.0079795181	0.0061732079
A60	numeric	0	98	0.0007	0.0053	0.0439	0.0065289157	0.0052317043

A.3.7 spambase

A.3.7.1 Summary, identities and class counts

Task `binary`; metric `auroc`; 3723 development rows; 878 sealed rows counted from assignments; 57 numeric and 0 categorical features. Development groups: 3381; all groups: 4207; largest group: 69. Original fold sizes:[696,781,710,789,747]; other-four-fold fit sizes:[3027, 2942, 3013, 2934, 2976]. Declared grouping columns: none.

Development file SHA-256: `f02d0bf948106d71bdc4dd57a1acad05e36935396d3ce2450aa326e863d93748`. Assignment SHA-256: `d740e152f7c9942115e88fe36274779a4187bc3cdafa050d75d75ca207804991`. Original source SHA-256: `735e8d43f7ee78ac54ca17404510e95f4c8b364a721e4d9532a1b3221ed70961`. Original license: unknown; repository license: MIT.

Frozen class order: `["0", "1"]`. Development counts: `{"0": 2261, "1": 1462}`. Counts by original fold: `{"0": {"0": 413, "1": 283}, "1": {"0": 486, "1": 295}, "2": {"0": 422, "1": 288}, "3": {"0": 489, "1": 300}, "4": {"0": 451, "1": 296}}`.

A.3.7.2 Feature statistics

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
0	numeric	0	136	0	0	4.34	0.10836691	0.31447626
1	numeric	0	158	0	0	14.28	0.23424926	1.4140581
2	numeric	0	208	0	0	5.1	0.28450175	0.50613967
3	numeric	0	39	0	0	42.81	0.076145582	1.53401
4	numeric	0	238	0	0	8.33	0.30181574	0.63247696
5	numeric	0	134	0	0	5.88	0.099076014	0.28457055
6	numeric	0	159	0	0	7.27	0.1084448	0.37299389
7	numeric	0	154	0	0	11.11	0.10242815	0.40766094
8	numeric	0	136	0	0	5.26	0.090405587	0.28227639
9	numeric	0	233	0	0	18.18	0.24231534	0.66893504

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
10	numeric	0	106	0	0	2.61	0.057700779	0.19908542
11	numeric	0	301	0	0.12	9.67	0.53767123	0.85542489
12	numeric	0	147	0	0	5.55	0.094139135	0.29447118
13	numeric	0	119	0	0	10	0.057160892	0.3424115
14	numeric	0	105	0	0	4.41	0.050021488	0.26409551
15	numeric	0	238	0	0	20	0.25256245	0.86568242
16	numeric	0	178	0	0	7.14	0.14358313	0.44016113
17	numeric	0	209	0	0	7.69	0.17846092	0.49965964
18	numeric	0	549	0	1.3	18.75	1.6304647	1.7444236
19	numeric	0	133	0	0	18.18	0.087045393	0.51437355
20	numeric	0	388	0	0.23	11.11	0.82221595	1.2353801
21	numeric	0	81	0	0	17.1	0.12898469	1.0758037
22	numeric	0	157	0	0	5.45	0.10662369	0.36628547
23	numeric	0	133	0	0	12.5	0.088568359	0.37568098
24	numeric	0	355	0	0	20.83	0.53969917	1.6402831
25	numeric	0	251	0	0	16.66	0.2710744	0.91077722
26	numeric	0	221	0	0	33.33	0.87795326	3.6989178
27	numeric	0	179	0	0	5.88	0.12094547	0.51186153
28	numeric	0	140	0	0	14.28	0.09994628	0.61141527
29	numeric	0	167	0	0	5.88	0.10789954	0.47740186
30	numeric	0	113	0	0	12.5	0.068775181	0.4277239
31	numeric	0	90	0	0	4.76	0.049648133	0.34482243
32	numeric	0	167	0	0	8.33	0.091326887	0.48361691
33	numeric	0	97	0	0	4.76	0.050577491	0.34585255
34	numeric	0	161	0	0	20	0.10762557	0.5614855
35	numeric	0	151	0	0	4.76	0.099151222	0.40425707
36	numeric	0	182	0	0	6.89	0.13856299	0.43320226

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
37	numeric	0	46	0	0	8.33	0.014488316	0.24219157
38	numeric	0	143	0	0	11.11	0.07558152	0.42405656
39	numeric	0	116	0	0	4.76	0.067894171	0.36984345
40	numeric	0	89	0	0	7.14	0.045412302	0.37629292
41	numeric	0	166	0	0	14.28	0.12858716	0.77234937
42	numeric	0	127	0	0	3.57	0.045417674	0.21994189
43	numeric	0	143	0	0	20	0.080515713	0.65927373
44	numeric	0	220	0	0	21.42	0.29374161	0.99304581
45	numeric	0	202	0	0	22.05	0.18154714	0.94847185
46	numeric	0	34	0	0	2.17	0.005962933 1	0.082484296
47	numeric	0	93	0	0	5	0.025753425	0.19009714
48	numeric	0	280	0	0	4.385	0.03976014	0.24401967
49	numeric	0	594	0	0.065	9.752	0.14165941	0.28547001
50	numeric	0	203	0	0	4.081	0.017885039	0.11686678
51	numeric	0	867	0	0	32.478	0.26717996	0.80078459
52	numeric	0	456	0	0	6.003	0.07593285	0.24670478
53	numeric	0	279	0	0	19.829	0.046236637	0.46825088
54	numeric	0	1867	1	2.285	1102.5	5.2333	32.564591
55	numeric	0	244	1	15	9989	53.011281	208.61383
56	numeric	0	827	1	97	15841	282.47059	604.67923

A.3.8 phoneme

Task `binary`; metric `auroc`; 4330 development rows; 1074 sealed rows counted from assignments; 5 numeric and 0 categorical features. Development groups: 4279; all groups: 5349; largest group: 2. Original fold sizes:[866,867,866,870,861]; other-four-fold fit sizes:[3464, 3463, 3464, 3460, 3469]. Declared grouping columns: none.

Development file SHA-256: `bbb0adca013fc84e710901a2a7ad7523000c351e38c9e041cbd9bef125315c3e`. Assignment SHA-256: `0b377d73783b9e26995fe4a821dfefc745d84316f7f457bcecf4985bb832e0f2`. Original source SHA-256: `2e1c31a2c560f3539be5bda1084939765c80ddd814bd6919fd8a3465f9b8aee6`. Original license: unknown; repository license: MIT.

Frozen class order: `["0", "1"]`. Development counts: `{"0": 3048, "1": 1282}`. Counts by original fold: `{"0": {"0": 610, "1": 256}, "1": {"0": 623, "1": 244}, "2": {"0": 583, "1": 283}, "3": {"0": 623, "1": 247}, "4": {"0": 609, "1": 252}}`.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
Aa	numeric	0	1851	-1.7	0.4865	4.107	0.82364804	0.86789971
Ao	numeric	0	2210	-1.327	1.068	4.378	1.2516467	0.85489827
Dcl	numeric	0	2277	-1.641	0.716	3.155	0.76292702	0.92614533
Iy	numeric	0	2152	-1.494	0.286	2.785	0.40085681	0.79622039
Sh	numeric	0	1607	-1.246	-0.044	2.719	0.079781524	0.57446239

A.3.9 tic_tac_toe

A.3.9.1 Summary and feature statistics

Task `binary`; metric `auroc`; 766 development rows; 192 sealed rows counted from assignments; 0 numeric and 9 categorical features. Development groups: 766; all groups: 958; largest group: 1. Original fold sizes:[154,153,153,153,153]; other-four-fold fit sizes:[612,613,613,613,613]. Declared grouping columns: none.

Development file SHA-256: `3d2766839ac876c2c3342d3340b2b10d78aae59d0663ec4ced605836f86fc643`. Assignment SHA-256: `5f25ffba77126e2c010512008c353731e2c65de4d33d7b47d2cad20a9366cf1b`. Original source SHA-256: `d00d1522a06b5b67a0aea355c268ffe575e4a28651fa4fbf0e204705be3ce47b`. Original license: unknown; repository license: MIT.

Frozen class order: `["0", "1"]`. Development counts: `{"0": 252, "1": 514}`. Counts by original fold: `{"0": {"0": 54, "1": 100}, "1": {"0": 53, "1": 100}, "2": {"0": 51, "1": 102}, "3": {"0": 47, "1": 106}, "4": {"0": 47, "1": 106}}`.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
t1	categorical	0	3	n/a	n/a	n/a	n/a	n/a
t2	categorical	0	3	n/a	n/a	n/a	n/a	n/a
t3	categorical	0	3	n/a	n/a	n/a	n/a	n/a
t4	categorical	0	3	n/a	n/a	n/a	n/a	n/a
t5	categorical	0	3	n/a	n/a	n/a	n/a	n/a
t6	categorical	0	3	n/a	n/a	n/a	n/a	n/a
t7	categorical	0	3	n/a	n/a	n/a	n/a	n/a
t8	categorical	0	3	n/a	n/a	n/a	n/a	n/a
t9	categorical	0	3	n/a	n/a	n/a	n/a	n/a

A.3.9.2 Category frequencies

Exact development category frequencies:

- t1: `{"0": 178, "1": 260, "2": 328}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- t2: `{"0": 201, "1": 266, "2": 299}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- t3: `{"0": 168, "1": 268, "2": 330}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- t4: `{"0": 204, "1": 262, "2": 300}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- t5: `{"0": 119, "1": 273, "2": 374}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- t6: `{"0": 196, "1": 261, "2": 309}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- t7: `{"0": 160, "1": 263, "2": 343}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- t8: `{"0": 214, "1": 262, "2": 290}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- t9: `{"0": 167, "1": 265, "2": 334}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.

A.3.10 horse_colic_surgery

A.3.10.1 Summary and feature statistics

Task `binary`; metric `auroc`; 292 development rows; 76 sealed rows counted from assignments; 7 numeric and 13 categorical features. Development groups: 283; all groups: 354; largest group: 4. Original fold sizes:[58, 61, 58, 58, 57]; other-four-fold fit sizes:[234,231,234,234,235]. Declared grouping columns: none.

Development file SHA-256: `4b92b18271d7a289565d1845df42b8d903b22c321954cc78dd9fba90800f92ff`. Assignment SHA-256: `1cd242e37b4b64ebc39ba83e834f7822ebe1bc19746b8970c74574deb99f3539`. Original source SHA-256: `5079ad94ed5d03057e754b97692f1675603d515ebd663fcf34b87c45103e449c`. Original license: CC BY 4.0; repository license: MIT.

Frozen class order: `["0", "1"]`. Development counts: `{"0": 180, "1": 112}`. Counts by original fold: `{"0": {"0": 38, "1": 20}, "1": {"0": 35, "1": 26}, "2": {"0": 31, "1": 27}, "3": {"0": 38, "1": 20}, "4": {"0": 38, "1": 19}}`.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
age_category	categorical	25	1	n/a	n/a	n/a	n/a	n/a
rectal_temperature	numeric	59	38	35.4	38.1	40.8	38.139056	0.73254976
heart_rate	numeric	20	53	30	62	184	71.330882	29.13295

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
respiratory_rate	numeric	62	38	8	28	96	31.617391	18.790878
extremity_temperature	categorical	53	4	n/a	n/a	n/a	n/a	n/a
peripheral_pulse	categorical	67	4	n/a	n/a	n/a	n/a	n/a
mucous_membrane_appearance	categorical	39	6	n/a	n/a	n/a	n/a	n/a
capillary_refill_time	categorical	36	2	n/a	n/a	n/a	n/a	n/a
pain	categorical	47	5	n/a	n/a	n/a	n/a	n/a
peristalsis	categorical	41	4	n/a	n/a	n/a	n/a	n/a
abdominal_distension	categorical	53	4	n/a	n/a	n/a	n/a	n/a
nasogastric_tube_stomach_gas	categorical	115	3	n/a	n/a	n/a	n/a	n/a
nasogastric_tube_reflux_volume	categorical	112	3	n/a	n/a	n/a	n/a	n/a
nasogastric_tube_reflux_ph	numeric	237	22	1	5	8	4.6490909	1.9697598
rectal_feces_presence	categorical	100	4	n/a	n/a	n/a	n/a	n/a
rectal_examination	categorical	112	5	n/a	n/a	n/a	n/a	n/a
red_blood_cell_count	numeric	31	50	4	44	75	44.961303	10.838738
blood_protein	numeric	37	78	3.3	7.5	86	25.715294	27.945219
abdominocentesis_appearance	categorical	154	3	n/a	n/a	n/a	n/a	n/a
abdominocentesis_protein	numeric	183	38	0.1	2	10.1	2.8963303	1.841366

A.3.10.2 Category frequencies

Exact development category frequencies:

- `age_category`: `{"0": 267}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `extremity_temperature`: `{"0": 74, "1": 33, "2": 107, "3": 25}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.

- peripheral_pulse: {"0": 5, "1": 118, "2": 91, "3": 11}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- mucous_membrane_appearance: {"0": 87, "1": 25, "2": 61, "3": 41, "4": 20, "5": 19}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- capillary_refill_time: {"0": 180, "1": 76}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- pain: {"0": 44, "1": 62, "2": 65, "3": 37, "4": 37}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- peristalsis: {"0": 41, "1": 16, "2": 123, "3": 71}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- abdominal_distension: {"0": 82, "1": 60, "2": 64, "3": 33}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- nasogastric_tube_stomach_gas: {"0": 61, "1": 92, "2": 24}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- nasogastric_tube_reflux_volume: {"0": 109, "1": 37, "2": 34}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- rectal_feces_presence: {"0": 11, "1": 56, "2": 50, "3": 75}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- rectal_examination: {"0": 27, "1": 17, "2": 18, "3": 40, "4": 78}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- abdominocentesis_appearance: {"0": 43, "1": 43, "2": 52}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.

A.4 Multiclass classification datasets

A.4.1 iris

Task multiclass; metric log_loss; 122 development rows; 28 sealed rows counted from assignments; 4 numeric and 0 categorical features. Development groups: 119; all groups: 147; largest group: 3. Original fold sizes:[25, 23, 25, 25, 24]; other-four-fold fit sizes:[97, 99, 97, 97, 98]. Declared grouping columns: none.

Development file SHA-256: 86b4b47bc14f4513838072b130aed1b9c4f5bd9056cc110215a8c66be3d9dcc3. Assignment SHA-256: 4322393c97bb172340ad840055cbc0edb09b8321ece9d7ce5b0aed1934aadfad. Original source SHA-256: ce4982cba30bd5aaa352e51d55e1cf56c56ae6d573d644f36257de1177717228. Original license: unknown; repository license: MIT.

Frozen class order: ["0", "1", "2"]. Development counts: {"0": 37, "1": 40, "2": 45}. Counts by original fold: {"0": {"0": 10, "1": 10, "2": 5}, "1": {"0": 5, "1": 11, "2": 7}, "2": {"0": 8, "1": 7, "2": 10}, "3": {"0": 7, "1": 8, "2": 10}, "4": {"0": 7, "1": 4, "2": 13}}.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
sepal-length	numeric	0	34	4.3	5.8	7.9	5.9008197	0.8357284
sepal-width	numeric	0	23	2	3	4.4	3.0491803	0.4435782
petal-length	numeric	0	40	1	4.5	6.9	3.8901639	1.7489767

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
petal-width	numeric	0	21	0.1	1.4	2.5	1.2540984	0.76170081

A.4.2 wine_recognition

Task `multiclass`; metric `log_loss`; 142 development rows; 36 sealed rows counted from assignments; 13 numeric and 0 categorical features. Development groups: 142; all groups: 178; largest group: 1. Original fold sizes:[29, 29, 28, 28, 28]; other-four-fold fit sizes:[113,113,114,114,114]. Declared grouping columns: none.

Development file SHA-256: `ff8de7ed12ef935c0d42be31c08bae6ca18ae909e60dee1d9c1887ce333e253f`. Assignment SHA-256: `0a5a84bbb63803ef0b31e7fde40c3fcd2d0e2246b62033cc820be3c907917614`. Original source SHA-256: `56f5ba3f3ba78f6e522a13e5a97e03ff28cf9fa4107021b22b932f6d4064c145`. Original license: unknown; repository license: MIT.

Frozen class order: `["1", "2", "3"]`. Development counts: `{"1": 47, "2": 57, "3": 38}`. Counts by original fold: `{"0": {"1": 10, "2": 11, "3": 8}, "1": {"1": 9, "2": 12, "3": 8}, "2": {"1": 10, "2": 11, "3": 7}, "3": {"1": 9, "2": 12, "3": 7}, "4": {"1": 9, "2": 11, "3": 8}}`.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
1	numeric	0	106	11.45	13.05	14.83	13.017746	0.78176704
2	numeric	0	114	0.89	1.88	5.8	2.3453521	1.140337
3	numeric	0	74	1.36	2.36	3.23	2.3671831	0.27320313
4	numeric	0	55	10.6	19.25	30	19.509859	3.3591227
5	numeric	0	48	70	98	162	99.43662	14.362247
6	numeric	0	80	0.98	2.38	3.88	2.3109155	0.62909045
7	numeric	0	111	0.34	2.135	5.08	2.0360563	1.0064125
8	numeric	0	39	0.13	0.335	0.66	0.36577465	0.12517213
9	numeric	0	89	0.42	1.545	3.58	1.6009859	0.58075874
10	numeric	0	111	1.28	4.69	13	5.0982394	2.3589459
11	numeric	0	71	0.48	0.96	1.45	0.94574648	0.21709943
12	numeric	0	106	1.27	2.78	3.82	2.6256338	0.69527932
13	numeric	0	106	278	673.5	1680	751.73239	317.35709

A.4.3 car_evaluation

A.4.3.1 Summary and feature statistics

Task `multiclass`; metric `log_loss`; 1382 development rows; 346 sealed rows counted from assignments; 0 numeric and 6 categorical features. Development groups: 1382; all groups: 1728; largest group: 1. Original fold sizes:[277,277,276,276,276]; other-four-fold fit sizes:[1105, 1105, 1106, 1106, 1106]. Declared grouping columns: none.

Development file SHA-256: `014c5c81b356e617cac7daa8901914c12837f5458c221882466f2536f047fc46`. Assignment SHA-256: `c36e08e3b50f2c560f4a0d06cf8d719da1a1352870246ad04f12ec7b7a855fab`. Original source SHA-256: `9c18fe91ba5401cab754456a9bfbfe54a6c307b0d4b880259fb4218edef77998`. Original license: CC BY 4.0; repository license: MIT.

Frozen class order: `["0", "1", "2", "3"]`. Development counts: `{"0": 967, "1": 310, "2": 49, "3": 56}`. Counts by original fold: `{"0": {"0": 194, "1": 64, "2": 10, "3": 9}, "1": {"0": 195, "1": 59, "2": 12, "3": 11}, "2": {"0": 200, "1": 53, "2": 9, "3": 14}, "3": {"0": 186, "1": 69, "2": 10, "3": 11}, "4": {"0": 192, "1": 65, "2": 8, "3": 11}}`.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
buying_price	categorical	0	4	n/a	n/a	n/a	n/a	n/a
maintenance_costs	categorical	0	4	n/a	n/a	n/a	n/a	n/a
doors	categorical	0	4	n/a	n/a	n/a	n/a	n/a
seats	categorical	0	3	n/a	n/a	n/a	n/a	n/a
luggage_boot	categorical	0	3	n/a	n/a	n/a	n/a	n/a
safety	categorical	0	3	n/a	n/a	n/a	n/a	n/a

A.4.3.2 Category frequencies

Exact development category frequencies:

- `buying_price`: `{"0": 345, "1": 346, "2": 346, "3": 345}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `maintenance_costs`: `{"0": 345, "1": 346, "2": 346, "3": 345}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `doors`: `{"0": 341, "1": 351, "2": 346, "3": 344}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `seats`: `{"0": 454, "1": 466, "2": 462}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `luggage_boot`: `{"0": 448, "1": 468, "2": 466}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `safety`: `{"0": 467, "1": 468, "2": 447}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.

A.4.4 balance_scale

A.4.4.1 Summary and feature statistics

Task `multiclass`; metric `log_loss`; 500 development rows; 125 sealed rows counted from assignments; 0 numeric and 4 categorical features. Development groups: 500; all groups: 625; largest group: 1. Original fold sizes:[100,100,100,100,100]; other-four-fold fit sizes:[400,400,400,400,400]. Declared grouping columns: none.

Development file SHA-256: `9267117d816d008803477a791ca9cf346fefc17aba62afc9a9b5d444e9530329`. Assignment SHA-256: `40efdd8487cb7076e85a00d07e1d42d4902a84e1e9aef740d291e341b2509764`. Original source SHA-256: `8c87be0e86f362154b34bd608caf8c5804e357e437b7db91b14974b6afaffe32`. Original license: unknown; repository license: MIT.

Frozen class order: `["0", "1", "2"]`. Development counts: `{"0": 40, "1": 211, "2": 249}`. Counts by original fold: `{"0": {"0": 4, "1": 49, "2": 47}, "1": {"0": 11, "1": 42, "2": 47}, "2": {"0": 8, "1": 39, "2": 53}, "3": {"0": 11, "1": 43, "2": 46}, "4": {"0": 6, "1": 38, "2": 56}}`.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
Left-Weight	categorical	0	5	n/a	n/a	n/a	n/a	n/a
Left-Distance	categorical	0	5	n/a	n/a	n/a	n/a	n/a
Right-Weight	categorical	0	5	n/a	n/a	n/a	n/a	n/a
Right-Distance	categorical	0	5	n/a	n/a	n/a	n/a	n/a

A.4.4.2 Category frequencies

Exact development category frequencies:

- `Left-Weight`: `{"1": 100, "2": 100, "3": 100, "4": 100, "5": 100}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `Left-Distance`: `{"1": 100, "2": 100, "3": 100, "4": 98, "5": 102}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `Right-Weight`: `{"1": 101, "2": 101, "3": 99, "4": 99, "5": 100}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `Right-Distance`: `{"1": 70, "2": 86, "3": 106, "4": 118, "5": 120}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.

A.4.5 dermatology

A.4.5.1 Summary, identities and class counts

Task `multiclass`; metric `log_loss`; 292 development rows; 74 sealed rows counted from assignments; 1 numeric and 33 categorical features. Development groups: 292; all groups: 366; largest group: 1. Original fold sizes:[59, 59, 58, 58, 58]; other-four-fold fit sizes:[233,233,234,234,234]. Declared grouping columns: none.

Development file SHA-256: `a447d3a279d1f4c11f3b330935d362e3d8ca4c5e99afe5276f01e411a8e0795e`. Assignment SHA-256: `88ce0ea91c15e418c566d1a455a7b01b6cbc4335be5fbfca3efcf4549267a379`. Original source SHA-256: `c8852ba546e92249debecd9a897cda3f94fd10872fa13c62eb14ab40e279166e`. Original license: unknown; repository license: MIT.

Frozen class order: `["1", "2", "3", "4", "5", "6"]`. Development counts: `{"1": 89, "2": 48, "3": 61, "4": 39, "5": 40, "6": 15}`. Counts by original fold: `{"0": {"1": 16, "2": 9, "3": 11, "4": 10, "5": 8, "6": 5}, "1": {"1": 16, "2": 13, "3": 13, "4": 9, "5": 6, "6": 2}, "2": {"1": 15, "2": 9, "3": 16, "4": 8, "5": 7, "6": 3}, "3": {"1": 22, "2": 9, "3": 10, "4": 6, "5": 10, "6": 1}, "4": {"1": 20, "2": 8, "3": 11, "4": 6, "5": 9, "6": 4}}`.

A.4.5.2 Feature statistics

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
erythema	categorical	0	4	n/a	n/a	n/a	n/a	n/a
scaling	categorical	0	4	n/a	n/a	n/a	n/a	n/a
definite_borders	categorical	0	4	n/a	n/a	n/a	n/a	n/a
itching	categorical	0	4	n/a	n/a	n/a	n/a	n/a
koebner_phenomenon	categorical	0	4	n/a	n/a	n/a	n/a	n/a
polygonal_papules	categorical	0	4	n/a	n/a	n/a	n/a	n/a
follicular_papules	categorical	0	4	n/a	n/a	n/a	n/a	n/a
oral_mucosal_involvement	categorical	0	4	n/a	n/a	n/a	n/a	n/a
knee_and_elbow_involvement	categorical	0	4	n/a	n/a	n/a	n/a	n/a
scalp_involvement	categorical	0	4	n/a	n/a	n/a	n/a	n/a
family_history	categorical	0	2	n/a	n/a	n/a	n/a	n/a
melanin_incontinence	categorical	0	4	n/a	n/a	n/a	n/a	n/a
eosinophils_in_the_infiltrate	categorical	0	3	n/a	n/a	n/a	n/a	n/a
PNL_infiltrate	categorical	0	4	n/a	n/a	n/a	n/a	n/a
fibrosis_of_the_papillary_dermis	categorical	0	4	n/a	n/a	n/a	n/a	n/a
exocytosis	categorical	0	4	n/a	n/a	n/a	n/a	n/a
acanthosis	categorical	0	4	n/a	n/a	n/a	n/a	n/a
hyperkeratosis	categorical	0	4	n/a	n/a	n/a	n/a	n/a
parakeratosis	categorical	0	4	n/a	n/a	n/a	n/a	n/a
clubbing_of_the_rete_ridges	categorical	0	4	n/a	n/a	n/a	n/a	n/a
elongation_of_the_rete_ridges	categorical	0	4	n/a	n/a	n/a	n/a	n/a

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
thinning_of_the_suprapapillary_epidermis	categorical	0	4	n/a	n/a	n/a	n/a	n/a
spongiform_pustule	categorical	0	4	n/a	n/a	n/a	n/a	n/a
munro_microabcess	categorical	0	4	n/a	n/a	n/a	n/a	n/a
focal_hypergranulosis	categorical	0	4	n/a	n/a	n/a	n/a	n/a
disappearance_of_the_granular_layer	categorical	0	4	n/a	n/a	n/a	n/a	n/a
vacuolisation_and_damage_of_basal_layer	categorical	0	4	n/a	n/a	n/a	n/a	n/a
spongiosis	categorical	0	4	n/a	n/a	n/a	n/a	n/a
saw-tooth_appearance_of_retes	categorical	0	4	n/a	n/a	n/a	n/a	n/a
follicular_horn_plug	categorical	0	4	n/a	n/a	n/a	n/a	n/a
perifollicular_parakeratosis	categorical	0	4	n/a	n/a	n/a	n/a	n/a
inflammatory_monoluclear_inflitrate	categorical	0	4	n/a	n/a	n/a	n/a	n/a
band-like_infiltrate	categorical	0	4	n/a	n/a	n/a	n/a	n/a
Age	numeric	0	58	0	26.5	60	27.921233	15.306591

A.4.5.3 Category frequencies

Exact development category frequencies:

- `erythema`: `{"0": 3, "1": 45, "2": 175, "3": 69}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `scaling`: `{"0": 6, "1": 84, "2": 160, "3": 42}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `definite_borders`: `{"0": 45, "1": 72, "2": 133, "3": 42}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `itching`: `{"0": 90, "1": 62, "2": 80, "3": 60}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.

- koebner_phenomenon: {"0": 176, "1": 58, "2": 42, "3": 16}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- polygonal_papules: {"0": 234, "1": 1, "2": 32, "3": 25}. Categories absent from the other four folds: {"0": [], "1": ["1"], "2": [], "3": [], "4": []}.
- follicular_papules: {"0": 265, "1": 10, "2": 14, "3": 3}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- oral_mucosal_involvement: {"0": 236, "1": 9, "2": 36, "3": 11}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- knee_and_elbow_involvement: {"0": 201, "1": 21, "2": 50, "3": 20}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- scalp_involvement: {"0": 210, "1": 23, "2": 44, "3": 15}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- family_history: {"0": 257, "1": 35}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- melanin_incontinence: {"0": 233, "1": 8, "2": 40, "3": 11}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- eosinophils_in_the_infiltrate: {"0": 263, "1": 22, "2": 7}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- PNL_infiltrate: {"0": 190, "1": 57, "2": 39, "3": 6}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- fibrosis_of_the_papillary_dermis: {"0": 250, "1": 7, "2": 18, "3": 17}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- exocytosis: {"0": 92, "1": 47, "2": 109, "3": 44}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- acanthosis: {"0": 8, "1": 59, "2": 164, "3": 61}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- hyperkeratosis: {"0": 181, "1": 73, "2": 33, "3": 5}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- parakeratosis: {"0": 70, "1": 94, "2": 105, "3": 23}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- clubbing_of_the_rete_ridges: {"0": 202, "1": 17, "2": 44, "3": 29}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- elongation_of_the_rete_ridges: {"0": 161, "1": 20, "2": 74, "3": 37}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- thinning_of_the_suprapapillary_epidermis: {"0": 203, "1": 16, "2": 48, "3": 25}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- spongiform_pustule: {"0": 240, "1": 28, "2": 21, "3": 3}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- munro_microabcess: {"0": 231, "1": 26, "2": 26, "3": 9}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- focal_hypergranulosis: {"0": 232, "1": 12, "2": 36, "3": 12}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- disappearance_of_the_granular_layer: {"0": 216, "1": 23, "2": 40, "3": 13}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.

- `vacuolisation_and_damage_of_basal_layer`: `{"0": 231, "1": 3, "2": 37, "3": 21}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `spongiosis`: `{"0": 155, "1": 23, "2": 81, "3": 33}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `saw-tooth_appearance_of_retes`: `{"0": 232, "1": 4, "2": 34, "3": 22}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `follicular_horn_plug`: `{"0": 276, "1": 8, "2": 5, "3": 3}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `perifollicular_parakeratosis`: `{"0": 276, "1": 4, "2": 9, "3": 3}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `inflammatory_monoluclear_inflitrate`: `{"0": 11, "1": 64, "2": 171, "3": 46}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `band-like_infiltrate`: `{"0": 226, "1": 3, "2": 19, "3": 44}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.

A.4.6 segmentation

Task `multiclass`; metric `log_loss`; 1848 development rows; 462 sealed rows counted from assignments; 18 numeric and 1 categorical features. Development groups: 1669; all groups: 2086; largest group: 3. Original fold sizes:[364,372,369,373,370]; other-four-fold fit sizes:[1484, 1476, 1479, 1475, 1478]. Declared grouping columns: none.

Development file SHA-256: `6691b654d75aa6ba4a66792fe7cbeef46738c9a02b5ca667cd41d9e141ecc67a`. Assignment SHA-256: `10bcfcc5634490f57e7b69f3b8071fd1e49d63e2a03a62e182772e6f3df321d6`. Original source SHA-256: `980ba4342d2fe489793cc3431678318f8d1467fe6bf6a687eb18bfc2267a049f`. Original license: unknown; repository license: MIT.

Frozen class order: `["0", "1", "2", "3", "4", "5", "6"]`. Development counts: `{"0": 261, "1": 262, "2": 265, "3": 262, "4": 269, "5": 260, "6": 269}`. Counts by original fold: `{"0": {"0": 47, "1": 49, "2": 51, "3": 54, "4": 51, "5": 59, "6": 53}, "1": {"0": 52, "1": 54, "2": 54, "3": 53, "4": 56, "5": 46, "6": 57}, "2": {"0": 51, "1": 52, "2": 51, "3": 53, "4": 56, "5": 53, "6": 53}, "3": {"0": 57, "1": 48, "2": 56, "3": 54, "4": 51, "5": 53, "6": 54}, "4": {"0": 54, "1": 59, "2": 53, "3": 48, "4": 55, "5": 49, "6": 52}}`.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
region-centroid-col	numeric	0	253	1	121	254	124.82955	72.877771
region-centroid-row	numeric	0	237	11	122	251	123.68019	57.350115
region-pixel-count	categorical	0	1	n/a	n/a	n/a	n/a	n/a
short-line-density-5	numeric	0	4	0	0	0.33333334	0.014490139	0.040503723
short-line-density-2	numeric	0	3	0	0	0.22222222	0.0046897546	0.024343745
vedge-mean	numeric	0	582	0	1.2777773	27.944443	1.8955628	2.6719697

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
vedge-sd	numeric	0	1459	0	0.83148183	991.7184	5.9348586	45.97572
hedge-mean	numeric	0	637	0	1.4444447	44.722225	2.4776034	3.7718629
hedge-sd	numeric	0	1472	-1.5894573e-08	0.99814636	1386.3292	8.7672312	61.930518
intensity-mean	numeric	0	1087	0	21.611111	143.44444	36.80293	38.072465
rawred-mean	numeric	0	622	0	19.555555	137.11111	32.616883	34.956813
rawblue-mean	numeric	0	714	0	27.277778	150.88889	43.863998	43.415929
rawgreen-mean	numeric	0	637	0	20.333334	142.55556	33.92791	36.242479
exred-mean	numeric	0	414	-49.666668	-10.666667	9.888889	-12.558141	11.498634
exblue-mean	numeric	0	604	-12.444445	19.444445	82	21.183201	19.520045
exgreen-mean	numeric	0	361	-33.88889	-10.666667	24.666666	-8.6250601	11.569724
value-mean	numeric	0	715	0	28.555555	150.88889	44.813493	42.815875
saturatoin-mean	numeric	0	1519	0	0.37455574	1	0.42759796	0.23090285
hue-mean	numeric	0	1552	-3.0441751	-2.0481283	2.9124804	-1.3641642	1.5416936

Exact development category frequencies:

- `region-pixel-count`: `{"9.0": 1848}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.

A.4.7 vehicle

Task `multiclass`; metric `log_loss`; 677 development rows; 169 sealed rows counted from assignments; 18 numeric and 0 categorical features. Development groups: 677; all groups: 846; largest group: 1. Original fold sizes:[135,136,135,136,135]; other-four-fold fit sizes:[542,541,542,541,542]. Declared grouping columns: none.

Development file SHA-256: `117699e46fbd73fdfaa9996fc671a7a88463cc768f0f1e859b77ad7120a49b18`. Assignment SHA-256: `44645392494e0be6c4a4d3e708db4af46b0d4c7af52bc6cf3666bf407109030b`. Original source SHA-256: `dad03273414ae7127e30b6e4e66df90319a148b37ead5dd376dd140be5baedc4`. Original license: unknown; repository license: MIT.

Frozen class order: `["1", "2", "3", "4"]`. Development counts: `{"1": 169, "2": 175, "3": 174, "4": 159}`. Counts by original fold: `{"0": {"1": 34, "2": 35, "3": 36, "4": 30}, "1": {"1": 33, "2": 36, "3": 33, "4": 34}, "2": {"1": 37, "2": 34, "3": 34, "4": 30}, "3": {"1": 34, "2": 36, "3": 32, "4": 34}, "4": {"1": 31, "2": 34, "3": 39, "4": 31}}`.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
COMPACTNESS	numeric	0	43	73	93	119	93.949778	8.1342405
CIRCULARITY	numeric	0	27	33	44	59	44.909897	6.1908953
DISTANCE CIRCULARITY	numeric	0	63	40	80	112	82.490399	15.889816
RADIUS RATIO	numeric	0	134	104	169	333	169.83161	33.806618
PR AXIS ASPECT RATIO	numeric	0	36	47	61	138	61.805022	8.2894697
MAX LENGTH ASPECT RATIO	numeric	0	19	2	8	55	8.6070901	4.8232398
SCATTER RATIO	numeric	0	126	112	157	265	169.65731	33.487679
ELONGATEDNESS	numeric	0	35	26	43	61	40.748892	7.8641824
PR AXISRECTANGULAR	numeric	0	13	17	20	29	20.645495	2.6103189
LENGTHRECTANGULAR	numeric	0	65	118	146	186	148.14032	14.560102
MAJORVARIANCE	numeric	0	125	130	179	320	189.4195	31.887562
MINORVARIANCE	numeric	0	384	184	366	1018	444.34269	178.10982
GYRATIONRADIUS	numeric	0	138	112	174	268	174.82422	32.80978
MAJORSKEWNESS	numeric	0	39	59	71	135	72.462334	7.7825212
MINORSKEWNESS	numeric	0	23	0	5	22	6.3353028	4.9982104
MINORKURTOSIS	numeric	0	41	0	11	41	12.82127	9.0213413
MAJORKURTOSIS	numeric	0	29	176	189	204	189.05761	6.1604295
HOLLOWS RATIO	numeric	0	31	181	197	211	195.79173	7.4621283

A.4.8 optdigits

A.4.8.1 Summary, identities and class counts

Task `multiclass`; metric `log_loss`; 4496 development rows; 1124 sealed rows counted from assignments; 51 numeric and 13 categorical features. Development groups: 4496; all groups: 5620; largest group: 1. Original fold sizes:[900,899,899,899,899]; other-four-fold fit sizes:[3596, 3597, 3597, 3597, 3597]. Declared grouping columns: none.

Development file SHA-256: `afaa9819c5ac7059a64f7457e73444b3ba7c28ccb4cd3de27cdf5fe13f7751f6`. Assignment SHA-256: `0407e498a26a4cea16fe28c170e626d1338dc4bf8148d0e1be17ba3d87056018`. Original source SHA-256: `b70b7ac68a54d1ccd41bd24f11324c3af96b67541791fb9d8684cbdf151533c3`. Original license: unknown; repository license: MIT.

Frozen class order: `["0", "1", "2", "3", "4", "5", "6", "7", "8", "9"]`. Development counts: `{"0": 444, "1": 456, "2": 444, "3": 476, "4": 462, "5": 433, "6": 437, "7": 444, "8": 439, "9": 461}`. Counts by original fold: `{"0": {"0": 84, "1": 90, "2": 96, "3": 82, "4": 100, "5": 96, "6": 86, "7": 85, "8": 87, "9": 94}, "1": {"0": 88, "1": 90, "2": 74, "3": 109, "4": 95, "5": 83, "6": 96, "7": 93, "8": 93, "9": 78}, "2": {"0": 77, "1": 99, "2": 106, "3": 92, "4": 89, "5": 83, "6": 95, "7": 80, "8": 72, "9": 106}, "3": {"0": 93, "1": 88, "2": 92, "3": 89, "4": 90, "5": 79, "6": 72, "7": 99, "8": 94, "9": 103}, "4": {"0": 102, "1": 89, "2": 76, "3": 104, "4": 88, "5": 92, "6": 88, "7": 87, "8": 93, "9": 80}}`.

A.4.8.2 Feature statistics

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
input1	categorical	0	1	n/a	n/a	n/a	n/a	n/a
input2	categorical	0	9	n/a	n/a	n/a	n/a	n/a
input3	numeric	0	17	0	5	16	5.3854537	4.6694807
input4	numeric	0	17	0	13	16	11.763345	4.2575085
input5	numeric	0	17	0	13	16	11.573176	4.4513964
input6	numeric	0	17	0	4	16	5.5584964	5.6073441
input7	numeric	0	17	0	0	16	1.3727758	3.349232
input8	numeric	0	16	0	0	15	0.13122776	0.99381714
input9	categorical	0	4	n/a	n/a	n/a	n/a	n/a
input10	numeric	0	16	0	0	15	1.9730872	3.0879654
input11	numeric	0	17	0	12	16	10.52758	5.4372549
input12	numeric	0	17	0	13	16	11.79782	4.01247
input13	numeric	0	17	0	11	16	10.540036	4.7693414
input14	numeric	0	17	0	9	16	8.261121	5.9790084
input15	numeric	0	17	0	0	16	2.0991993	3.9437706
input16	numeric	0	14	0	0	13	0.13411922	0.91377922
input17	categorical	0	5	n/a	n/a	n/a	n/a	n/a

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
input18	numeric	0	17	0	1	16	2.5849644	3.4827212
input19	numeric	0	17	0	11	16	9.6209964	5.8251162
input20	numeric	0	17	0	6	16	6.8256228	5.8753604
input21	numeric	0	17	0	6	16	7.1846085	6.1510996
input22	numeric	0	17	0	9	16	7.9857651	6.2479372
input23	numeric	0	17	0	0	16	1.9759786	3.498392
input24	categorical	0	8	n/a	n/a	n/a	n/a	n/a
input25	categorical	0	2	n/a	n/a	n/a	n/a	n/a
input26	numeric	0	17	0	1	16	2.363879	3.0806441
input27	numeric	0	17	0	11	16	9.1581406	6.1416147
input28	numeric	0	17	0	10	16	9.0745107	5.9029827
input29	numeric	0	17	0	12	16	9.7602313	6.2595395
input30	numeric	0	17	0	8	16	7.7955961	5.9558196
input31	numeric	0	17	0	0	16	2.3709964	3.6629328
input32	categorical	0	3	n/a	n/a	n/a	n/a	n/a
input33	categorical	0	2	n/a	n/a	n/a	n/a	n/a
input34	numeric	0	16	0	0	15	2.1419039	3.3179478
input35	numeric	0	17	0	8	16	7.67371	6.2749206
input36	numeric	0	17	0	11	16	9.155694	6.1866184
input37	numeric	0	17	0	12	16	10.287811	5.9272272
input38	numeric	0	17	0	10	16	9.0538256	5.8700142
input39	numeric	0	15	0	1	14	2.9219306	3.5062095
input40	categorical	0	1	n/a	n/a	n/a	n/a	n/a
input41	categorical	0	8	n/a	n/a	n/a	n/a	n/a
input42	numeric	0	17	0	0	16	1.4971085	3.0053319
input43	numeric	0	17	0	5	16	6.6127669	6.5197965
input44	numeric	0	17	0	7	16	7.1683719	6.4851243
input45	numeric	0	17	0	8	16	7.8153915	6.2973257
input46	numeric	0	17	0	10	16	8.5473754	5.7808416
input47	numeric	0	17	0	1	16	3.518016	4.3473655
input48	categorical	0	7	n/a	n/a	n/a	n/a	n/a
input49	categorical	0	9	n/a	n/a	n/a	n/a	n/a
input50	numeric	0	17	0	0	16	0.80649466	1.9683692

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
input51	numeric	0	17	0	8	16	7.7595641	5.6567184
input52	numeric	0	17	0	10	16	9.7504448	5.1729862
input53	numeric	0	17	0	10	16	9.6554715	5.2848009
input54	numeric	0	17	0	11	16	9.1548043	5.9832257
input55	numeric	0	17	0	1	16	3.7446619	4.8885914
input56	numeric	0	13	0	0	13	0.15814057	0.81911142
input57	categorical	0	2	n/a	n/a	n/a	n/a	n/a
input58	numeric	0	11	0	0	10	0.27935943	0.92382518
input59	numeric	0	17	0	5	16	5.7382117	5.007576
input60	numeric	0	17	0	13	16	11.981317	4.3434726
input61	numeric	0	17	0	13	16	11.577625	4.9533614
input62	numeric	0	17	0	6	16	6.7164146	5.7925554
input63	numeric	0	17	0	0	16	2.0507117	3.990158
input64	numeric	0	17	0	0	16	0.24154804	1.3632569

A.4.8.3 Category frequencies

Exact development category frequencies:

- `input1`: `{"0.0": 4496}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `input2`: `{"0.0": 3810, "1.0": 343, "2.0": 175, "3.0": 89, "4.0": 41, "5.0": 19, "6.0": 10, "7.0": 6, "8.0": 3}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `input9`: `{"0.0": 4489, "1.0": 4, "2.0": 2, "5.0": 1}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": ["5.0"], "3": [], "4": []}`.
- `input17`: `{"0.0": 4484, "1.0": 6, "2.0": 3, "3.0": 2, "5.0": 1}`. Categories absent from the other four folds: `{"0": [], "1": ["5.0"], "2": [], "3": [], "4": []}`.
- `input24`: `{"0.0": 4418, "1.0": 24, "2.0": 17, "3.0": 11, "4.0": 12, "5.0": 7, "6.0": 5, "8.0": 2}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": ["8.0"], "3": [], "4": []}`.
- `input25`: `{"0.0": 4491, "1.0": 5}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `input32`: `{"0.0": 4484, "1.0": 11, "2.0": 1}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": ["2.0"], "4": []}`.
- `input33`: `{"0.0": 4492, "1.0": 4}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `input40`: `{"0.0": 4496}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.

- input41: {"0.0": 4455, "1.0": 14, "2.0": 11, "3.0": 6, "4.0": 5, "5.0": 3, "6.0": 1, "7.0": 1}. Categories absent from the other four folds: {"0": [], "1": ["7.0"], "2": [], "3": ["6.0"], "4": []}.
- input48: {"0.0": 4441, "1.0": 36, "2.0": 11, "3.0": 2, "4.0": 3, "5.0": 1, "6.0": 2}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": ["5.0"]}.
- input49: {"0.0": 4465, "1.0": 18, "10.0": 1, "2.0": 4, "3.0": 4, "4.0": 1, "5.0": 1, "7.0": 1, "8.0": 1}. Categories absent from the other four folds: {"0": ["5.0"], "1": ["4.0"], "2": ["10.0", "7.0"], "3": [], "4": ["8.0"]}.
- input57: {"0.0": 4494, "1.0": 2}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.

A.4.9 penguins

Task multiclass; metric log_loss; 275 development rows; 69 sealed rows counted from assignments; 5 numeric and 2 categorical features. Development groups: 275; all groups: 344; largest group: 1. Original fold sizes:[55, 55, 55, 55, 55]; other-four-fold fit sizes:[220,220,220,220,220]. Declared grouping columns: none.

Development file SHA-256: cc9e242f3aa638cbaf2fc6951c37631b5928239aaa0c1dba03cc176629d0a4ed. Assignment SHA-256: 81a892fa47114e6462ad030e74c1dad1e3936fdd3c0aadfa8cacd354db82b332. Original source SHA-256: 416c13c9349eba384c5af4711f77a7761ffebcef23cb7e780f65172827b29087. Original license: unknown; repository license: MIT.

Frozen class order: ["0", "1", "2"]. Development counts: {"0": 121, "1": 55, "2": 99}. Counts by original fold: {"0": {"0": 25, "1": 11, "2": 19}, "1": {"0": 24, "1": 11, "2": 20}, "2": {"0": 24, "1": 11, "2": 20}, "3": {"0": 24, "1": 11, "2": 20}, "4": {"0": 24, "1": 11, "2": 20}}.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
island	categorical	0	3	n/a	n/a	n/a	n/a	n/a
bill_length_mm	numeric	2	150	33.1	44.5	59.6	43.959341	5.5528954
bill_depth_mm	numeric	2	75	13.1	17.3	21.2	17.141026	1.9388646
flipper_length_mm	numeric	2	55	172	197	231	200.89744	14.092984
body_mass_g	numeric	2	89	2700	4050	6300	4188.4615	815.6716
sex	categorical	11	2	n/a	n/a	n/a	n/a	n/a
year	numeric	0	3	2007	2008	2009	2008.0218	0.81768874

Exact development category frequencies:

- island: {"0": 135, "1": 100, "2": 40}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- sex: {"0": 130, "1": 134}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.

A.4.10 tae

Task `multiclass`; metric `log_loss`; 124 development rows; 27 sealed rows counted from assignments; 3 numeric and 2 categorical features. Development groups: 86; all groups: 106; largest group: 4. Original fold sizes:[20, 25, 27, 30, 22]; other-four-fold fit sizes:[104, 99, 97, 94,102]. Declared grouping columns: none.

Development file SHA-256: `96b5a504b8ee490a66b5d4b01f8fed9f7753885bfc0b49e17c42099cc441517e`. Assignment SHA-256: `7e83b2b19fc5acb93d2afc04ce0a6430f5609b7380c806978d603562186e08ea`. Original source SHA-256: `5eb60cdd947ecb4ea476ba854299dbe18d522df3ed684fb8f60657fdd4132910`. Original license: unknown; repository license: MIT.

Frozen class order: `["1", "2", "3"]`. Development counts: `{"1": 41, "2": 40, "3": 43}`. Counts by original fold: `{"0": {"1": 6, "2": 10, "3": 4}, "1": {"1": 9, "2": 9, "3": 7}, "2": {"1": 10, "2": 5, "3": 12}, "3": {"1": 10, "2": 9, "3": 11}, "4": {"1": 6, "2": 7, "3": 9}}`.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
Whether_of_not_the_TA_is_a_native_English_speaker	categorical	0	2	n/a	n/a	n/a	n/a	n/a
Course_instructor	numeric	0	24	1	13	25	13.822581	6.9607753
Course	numeric	0	23	1	3.5	25	8.0806452	6.9683579
Summer_or_regular_semester	categorical	0	2	n/a	n/a	n/a	n/a	n/a
Class_size	numeric	0	42	5	27	66	28.193548	12.968123

Exact development category frequencies:

- `Whether_of_not_the_TA_is_a_native_English_speaker`: `{"1": 23, "2": 101}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `Summer_or_regular_semester`: `{"1": 19, "2": 105}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.

A.5 Explicitly omitted or unknown statistics

Sealed target values, target distributions, class balance, feature distributions, missingness and category cardinalities are deliberately unread. Their row counts and expected file identities come from frozen metadata only. Raw full-dataset distribution statistics are therefore unavailable; development summaries must not be labeled full-dataset summaries. Checkpoint dataset exposure, unrecorded subject/time identifiers and original licenses marked unknown remain unknown. The broad S3 catalog is a stored metadata inventory, not fresh payload validation, admissible-teacher inventory or a count of trained models. Exact current panel ranks, per-category wins and Elo await the unchanged panel, whose matrix has no completed cells as of 2026-09-23 about 23:55 UTC (§C.2.1 Retained snapshots and plot data). Final cold installation, final warm target performance, actual GPU training hours and trained-checkpoint consumption await their own receipts. Under the owner's family-level domains

(teacher/train, adaptive development, and evaluation outside training and selection), these thirty datasets remain adaptive-development data (§11.2.2 Adaptive development exposure); whether a pretrained encoder was exposed to them also remains unknown.

APPENDIX B

Historical training, checkpoint lineage and original loss trajectories

IN THIS APPENDIX

B.1 Dated lineage and evidence coverage 49
B.2 Legacy Python training: gen31 and round 1 50
B.2.1 Gen31: fitting the objectives versus using the context 50
B.2.1.1 Epoch losses and the routing warning 50
B.2.1.2 Context advantage and ranking diagnostics 52
B.2.1.3 Metric definitions and the v3.2 objective 52
B.2.2 Round 1 and the reported self-play ladder 53
B.2.2.1 Round-1 epochs and the gate receipt 53
B.2.2.2 The reported generation-level loss and internal Elo 55
B.3 Native training: the 108-block audit and the A100 recovery lineage 56
B.3.1 The September 11 native objective and 108 reset blocks 56
B.3.1.1 Native architecture and objective 56
B.3.1.2 Endpoint measurements and parent regimes 57
B.3.1.3 Adaptive validation observations 58
B.3.2 Fixed monitoring and actual promotion evidence 59
B.3.3 The complete A100 loss series: steps 6,001–69,000 60
B.3.3.1 Source and extraction 60
B.3.3.2 Within-block and boundary patterns 61
B.3.3.3 Telemetry limits and the recovery boundary 62
B.3.4 Carried state, accepted step 60,000 and RTX step 60,100 62
B.3.4.1 A100 checkpoints and the accepted step 60,000 62
B.3.4.2 The RTX step 60,100 qualification run 63
B.4 Current publication regeneration and provenance 64

The supplied CSVs retain original values, source file SHA-256, source path and line, and JSON Pointer where appropriate. There is no interpolation. Separate figures show different populations; joining them into one curve would erase changes in architecture, objective, corpus, sampler and validation protocol. The later recovery lineage has an independently audited, complete retained series of 63,000 per-step observations; the older 108-block capture retains only block endpoints.

The new core-technology chapter adds a separately verified 24-record A100 fixed-monitor series and historical throughput. It does not replace the different populations retained below.

B.1 Dated lineage and evidence coverage

Observation or historical interval	Retained state	What is established; what is not
Jope Prime gen31, preserved by the provenance inventory dated 2026-07-12	Fourteen complete epoch-summary rows, explicitly `policy_v3.2`; immutable warm-start checkpoint identified as `6a204434…`; 125,970 training and 2,169 validation rows	Real logged training and validation measurements. The log itself has no wall-clock timestamp or exact trainer-source hash; its event date is not inferred from file modification time. The provenance inventory's date is a historical declaration, not a newly verified execution date.
Lineage A, historical xbabe2 Python control plane	Four surviving frozen corpus snapshots and original warm-start state; substantial live control-plane state and later checkpoint descendants declared lost	Historical training and teaching data existed. The lost accepted-pointer and sealed-use history cannot be reconstructed merely from aggregate scores.
September 9–11 native training audit	108 completed 3,000-step blocks, 108 different corpus identities and 62 starting checkpoint identities	Original initialization records explicitly reset AdamW. The capture retains each block's first and last metric row, not its complete per-step log.
September 12–13 recovery lineage (lineage B, §D.5.1 Lineages A and B and the staging store), retained A100 metrics and checkpoint manifests	63,000 complete metric rows at steps 6,001–69,000 in 21 blocks/corpora; 78 checkpoints at steps 4,000–69,000 across 22 corpus identities	This later lineage carries FP32 AdamW ancestry. Complete logged losses accompany the metadata. The accepted serving checkpoint remains 60,000; 69,000 is unaccepted. Validation remains adaptive and its corpus/panel changes.
September 14, 22:19:52–22:27:47 UTC	Complete 100-row RTX qualification trajectory, steps 60,001–60,100, interrupted and resumed after 60,006	Original metadata records carried optimizer state and changed model/parameter identities. It explicitly describes unchanged-corpus hardware qualification, with zero fresh programs: a historical hardware continuation, not a production improvement or promotion run.
September 18, 20:17:48 UTC publication manifest	Product bundle `policy-step-60000`, source checkpoint `8ac80c269cc47c253b6f53fa3bc1eb75bb2fcda0bdace3ff8554be0175c400d6`	The current product's accepted source checkpoint is pinned. Publication time is distinct from training time.
Current bounded ASM work, as of 2026-09-23 about 23:55 UTC	Verified one-dataset regression neural-origin proof at source `68f56404`, merged to main as `a75e543d2` (PR83); the training64 follow-on remains a stopped, unmerged patch (pending integration), and no eligible real corpus has been demonstrated for its admission guard (§7.4 Present teacher readiness and the smallest real path)	New category training and subsequent consumption of its learned export remain pending; no checkpoint has been trained by this work. No new GPU-hour budget is inferred from historical training. Experimental regression lineage v1 (pending integration) is not part of this history.

Sources: `docs/adr/ADR-0005-canonical-learning-loop.md:12-24,72`, `docs/learning.md:280-285`; historical native audit `learning-hpc.md:37`; original checkpoint manifests and metrics pinned in the CSVs; `models/bundles/policy-step-60000.json:7,73`; `/home/ubuntu/jope-prime/provenance.json:3,13,129`. The retained legacy source declares `policy_v3.2` at `/home/ubuntu/jope/src/jope/policy/model_v3.py:40`; it is useful for interpreting the old log, but the log does not attest that exact source build.

The accepted serving model is the native 512×6 decoder at step 60,000 (§2.8.1 Checkpoint binding and configuration). Experimental regression lineage v1 (256×2, new initialization) is the separate line that the pending bounded campaign would start: a Python `JopePolicyModelV3` with width 256 and two layers, a new, smaller model family rather than a continuation of step 60,000 (§8.4.1 What the pending campaign trains). Status at the PR83/stopped-overlay snapshot: pending integration; the training64 patch was then outside main. PR88 later integrated it (§1.8.1 Implementation snapshot and this manual's own identity); no real campaign checkpoint is established. Its initial model `M_init` is named `M0` only after it has actually trained on its first corpus `C0`; as of 2026-09-23 about 23:55 UTC it has no trained checkpoint. None of the records in this appendix belongs to it, and its measurements must not be joined to these historical trajectories.

B.2 Legacy Python training: gen31 and round 1

B.2.1 Gen31: fitting the objectives versus using the context

B.2.1.1 Epoch losses and the routing warning

The gen31 log contains epochs 1–14, with phase A at epochs 1–6 and phase B at 7–14. Both phases train the ranking branch. The retained v3.2 trainer lowers the context-encoder learning rate at the phase boundary, while phase B adds reward-weighted sequence imitation. It logs two GPUs, requested batch size 2,048 and learning rate 0.00015. Its final elapsed timer is 1,135.158 seconds; that timer is a run-local measurement, not a cold-install or end-to-end campaign duration.

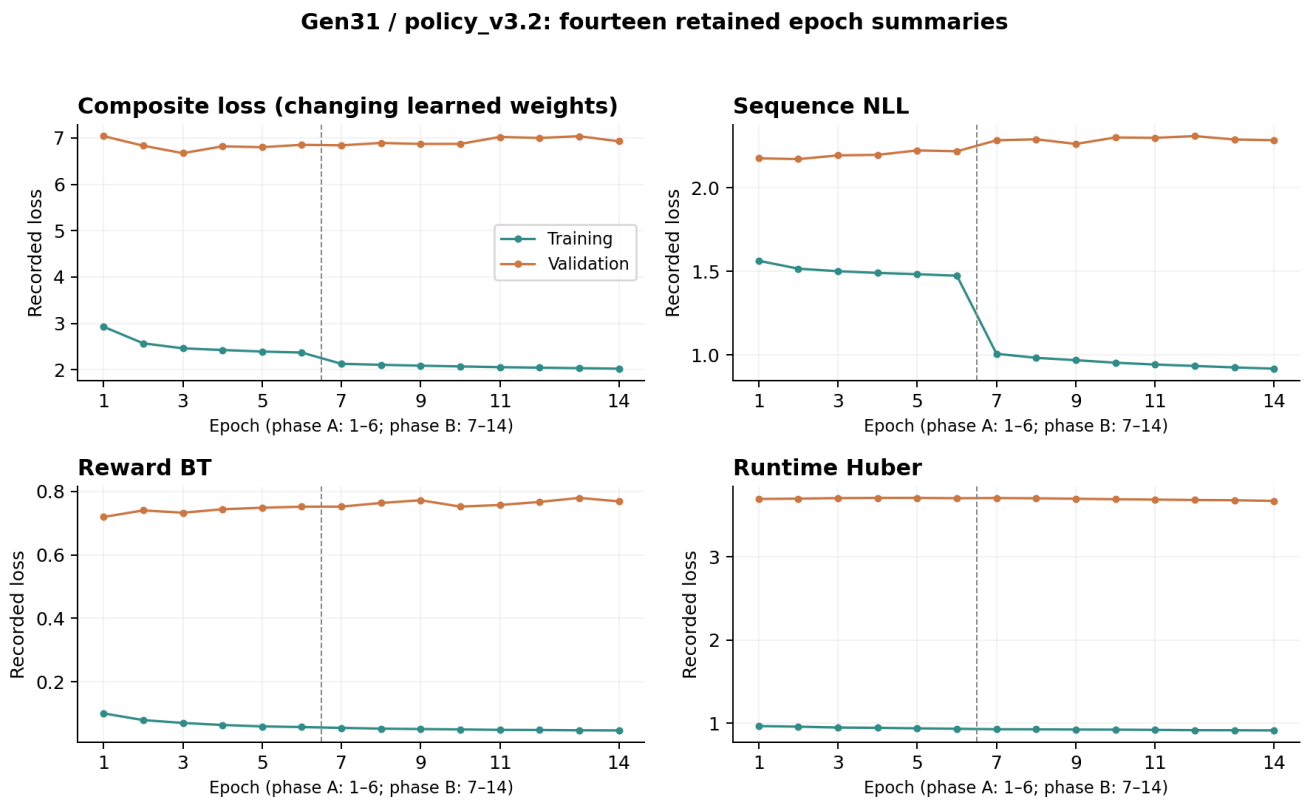


Figure B.1 Fourteen original gen31 epoch summaries, including training and validation components

Every point is an original epoch summary. The training sequence component (`train_loss_components.seq_nll`) fell from 1.563002 to 0.918725, but 0.466185 of that fall occurred in one step at the phase A→B boundary (1.473704 at epoch 6, 1.007519 at epoch 7), where phase B adds reward-weighted sequence imitation; within each phase it fell by only about 0.089. Held-out sequence NLL (`val_seq_nll`, no label smoothing) rose in every phase-A epoch and overall from 2.055572 to 2.182772 (+6.2%). The full training composite fell from 2.926419 to 2.022394, while validation composite changed from 7.039105 to 6.926823 and reached its minimum of 6.671162 at epoch 3. The final epoch therefore was not the best by every monitored objective. The train/validation composite gap grew from 4.112686 to 4.904429. The model fitted its training objective while held-out likelihood got worse; these measurements do not show that a newly generated algorithm outperformed its ancestor. The figure's validation sequence line is the 0.05-smoothed `val_loss_components.seq_nll`, not `val_seq_nll`.

Routing is a particularly visible warning: training routing cross-entropy fell from 0.642189 to 0.005816, while validation routing cross-entropy remained 3.498959 at the end. The retained labels and grouping still need to be held constant before assigning causality, but a near-zero training classifier loss alone is plainly insufficient evidence of useful routing.

B.2.1.2 Context advantage and ranking diagnostics

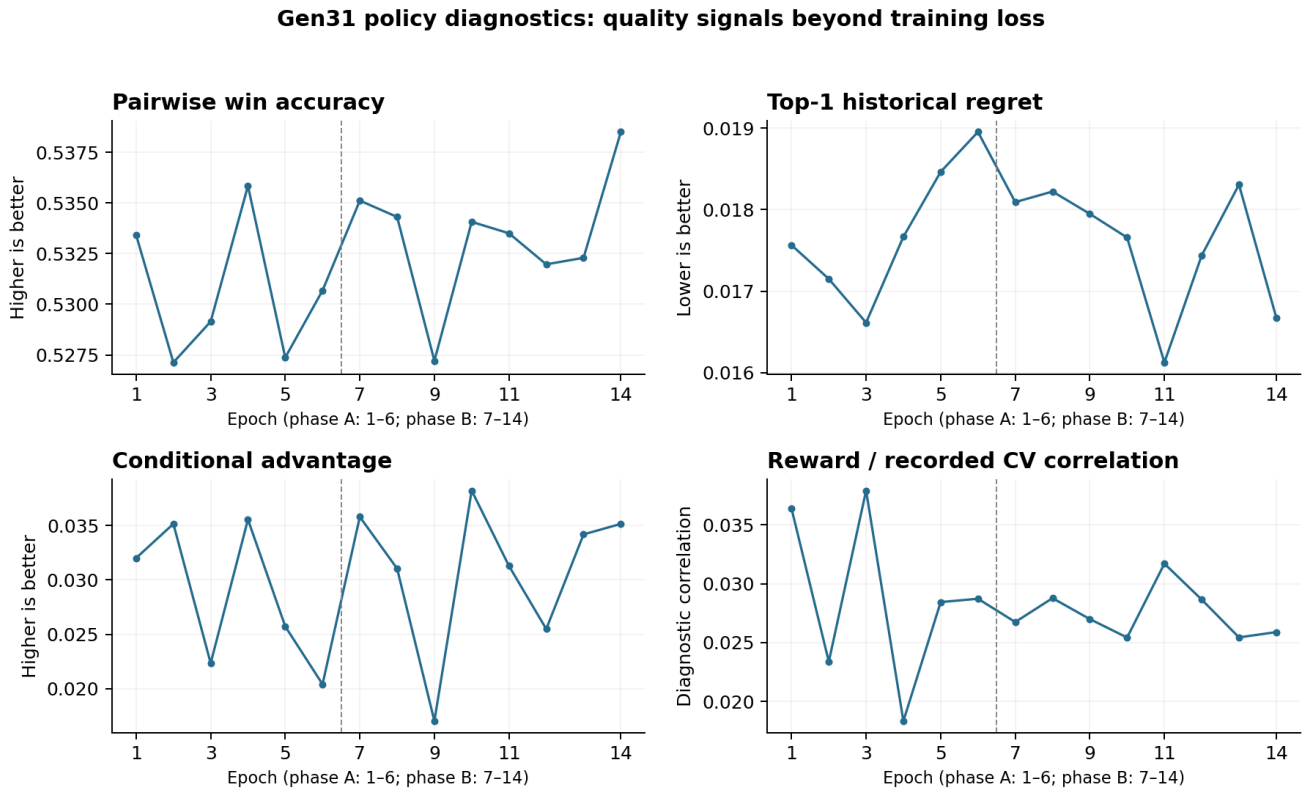


Figure B.2 Gen31 recorded context advantage and ranking diagnostics

Pairwise validation ordering accuracy rose from 0.533419 to 0.538499. The genome-only control rose from 0.501411 to 0.503346, so recorded conditional advantage changed from 0.032008 to 0.035153: approximately 3.20 to 3.52 percentage points. This is a modest ranking diagnostic on retained candidates, not a 53.8% downstream tournament win rate. Historical top-1 regret decreased from 0.017565 to 0.016676 in the retained target's units. Predicted-reward correlation remained small, 0.036390 to 0.025889. The uncertainty/error statistic was negative, −0.196674 to −0.228137; it does not establish calibrated uncertainty.

B.2.1.3 Metric definitions and the v3.2 objective

The following definitions explain why similarly named numbers cannot be substituted for one another:

Logged metric	Source meaning and limitation
`train_loss`, `val_loss`	Combined objectives, including learned uncertainty weights and regularizing log-variance terms. They are not sums of the plotted raw component values. Training aggregation averages executed batch losses across ranks.
`train_loss_components.seq_nll` / `val_loss_components.seq_nll`	Legal-token, 0.05-smoothed sequence cross-entropy with applicable provenance/phase row weights.
`val_seq_nll`	A separate validation evaluator uses zero label smoothing and averages batch NLL. It is deliberately distinct from `val_loss_components.seq_nll`; the first epoch logs 2.055572 versus 2.175046.

Logged metric	Source meaning and limitation
Pairwise ordering accuracy	Agreement on unequal-value pairs within a dataset. This implementation does not give explicit half-credit to tied predictions; its Boolean comparison has an order-dependent tie convention.
Conditional advantage	Conditional reward ranking accuracy minus the genome-only control's accuracy on the same retained population. It probes context use; it does not fit new programs.
Historical top-1 regret	Per dataset, best retained target value minus the target value of the retained program selected by predicted reward, averaged over datasets. It is not regret relative to an unknown optimum.
Reward CV correlation	Pearson correlation with retained `robust_delta` labels. The name does not make it a new downstream cross-validation score.
Head accuracy	Match at the first head-role grammar token under teacher-forced decoding, grouped by engine. It is not fitted-prediction accuracy.
Uncertainty/error Spearman	Correlation between sorted positional ranks of ensemble dispersion and absolute reward error. The retained implementation does not average tied ranks, so preserve the implementation identity when comparing it.

The old v3.2 objective has seven components: sequence NLL, reward Bradley–Terry, reward listwise, runtime Huber, stability BCE, ablation Bradley–Terry and routing cross-entropy. Its profile scales are respectively 1, 1, 0.5, 0.2, 0.5, 1 and 0.5. Reward/ranking/ablation/routing terms use fixed profile scales; sequence, runtime and stability use the retained Kendall–Gal-style learned weighting. The log's `loss_weights` values are `exp(-log_var)`, not complete effective coefficients. This differs materially from the five-term native objective below. Source interpretation is grounded in retained v3.2 code; absence of an exact run-source binding remains explicit.

Sources: original `train_metrics.jsonl` SHA `d57de1b28a8678327c984a82732cd774520db2cb0994d893a45f8a5ea428e312`, lines 1–14; `/home/ubuntu/jope/src/jope/policy/train_v3.py:360,364,438,444,474,580,613`; `policy/model_v3.py:40,286,300,305,372,376`; `policy/losses.py:23,35,48,86,100`; `model.py:981,997,1015`. Full hashes and exact definitions are retained in this chapter's metadata.

B.2.2 Round 1 and the reported self-play ladder

B.2.2.1 Round-1 epochs and the gate receipt

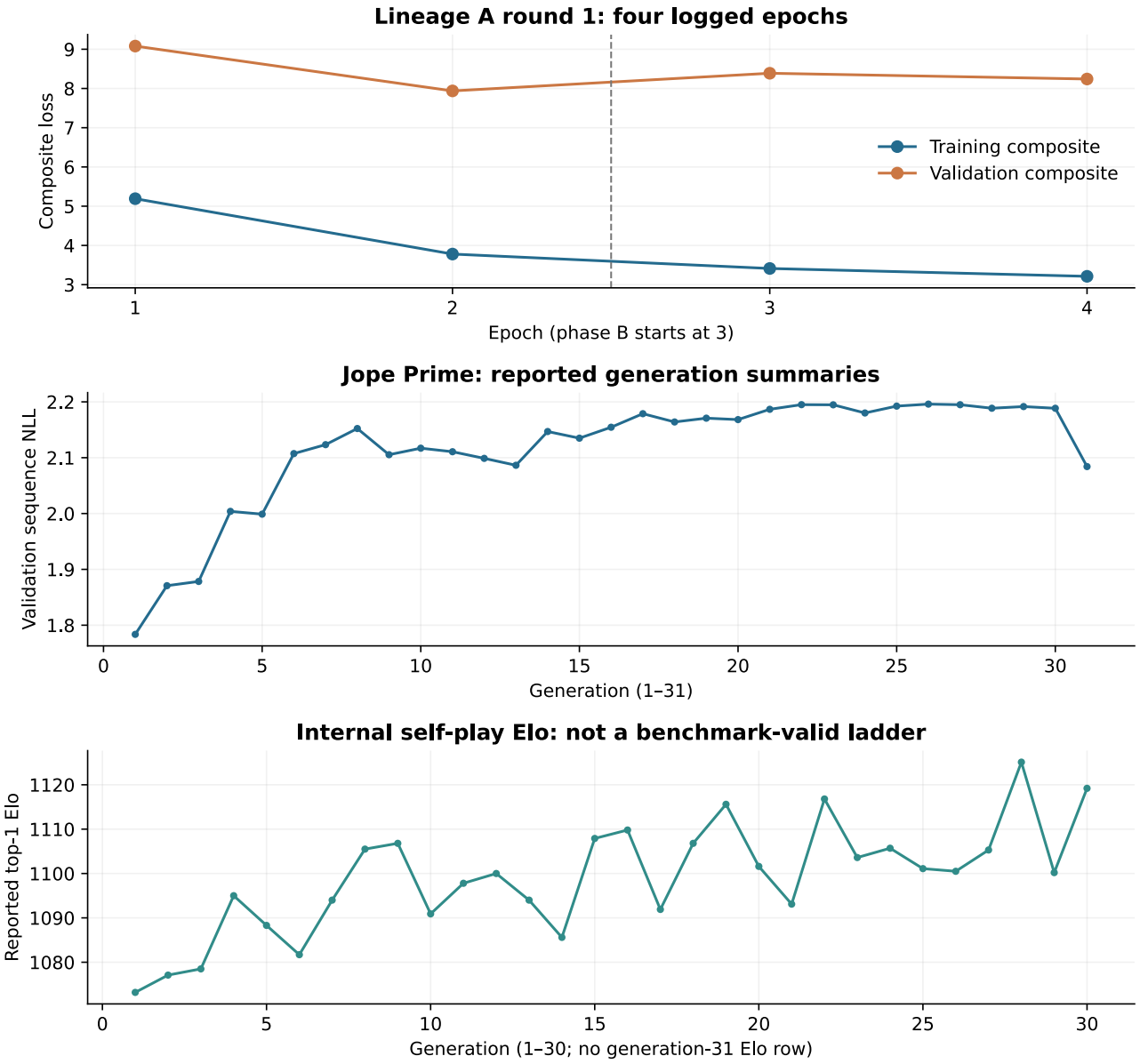
Four original `round1.train_metrics.jsonl` rows survive for the lineage-A Python model `policy_v3.3-fused-heads`. They log phase A at epochs 1–2 and phase B at 3–4, with learning rate 0.0005. This is a separate model/run from gen31 and the native recovery lineage; their composite losses must not be joined into one trajectory.

Logged measure	Epoch 1	Epoch 4	Interpretation
Training composite	5.191046	3.212379	Better fit to the run's changing phase objective

Logged measure	Epoch 1	Epoch 4	Interpretation
Validation composite	9.080722	8.241397	Minimum 7.938042 at epoch 2; the final epoch is worse than that minimum
Validation minus training composite	3.889675	5.029017	The recorded gap widens; phase changes limit a simple overfitting interpretation
Validation pairwise accuracy	0.577940	0.624193	A ranking diagnostic, not downstream predictive performance
Validation sequence NLL	1.686433	1.773742	The final value is worse, despite the lower composite

A separate gate receipt dated **2026-07-25 10:24:49.812 UTC** compares 557 validation rows per model: pairwise accuracy 0.534717 → 0.663611, sequence NLL 2.110434 → 1.838604, and reward/CV correlation −0.097427 → 0.445109. It records `pass: true` and explicitly says “gate only; nothing deployed or rotated.” These are real retained gate measurements, not proof of current teacher eligibility, an independent unseen-family evaluation, or a subsequent serving promotion. The gate's timestamp does not supply a missing event timestamp for every epoch.

B.2.2.2 The reported generation-level loss and internal Elo



Separate populations/objectives. Paper CSVs retain reported ladder values, not a replayed tournament. Generation-31 summary matches epoch 3, not the final epoch. Higher Elo does not imply lower NLL.

Figure B.3 Legacy round-1 epoch measurements and reported generation-level loss and internal Elo; the generation panels reproduce reported paper summaries, not a replayed tournament

The two lower panels retain the **reported paper CSV summaries** from `/home/ubuntu/jope-prime/docs/paper/data/`, rather than replaying their historical tournaments. The loss table has generations 1–31; the internal self-play Elo table has generations 1–30. Across the loss summaries, validation sequence NLL changes from 1.783587 to 2.084099, and pairwise accuracy from 0.594856 to 0.529146. Internal top-1 Elo changes from 1,073.2 to 1,119.2. Over their thirty common generations, the descriptive Pearson correlation between Elo and sequence NLL is +0.728336: higher internal Elo coexists with worse likelihood. Correlation across this evolving history is not a causal result, and the underlying populations/objectives have not been frozen for a new matched evaluation.

The generation-31 summary exactly matches **epoch 3**, not epoch 14, on validation sequence NLL, validation pairwise accuracy and the training sequence component. That explains why a generation-level table and the final gen31 log row can disagree without either number being fabricated. There is no generation-31 Elo row in this retained table. Neither the internal ladder nor its `shipped` column establishes a current release or the benchmark-valid Elo described in §C.2.3.1 Historical benchmark-valid Elo. Full source-to-tournament provenance and exposure qualification remain separate from this arithmetic/source-file check.

The companion `data/legacy-training/` retains all four original files byte-for-byte, a hash-pinned recomputation receipt and the plot input/output hashes. Their source paths are `/home/ubuntu/jope-flywheel/training/round1/{round1.train_metrics.jsonl,gate-result.v1.json}` and `/home/ubuntu/jope-prime/docs/paper/data/{loss_components_by_generation.csv,elo_trajectory.csv}`. No model or dataset payload was opened to assemble this supplement.

B.3 Native training: the 108-block audit and the A100 recovery lineage

B.3.1 The September 11 native objective and 108 reset blocks

B.3.1.1 Native architecture and objective

The audited native decoder consumes a 1,536-dimensional dataset vector and a 512-dimensional target vector. The configuration uses width 512, six decoder layers, four attention heads, FFN width 2,048, dropout 0.1, 401 vocabulary tokens, 657 embedding rows and maximum sequence length 128. The reward path has separate context processing and four reward heads. These are the observed historical dimensions, not a claim that every historical model version used identical architecture.

The active native objective was:

`L = sequence_nll + 0.25 × ranking_bt + 0.10 × ranking_listwise + 0.10 × runtime_huber + 0.05 × contextfree_bt`.

Sequence loss is legal-token cross-entropy with 0.05 smoothing and normalized program weights `0.05 + percentile^4`. Bradley–Terry uses weighted within-comparison winner/loser pairs. Listwise uses a `softmax(percentile / 0.25)` target; its nonzero entropy floor matters when interpreting an apparent plateau. Runtime loss compares `log1p` seconds. The program-only ablation term checks whether ranking can be achieved without dataset context. Stability and the seven stored log-variance parameters were inactive in this native build. Seven parameter names do not imply seven optimized losses.

B.3.1.2 Endpoint measurements and parent regimes

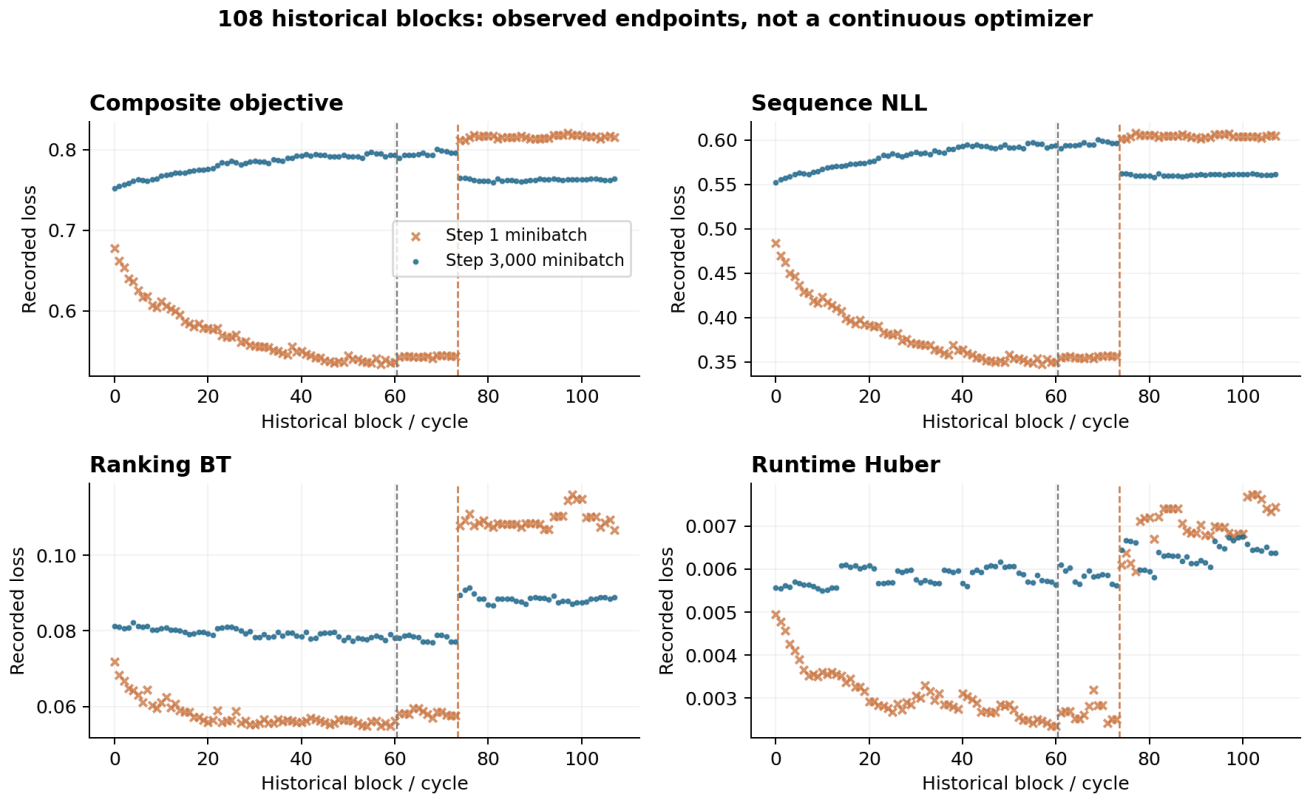


Figure B.4 First and last sampled minibatch measurements for each of 108 reset training blocks

The audit's original capture includes exactly two metric records per block: first and last, plus the original full log's path, SHA and reported 3,000-row count. The intermediate 2,998 rows per block are not reconstructed. These endpoints are sampled minibatches, not averages over the whole block. All 108 blocks reset AdamW, so their 324,000 summed steps are **block work**, not one optimizer's age. At global batch 2,048, the corresponding 663,552,000 nominal program-example presentations are not unique programs or discoveries. Replayed or uncommitted work outside this completed block census is not silently included.

The starting checkpoint changes in three regimes. Cycle 0 starts from the earlier campaign checkpoint `20dca670…`. Cycles 1–60 warm-start their model from the preceding block's checkpoint, while still resetting AdamW. Cycles 61–73 repeatedly start from cycle 59 (`61ba6043…`), and cycles 74–107 repeatedly start from the recovery checkpoint `c2f229c9…`. These are model-parent relationships, not optimizer-state continuation.

All 34 first-to-last reductions in composite, sequence NLL, Bradley–Terry and listwise loss occur exactly in cycles 74–107, the recovery-parent regime. Every block in cycles 0–73 has a higher final sampled value for those four losses than its first value. Runtime Huber decreases exactly in cycles 78–107. The counts therefore align with the parent regime; they are not scattered improvements across 108 comparable starts. The visible jumps at cycle 74 coincide with the rollback to the recovery parent and must not be interpreted as a continuous learning event. Endpoints still sample different minibatches on different corpora, so this is descriptive evidence, not a paired significance test or causal isolation of the parent change. The last block's training NLL was 0.561710, while its adaptive validation NLL was 1.410741; that gap is not a sealed generalization estimate.

B.3.1.3 Adaptive validation observations

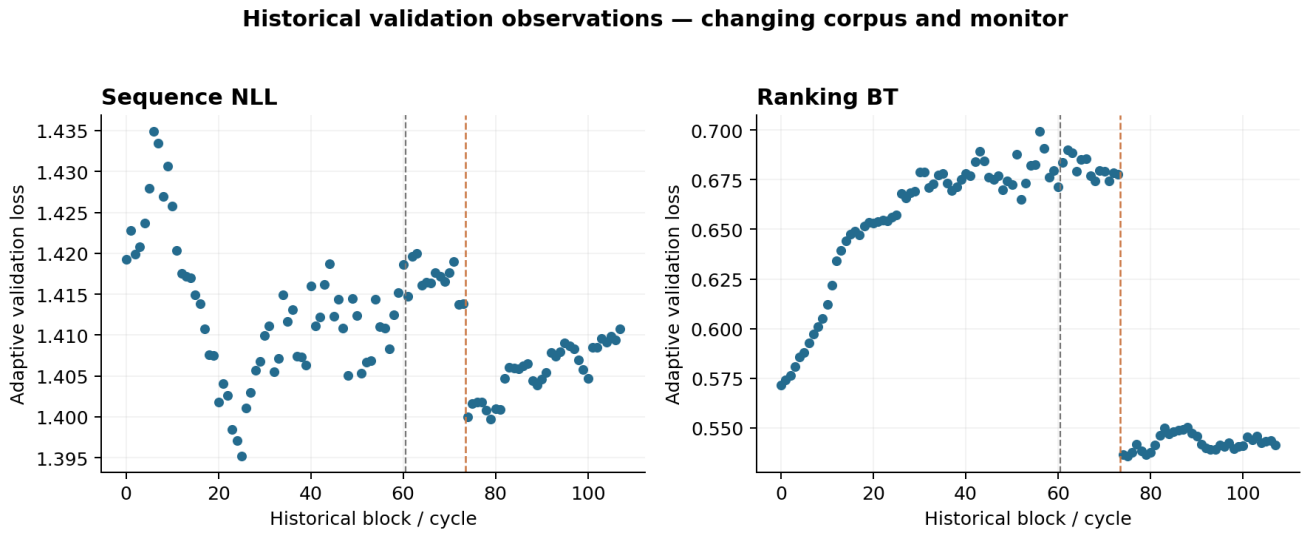
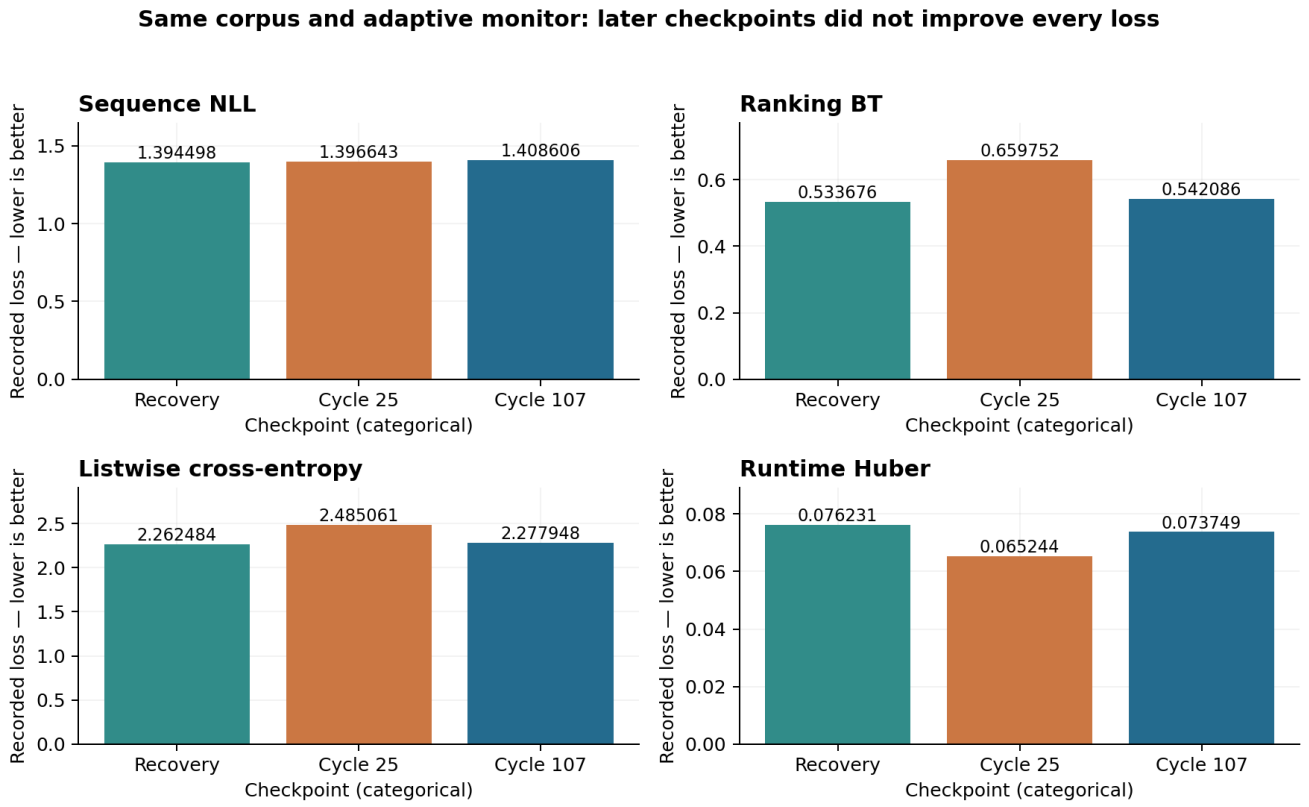


Figure B.5 Adaptive validation observations from changing corpus and panel identities

The plotted validation points are deliberately not a continuous generalization curve. Dashed boundaries identify the repeated-cycle-59 regime from cycle 61 and the recovery-parent regime from cycle 74. The validation Bradley–Terry jump near cycle 74 coincides with that rollback; it is not evidence of smooth accumulated improvement. Each source identity travels with its value. Old native telemetry exposed total loss and four component losses but omitted the active ablation component; its contribution cannot be inferred exactly by subtracting rounded or differently aggregated values. Per-term gradient norms were also unavailable. The later RTX log does expose `contextfree_bt` and weighted terms, so that historical limitation must not be presented as universal.

Sources: `docs/legacy/root-docs/audits/worldclass-redteam-20260911/learning-hpc.md:9,21,33,37`; retained `evidence/learning-native-v3/neural.rs:447,512` and `training.rs:692,712,720,863`; research capture `/home/ubuntu/jain-split/jope-dime/docs/audits/worldclass-redteam-20260911/learning-remote-metadata.json` SHA `1d2ab6ad9a8df4749bd1da000c6b1ea04d00395a777199907f3b599ad9af528b`.

B.3.2 Fixed monitoring and actual promotion evidence



Recovery was measured under both relevant executables with identical outputs; each challenger has its control.
2,048 examples per evaluation. Adaptive teaching monitor, not sealed or independent evaluation.
Source: fixed-monitor-metrics.csv; original record and executable hashes retained.

Figure B.6 Fixed-corpus and fixed-panel historical checkpoint comparison

The fixed monitor is the stronger loss comparison in this audit: all records bind corpus `11e60a43…`, panel `fbff7a048…` and 2,048 sampled examples. Recovery was evaluated by both relevant executable builds with identical recorded outputs. Cycle 25 can be paired with recovery under executable `8ca21024…`; cycle 107 can be paired with recovery under `3382cdcb…`. The CSV retains both recovery rows even where the visual combines their identical values.

Checkpoint	Sequence NLL	Bradley–Terry	Listwise	Runtime Huber
Recovery checkpoint `c2f229c9…` (step 3,000)	1.394498	0.533676	2.262484	0.076231
Cycle 25	1.396643	0.659752	2.485061	0.065244
Cycle 107 (one reset block started from recovery)	1.408606	0.542086	2.277948	0.073749

Sequence and reward-ranking losses did not improve over recovery in these fixed-monitor observations; runtime error was lower. This is an objective tradeoff, not an overall win. The monitor explicitly marks itself adaptive teaching validation, not independent evaluation; `training_selection_used` is false, but that does not turn previously used teaching labels into sealed data.

All 48 promotion decisions indexed in the September 11 capture were **inconclusive**, with qualification evidence not supplied. This is not a measured 0/48 challenger win rate. It establishes missing evidence at that historical observation. The later accepted 60,000 checkpoint belongs to a subsequent recovery lineage; the September 11 result must not be extended to claim that no later checkpoint was ever accepted. A full joined chain of matched executable comparisons and successive qualified target improvements is still required for a quantitative “the flywheel kept improving” claim.

B.3.3 The complete A100 loss series: steps 6,001–69,000

B.3.3.1 Source and extraction

A second retained source supplies the complete per-step loss series for this lineage. Original `metrics.jsonl` files in cycles `000001` through `000021` of `eight-a100-recovery-20260912/learning`, under `/home/ubuntu/jope-migration-evidence/claude/node-high-value/files/home/ubuntu/jope-dime/parallel/runs`, contain **63,000 contiguous reported optimizer steps**, exactly 3,000 per block. Their embedded timestamps span September 12, 18:43:27 through September 13, 18:09:49 UTC. These dates come from the rows, not filesystem modification time. All rows record rank 0, world 8, microbatch 256, accumulation 1 and global batch 2,048. Every row binds the block's corpus and a carried FP32 AdamW parent; parent identities match the retained checkpoint lineage. A100 attribution is historical run lineage, not a new device probe.

The compressed `data/per-step-history/a100-step-metrics.csv.gz` preserves all 63,000 rows and **378,000 exact numeric metric tokens**: `loss`, `sequence_nll`, `ranking_bt`, `ranking_listwise`, `runtime_huber` and `gradient_norm`. Step, local step and timestamp tokens are also preserved. Cycle plus one-based `source_line` resolves each value to a pinned original log in `source-manifest.json`. This is a selected-column extraction, not a replacement for all raw telemetry: timing breakdowns, counters and other nested fields remain in the source logs. A second read compares every selected token against its original and rechecks each full log hash.

B.3.3.2 Within-block and boundary patterns

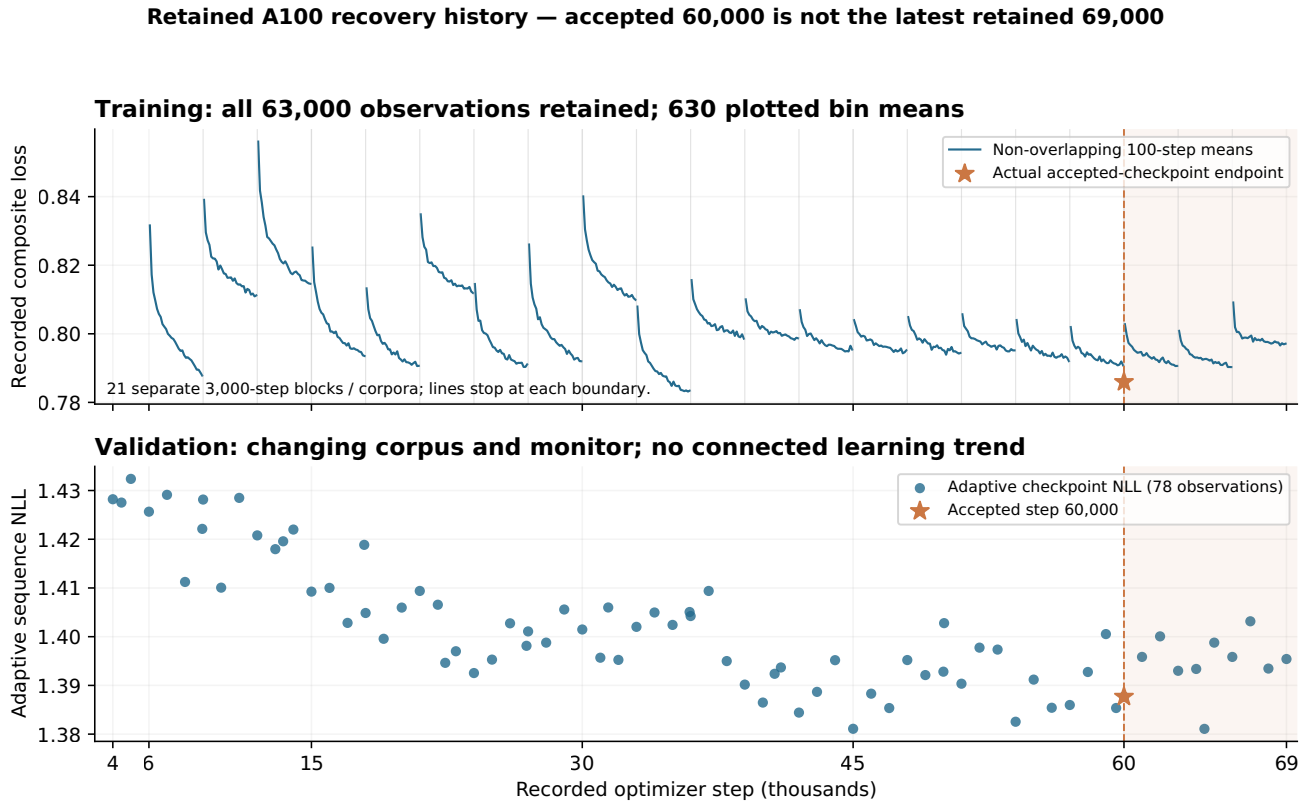


Figure B.7 Complete retained A100 history: 100-step training averages (upper) and adaptive checkpoint validation on changing panels (lower); the lower panel's decline mixes learning with panel changes and is not a generalization curve

The upper panel shows 630 **non-overlapping 100-step arithmetic means**, with a separate line per corpus; it does not display every noisy minibatch point or interpolate absent data. The underlying CSV retains every selected point. The lower panel shows the original 78 checkpoint sequence-NLL observations without joining changed evaluation populations. Dashed orange marks the accepted step 60,000; shaded later steps remain unaccepted.

The within-block pattern is consistent, but its interpretation is bounded. All 21 composite-loss OLS slopes are negative, all 21 final sampled losses are below their first samples, and all 21 last-100 means are below their first-100 means. The median last-100 minus first-100 change is **−0.017066**, with range **−0.043771 to −0.008792**. Every single-point block boundary jumps upward; 19 of 20 boundary comparisons also rise when comparing adjacent 100-step means. Thus “sawtooth” describes the recorded pattern, not a claim that every individual optimization step improves. Each of the 21 blocks has a distinct corpus. Changed sampling and teaching data prevent attributing a boundary jump, or an across-block decline, to parameters alone.

Block and optimizer range	First sampled loss	Last sampled loss	First-100 mean	Last-100 mean
1: 6,001–9,000	0.855045	0.792636	0.831595	0.787825
18: 57,001–60,000, accepted endpoint	0.819152	0.785951	0.801961	0.790811

Block and optimizer range	First sampled loss	Last sampled loss	First-100 mean	Last-100 mean
21: 66,001–69,000, unaccepted endpoint	0.830200	0.792011	0.809123	0.797115

A descriptive OLS over all 54,000 observed steps from 6,001 through 60,000 has slope **-3.3860×10^{-7} loss per step**, R^2 **0.1885**, and fitted first-to-last change **−0.018284**. These are arithmetic summaries of serially dependent observations from changing corpora, not confidence estimates or a measured generalization gain. `a100-block-statistics.csv` retains all 126 block/metric summaries, including each slope and endpoints; `a100-corpus-boundaries.csv` makes the boundary comparisons explicit.

B.3.3.3 Telemetry limits and the recovery boundary

The composite loss is retained as reported. These 63,000 rows expose four named objective components and gradient norm, but omit top-level `contextfree_bt`, effective objective weights and per-step learning rate. Absence of a telemetry field does not establish that its objective was inactive. Do not fill it with zero, reconstruct the full objective from an incomplete component list, or directly compare its composite scale with the separately recorded RTX qualification run. Loss reduction is evidence of objective fitting; it does not by itself establish better encoding-conditioned GP proposals or cheaper, better fitted algorithms.

The inspected cycle-`000000/train-2` directory has checkpoint/run metadata but no `metrics.jsonl`: steps 3,001–6,000 therefore lack per-step rows **in this inspected source**. The antecedent steps 1–3,000 are outside this per-step source set. This is a precise recovery boundary, not a claim that no other log could exist. It also does not erase the earlier fixed 2,048-example monitor: that separate recovery/cycle-25/cycle-107 comparison remains in the preceding section. The later 78-checkpoint series spans 22 corpora and 21 panels and does not provide one unchanged evaluation population.

B.3.4 Carried state, accepted step 60,000 and RTX step 60,100

B.3.4.1 A100 checkpoints and the accepted step 60,000

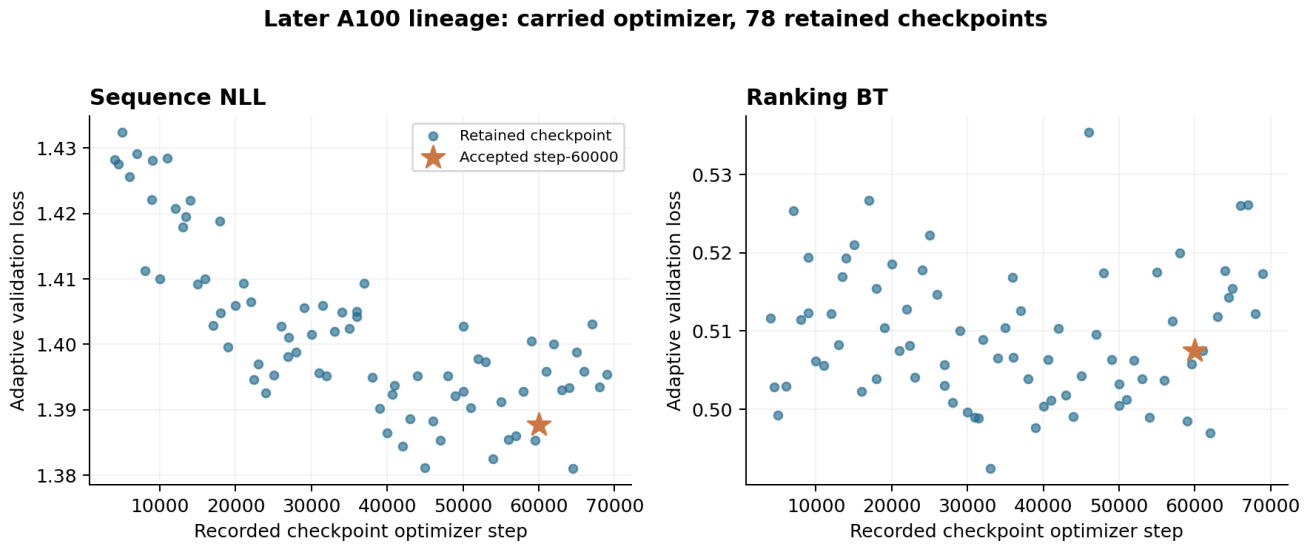
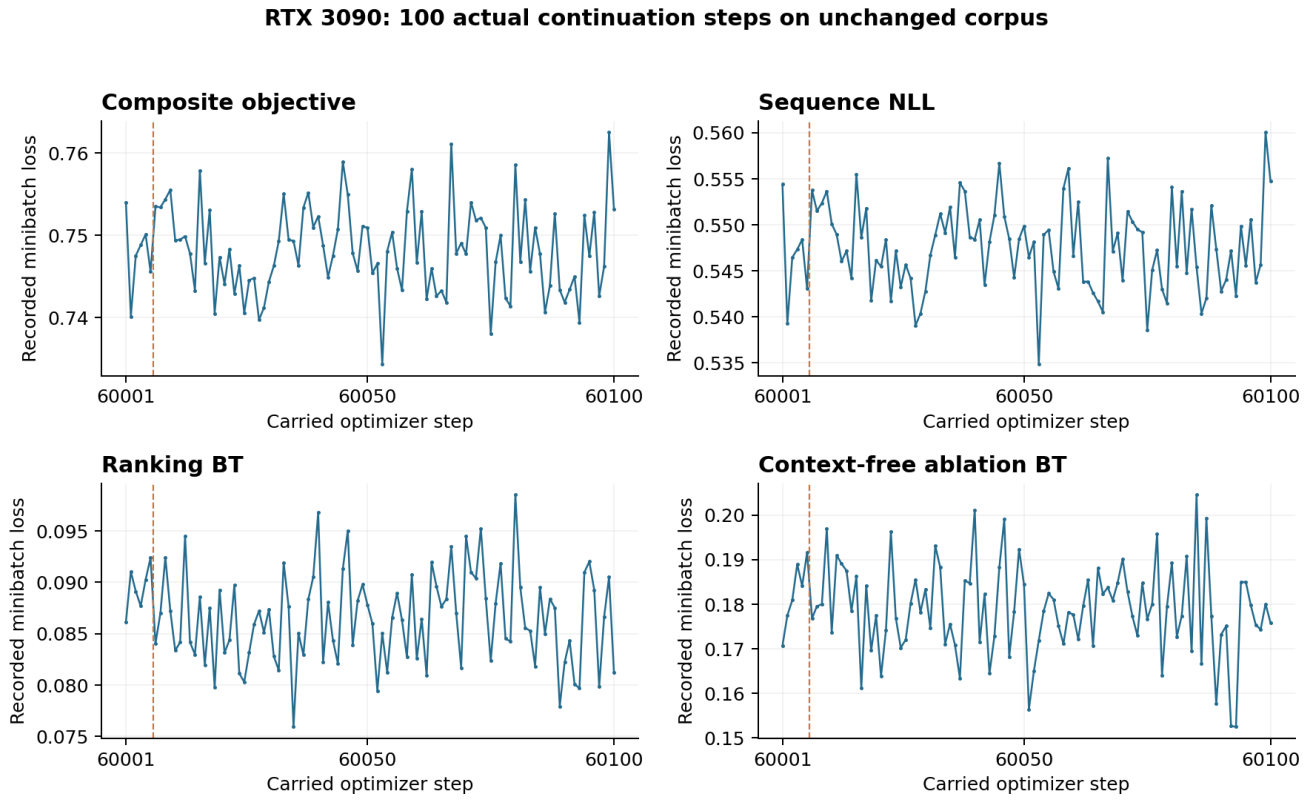


Figure B.8 Original A100 checkpoint validation measurements, with accepted step 60000 marked

The 78 retained A100 state manifests live under `/home/ubuntu/jope-node-mirror/weights/runs/eight-a100-recovery-20260912/learning/cycle-*/train-2/checkpoints/step-*/state.json`. The retained metadata covers 78 checkpoints, all with “carried FP32 AdamW with bias-correction step.” This is materially different from the 108 reset blocks. It spans 22 corpus identities and recorded steps 4,000–69,000. The graph keeps these as adaptive validation observations. A higher retained step is not automatically the serving model. The accepted 60,000 state's exact hash is `8ac80c26…`, corpus `1c840597…`, model `053df4f5…`, and parameter digest `9c7057e7…`; it records ancestry from step 57,000 and requests fresh-balanced sampling. Its request for 50% fresh comparisons does not mean 50% fresh program slots or measured downstream improvement. The block-18 per-step log ends at this exact step/corpus; block 19 explicitly starts from accepted state `8ac80c26…`. Of the 78 retained checkpoints, 74 also have validation entries inside the per-step logs; all 296 overlapping metric cells match the separately extracted checkpoint CSV. Sequence NLL is 1.428209 at step 4,000, 1.387687 at accepted 60,000, and 1.395417 at unaccepted 69,000; different adaptive panels limit this to descriptive observations. The product explicitly refuses the retained step-69,000 state/model identities rather than selecting the highest numbered checkpoint. The accepted state records no `execution_partition`, while the native trainer's exact resume requires one equal to the requested topology (`crates/jope-engine/src/training.rs:1106`), so step 60,000 cannot be resumed as the same experiment; §8.2.4 Native lineage B: the held command and its blockers describes the held continuation routes.

B.3.4.2 The RTX step 60,100 qualification run



14 September 2026; world 1 × microbatch 256 × accumulation 8 = 2,048. Dashed line marks resumed attempt.
204,800 committed program-example presentations; zero fresh presentations. Hardware qualification, not promotion.
Source: rtx60100-step-metrics.csv; all 100 logged steps shown, without smoothing or reconstructed points.

Figure B.9 All 100 original step measurements of the world-1 qualification run, a historical hardware continuation rather than a promotion, with the recorded resume boundary; the RTX 3090 attribution comes from ADR-0005, not from the run metadata

The RTX run retains every metric record at steps 60,001–60,100. The exact 100-row source is `/home/ubuntu/jope-evidence/xbabe2-migration-20260914/accepted60000-restored/parallel/runs/xbabe2-hardware-qualification-20260914/final-runtime-v2-eof/train/metrics.jsonl`, distinct from the earlier recovery attempts. RTX 3090 attribution comes from ADR-0005; the retained run metadata itself records world size 1. Its first attempt starts from 60,000; later rows record initial optimizer step 60,006 after recovery. Therefore the graph uses optimizer step, not the run-local elapsed timer, which resets at resume. The timestamp span includes interruption/recovery and differs from the final attempt's 73.561-second elapsed value; neither should replace the other without labeling its scope.

The final state records **world 1 × microbatch 256 × accumulation 8 = global batch 2,048**. ADR-0005 at `a75e543d2` describes this run as microbatch 32 × accumulation 8; the retained state corrects that. Documentation-correction PR86 (head `0079ceddb`), merged at 2026-09-24T00:04:12Z as main `2b375ef11`, appended the same correction to ADR-0005 as its “Implementation correction — 2026-09-23” section without rewriting its historical decision. Model digest changes from `053df4f5…` to `1f202cce…`; parameter digest changes from `9c7057e7…` to `762094a0…`; optimizer identity also changes. The final state is `745be4b5e998bce85615349a5b03b901cd8ad2cf605e6dce3b86fbf777cc7388`. These are retained training-update and continuation evidence, not a fresh numerical replay performed for this manual.

The receipt commits 204,800 program presentations and zero fresh programs. Its purpose is explicitly “isolated unchanged-corpus hardware qualification, not production continuation outcomes.” Final adaptive validation NLL is 1.409792 and Bradley–Terry 0.475421, on 256 examples. Step 60,000's retained validation used a different panel and 2,048 examples; comparing those numbers as a paired improvement would be invalid. The checkpoint is not promoted by being 100 steps newer. Training further on one unchanged corpus is continuation, not a new generation. The recorded interruption and recovery after step 60,006 is also not evidence of an exact interrupted resume as the owner now defines it: that must reject optimizer fallback before any update and prove that the interrupted and uninterrupted runs reach the same state, and this manual has no such parity comparison for this run.

B.4 Current publication regeneration and provenance

Edition 3 regenerated the seven retained-history figures and the 63,000-step SVG from the same authenticated observations. `data/historical-plot-provenance.json` records each figure’s source hashes, extraction/replay and plotting commands, coverage, units and interpretation limits. `tools/reextract_training_metrics.py --output NEW_CUSTODY_DIRECTORY` reproduces the 3,218 original metric values and 78 checkpoint lineage rows using frozen row-selection metadata; it does not overwrite original records or invent missing observations. The A100 per-step extraction command remains in `data/per-step-history/README.md`. This rebuild rechecked all 21 original A100 logs and all 82 retained historical sources. For the 108-block history, the original remote logs are represented only by their retained endpoint capture and recorded hashes; this is not a new independent recovery of those remote logs.

All 63,000 step rows and 378,000 six-metric tokens matched their retained original logs. These are metadata reconstruction and plotting checks, not optimizer execution or new measurements. The new observed-transferable-lift KPI remains **not yet measured** (§9.1 Observed transferable lift: reporting contract, not yet measured).

APPENDIX C

Historical GP targets, rankings, Elo and attribution limits

IN THIS APPENDIX

C.1 Did GP targets get better? 65
C.1.1 Carried-forward scores and the admitted receipt inventory 65
C.1.2 First-to-last comparison at 1,000 evaluations 66
C.1.3 The pine-chromium example, attribution limits and published data 67
C.1.4 Paired final-search outcomes within the admitted development inventory 69
C.2 Existing historical rankings and Elo 69
C.2.1 Retained snapshots and plot data 69
C.2.2 Ranking figures: interpretation contract 70
C.2.3 SOTArena published and corrected ladders 70
C.2.3.1 Historical benchmark-valid Elo 70
C.2.3.2 Why the two historical ladders differ 71
C.2.4 Historical same-cut screening and deepening 72
C.2.4.1 Same-cut v1 and v2 snapshots with screen and deep Elo 72
C.2.4.2 Historical relative loss and per-dataset rank 73
C.2.5 Historical production time and the supplied reference scenario 75

This chapter answers two questions: which historical corpora and target records survive, and how the historical target collections and their observed scores changed. What one training example contains is described in §7.1 A dataset is not one training example, and how the implementation stores and loads packed records in §D.7.1 Record layout. This manual's audit read metadata, manifests and development-score receipts. It did not read sealed labels, model tensors, packed target-record bodies or the large audit-label sidecars, and it ran no training or fits. External packing, inventory and seed audits report results that depend on payloads this audit did not open; those results are attributed separately below and were checked here through retained metadata, scripts and arithmetic, without replaying the payload reads. File-metadata observations, externally reported payload checks and source-derived calculations remain distinct evidence levels.

C.1 Did GP targets get better?

C.1.1 Carried-forward scores and the admitted receipt inventory

Some recorded carried-forward best scores became lower. These are historical development outcomes from searches sharing incumbent and teacher seeds. They are not independent discoveries by each arm, and their no-worsening pattern is partly structural. Corpus growth alone would not establish even this narrower result.

The audit enumerated 29,744 `search.json` paths under `/home/ubuntu/jope-node-mirror/results`, which resolves into `/home/ubuntu/jain-split/jain-datasets/.runtime/node-archive-mirror-20260914/results`. This source covers generations 0–70. The audit rejected 3,387 paths outside the accepted context index's `meta_train` dataset/cut allowlist **before opening their score files**. The remaining 26,357 receipts identify adaptive development-validation RMSE, prepared data, vocabulary, evaluator, execution

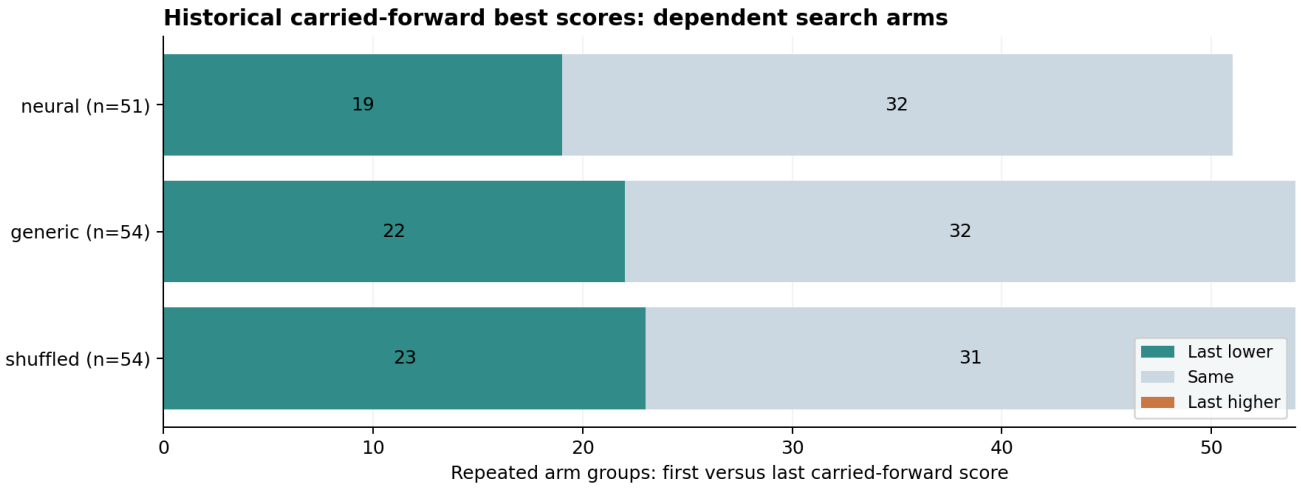
protocol and completion metadata. Each source file and observation ledger has a distinct recorded digest. Their recorded completion times span September 12, 2026 18:37:13 UTC to September 14, 2026 13:37:43 UTC. These times are receipt metadata, not a new replay of the fits.

Only 3,530 of the 26,357 permitted receipts are also present in the separately preserved `accepted60000-restored` tree, all byte-identical; the other 22,827 receipts, including every permitted generation 10–70 receipt, are absent from that tree. A custody archive preserves all **26,357 original search receipts**, totaling **62,271,855 uncompressed bytes**. A separate streaming verification checked every member name, size and original SHA256 without reopening the source bodies. `original-target-receipts-custody.json` pins the archive and verification. The archive is a second local copy on the same filesystem, not an off-host backup. The separate seed bodies used by the external carry-forward analysis are outside that archive.

C.1.2 First-to-last comparison at 1,000 evaluations

For a readable comparison, the table restricts each repeated comparison to exactly 1,000 completed evaluations, matching its recorded dataset/family, cut, prepared SHA, vocabulary SHA, evaluator SHA/protocol, execution-order protocol and run. It compares the first retained generation with the last retained generation within that group. Lower RMSE is better. Improvement and worsening use exact < and > comparisons of the parsed reported scores; unchanged means exact equality, with no epsilon. The analysis script applies no additional cumulative minimum, but the underlying search already carries incumbent/teacher programs forward. Thus the reported per-generation best usually behaves like an inherited incumbent rather than a fresh independent draw, but it is not strictly monotone. In the 9,536 adjacent 1,000-evaluation transitions where the previous best was re-seeded, the recorded best stayed equal 9,406 times, fell 130 times and never rose; all 27 rises occur in the 76 transitions where it was not re-seeded. The score field is `best_validation_rmse` under the historical `featmath_training_fold_normalized_v1` evaluator; the plots do not claim original target units.

Historical search arm	Repeated arm groups	Last lower	Same	Last higher	Median relative reduction, all groups (unchanged included)
Neural	51	19	32	0	0%
Generic	54	22	32	0	0%
Shuffled	54	23	31	0	0%



1,000 evaluations per search; shared incumbent/teacher seeds; unequal elapsed time. Arms are dependent.
64 lower arm groups represent 23 of 55 distinct dataset/cuts; median relative reduction is zero in each arm.
Source: target-quality-1000-summary.csv and all 159 comparison rows; carry-forward-audit.json.

Figure C.1 First-to-last carried-forward best scores at 1,000 evaluations; the arms share seeds. The 64 lower arm groups cover 23 distinct dataset/cuts. No arm group ended higher, so the “Last higher” legend entry has no bar.

The 159 arm groups cover **55 distinct dataset/cut pairs**. The 64 lower arm groups (19 + 22 + 23) cover only **23 distinct pairs**; 22 of those 23 end at the same score in every available arm. These counts were independently recomputed from the published comparison CSV. The denominators and shared seed pools make the arm counts dependent; one discovery can later appear in multiple arms.

The separately released external seed audit reports `seed_population_supplied=true` and `teacher_first_v1` for all 26,357 receipts. The previous recorded best appears in the next seed set in 25,790 of 25,985 transitions; 26,051 reported winners (98.84%) are already in that search's seed population. Incumbent, retained-teacher and exploration seed sets are identical across arms in all 8,785 multi-arm dataset/cut/generation groups checked. Those seed-body counts are externally reported and metadata-checked here, not a new local seed-payload replay. They explain why zero first-to-last worsening is not independent evidence of stability. A lower inherited score is also not a new discovery by the receiving arm. `carry-forward-audit.json` preserves these evidence boundaries.

This table is therefore not a paired policy-ranking result. Random seeds change between generations, elapsed time varies, and this audit did not requalify the actual CPU/memory/hardware enforcement or complete build closure of the historical searches. Matching these recorded fields is narrower than the complete current experiment identity. The recorded generation manifests declare 600 seconds and 1,000 evaluations for the illustrative example; those declarations do not establish equal consumed wall time. Checkpoint association is available by generation, but exact exposure of each improved target to later optimizer batches remains unjoined.

C.1.3 The pine-chromium example, attribution limits and published data

A concrete example is `mendeley-6bspjkt7bg-pine-chromium`, dataset `b68222ad60398ea8`, cut 1. Across generations 0–70 the recorded best development RMSE changes from **0.009749860799638275 to 0.003185684481133936**, a **67.3258% relative reduction**. All three arms finish at the same score. The neural arm first records that final value at generation 43; generic and shuffled do so at generation 42. The external seed audit identifies the final program outside the generic and shuffled seed sets at generation 42, then as an `incumbent_cut_teacher` seed in all three arms at generation 43. The neural arm's later equality is carry-forward, not evidence that it independently discovered that program. This identifies generation-level observations; it does not establish which of two same-generation arms

discovered the program first in wall-clock time. The recorded best is not monotone here either: generic reports 0.007209 at generation 29 and 0.009750 at generation 30, and generic and shuffled both rise from 0.004698 at generation 39 to 0.006747 at generation 40; the neural arm first reports 0.007209 at generation 39. The plot data include all 213 generation/arm pairs, with no interpolated missing generation. This example was selected after inspection as the largest first-to-last relative improvement in the neural 1,000-evaluation subset; it is illustrative, not representative.

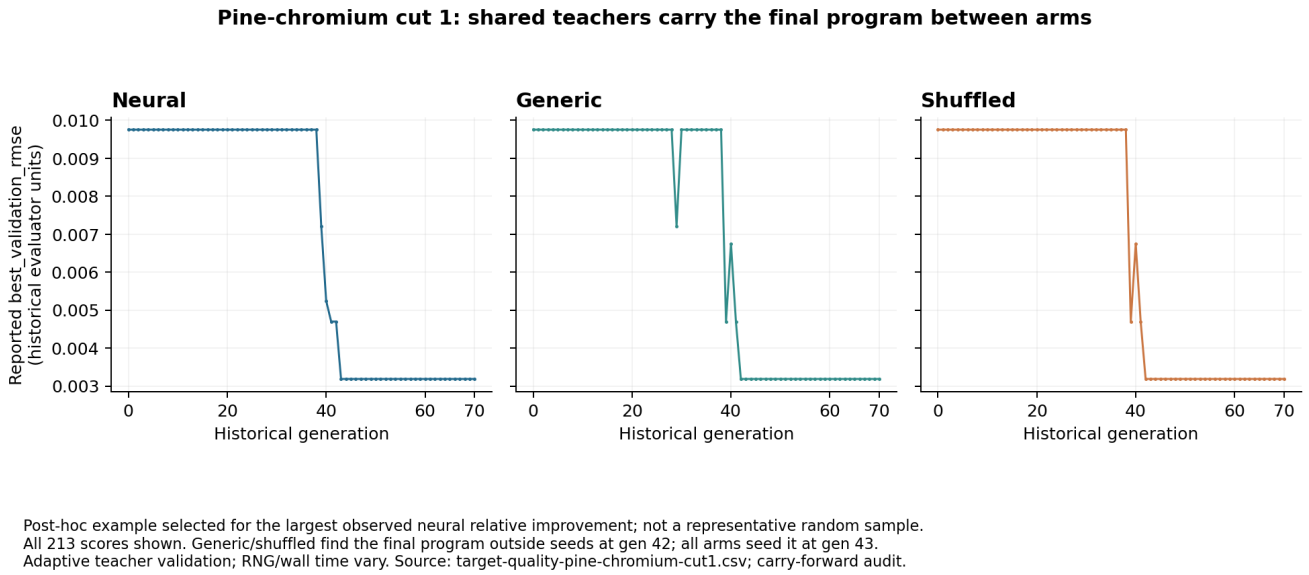


Figure C.2 Post-hoc pine-chromium cut 1 history with shared teachers: generic/shuffled record the final program outside their seeds at generation 42; all arms inherit it at generation 43.

Relative improvement is `(first_RMSE - last_RMSE) / first_RMSE`. The percentage is meaningful within that dataset/cut. Raw RMSEs from unlike target scales are not averaged to claim overall progress. A running-best curve, if separately shown, must be labelled as such: its monotonicity follows partly from selection. Better retained minima do not prove that a trained neural proposal policy caused the gain. Shared incumbent/teacher seed pools and carry-forward are established in the external audit; search history and ordinary exploration also prevent attribution to policy learning. Count a new search discovery separately from a copied incumbent: authenticate the full seed set, identify the first measured program absent from it, and retain the discovering arm, generation, context, program and parent identities. The aggregate table above does not provide that full per-program discovery ledger. The owner's acceptance rules (proposed) keep four claims distinct: an eligible teacher, a new discovery, information actually consumed by training, and an improved policy. Initial populations must be authenticated, and a discovery must not silently seed another arm. The historical records in this appendix support at most lower retained development scores and, for single programs such as the pine-chromium example, a generation-level first appearance outside a seed set. They do not establish teacher eligibility, consumption by a later optimizer or an improved policy, and a receipt hash alone does not make a program eligible.

The small publication files are `target-quality-1000-summary.csv`, `target-quality-1000-comparisons.csv` and `target-quality-pine-chromium-cut1.csv`. Full 26,357-row extracted evidence stays in custody, with its hash in `historical-target-quality-summary.json`; the manual need not embed the 19.5 MB full extraction. This analysis reads development scores only and creates no new teacher eligibility, promotion or benchmark-valid result.

C.1.4 Paired final-search outcomes within the admitted development inventory

A metadata-only comparison pairs arms **within the same generation**, keeping the recorded run, dataset/family, cut, prepared SHA, vocabulary SHA, evaluator SHA/protocol, execution-order protocol and exactly 1,000 completed evaluations equal. It uses the existing 26,357 admitted receipts, not a newly expanded input boundary. Scores are finite positive `best_validation_rmse`; lower is better. Each comparison counts exact score equality, with no epsilon.

Neural compared with	Paired observations	Neural lower	Equal	Neural higher	Geometric mean neural/ control RMSE
Generic	3,268	24	3,227	17	1.0001869094
Shuffled	3,253	24	3,213	16	1.0000854976

These are repeated adaptive **final-search** observations across generations 0–70, not measurements of initial neural seed quality. The generic comparison covers 52 dataset/cut pairs from 30 families; the shuffled comparison covers 52 pairs from 31 families. Reported wall times differ in every pair. Families, contexts and generations repeat; the arms share incumbent, retained-teacher and exploration seeds, and the search can recover from poor initial proposals. The paired rows are not independent arm discoveries. No independent-sample confidence interval, causal advantage or failure-inclusive benchmark claim follows. The ratios are descriptive and close to one; they do not establish that training improved routing.

The external report's broader routing extract has 42,863 rows. Only 3,530 match this chapter's admitted receipt inventory by retained path and all checked score, prepared-data, vocabulary, evaluator, protocol, completed-evaluation and wall-time fields. The other 39,333 are outside that existing inventory; their eligibility remains unqualified here, **not classified as all sealed**. No original score path was reopened for this join. The broader campaign ratios and p-values are therefore not imported. The compact `packing-routing-audit.json` pins inputs, matching fields, arithmetic and exclusions.

C.2 Existing historical rankings and Elo

C.2.1 Retained snapshots and plot data

Six committed snapshots are available and their ladder and score SHA-256 values were verified against retained bytes. No new Elo computation occurred in this audit. The snapshots use their own method pools and dataset protocols, so their Elo numbers cannot be compared across snapshots as a universal score.

Snapshot	Rankable rows	Methods	Datasets	Interpretation
same-cut-v1-screen	110	11	10	historical_same_cut_reproducible_snapshot_not_current_ASM30
same-cut-v1-deep	50	11	10	historical_same_cut_reproducible_snapshot_not_current_ASM30

Snapshot	Rankable rows	Methods	Datasets	Interpretation
same-cut-v2-screen	150	15	10	historical_same_cut_reproducible_snapshot_not_current_ASM30
same-cut-v2-deep	50	15	10	historical_same_cut_reproducible_snapshot_not_current_ASM30
historical-sotarena-published	8763	305	45	historical_published_best_known_test_selected_not_benchmark_valid
historical-sotarena-benchmark-valid	8718	304	45	historical_corrected_public_development

For plots, use `performance-evidence.json.gz`: 6 complete ladder row sets; 520 same-cut score cells, including missing/rejected statuses; 20 per-dataset reports with exact cut identities, source heads and screen/deep timings; 8,763 historical SOTArena score rows; and the 2 current-origin refit loss rows. Do not blend these into one chart without explicit separate protocol panels. The frozen panel (asm-private-30-v1) matrix had no completed cells as of 2026-09-23 about 23:55 UTC; represent it as pending, not a fabricated ranking.

C.2.2 Ranking figures: interpretation contract

Historical ladders retain their original snapshot names. Benchmark-valid and public-development views are separated. Same-cut screening includes all admitted methods; deepening includes only finalists, so empty cells have a deliberate meaning. A rank heatmap compares methods within a dataset and stage; it is not a plot of absolute loss across datasets. Any chart with only the top methods states that limit, while its CSV preserves the full underlying inventory.

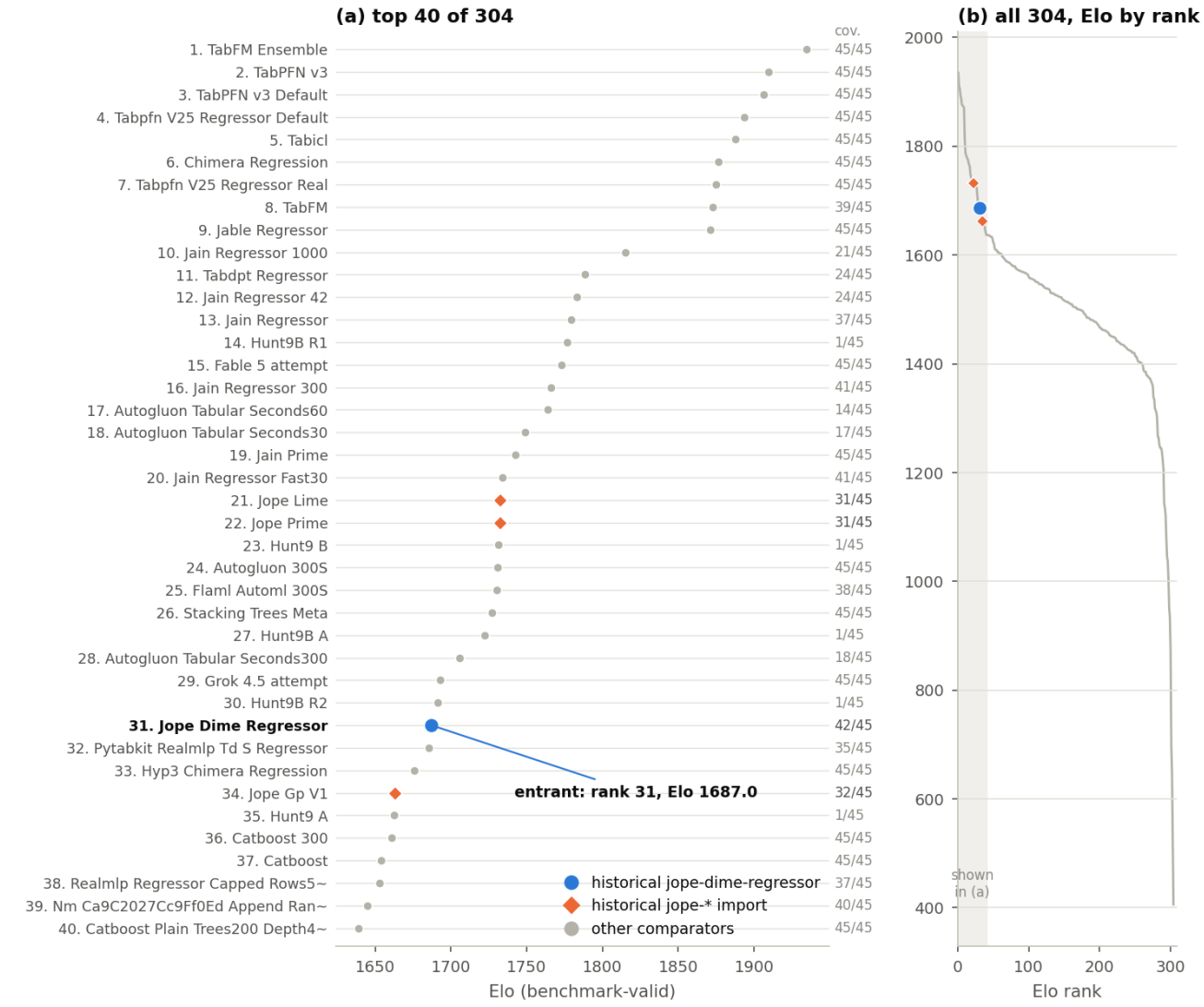
The historical ranking plots that follow use their separately pinned snapshots. They are not new model runs and are not a frozen-panel (asm-private-30-v1) campaign. Benchmark-valid and public-development views retain their original eligibility distinctions. Same-cut deep-stage coverage varies because only finalists were deepened; absent cells are not zero losses. See §11.5.1 The Bradley–Terry model, tie rule and defaults for the exact Elo model and tie rule, §11.5.2 Interpreting a rating for coverage limits and §11.2.2 Adaptive development exposure for the adaptive-development interpretation.

All machine-readable rows are retained in `data/performance-evidence.json.gz` and `data/plot-historical-elo.csv`. A displayed top-method subset does not remove the other methods from those inventories. Dataset selection and historical protocol caveats travel with each figure.

C.2.3 SOTArena published and corrected ladders

C.2.3.1 Historical benchmark-valid Elo

This view retains 304 methods and the historical corrected eligibility filter. The historical `jope-dime-regressor` entrant is rank 31 with Elo 1687.0 in this snapshot. “Benchmark-valid” means no retained artifact contradicts a row's flag; comparator protocols were not independently verified. It does not certify that every comparator obeyed today's frozen ASM protocol.



Source: assets/regression/leaderboard-tabular-regression-v2.benchmark-valid.csv (sha256 4c39f51c2e269114), cross-checked against assets/regression/elo-tabular-regression-v2.benchmark-valid.json; the published scores.csv with assets/regression/protocol-corrections-tabular-regression-v2.json applied. Committed arena snapshot; nothing re-run. Elo: regularized Bradley-Terry on the Elo scale, protocols ranked together, partial coverage allowed.
Caveat: 'benchmark-valid' means no artefact in this repository contradicts the row's published flag; comparator protocols were not verified. Entrant = the fixed-seed reproducible run (public-development). Orange = other historical jope-* imported run reports, not claimed by this repository. Coverage n/45 at right. As of 2026-09-23.

Figure C.3 Historical benchmark-valid Elo ladder and complete rank distribution

The top-method view is paired with the distribution of the full method universe. Dataset coverage and the identity of the historical entrant remain visible. These numbers describe the retained historical ladder, not the current accepted step-60000 model's thirty-dataset performance.

C.2.3.2 Why the two historical ladders differ

The published SOTArena snapshot puts historical `jope-dime` at rank 1/Elo 2259.8 with 11 wins and 45/45 coverage. All 45 rows are test-selected: 42 best-test evolution-envelope rows plus 3 test-adaptive accepted replays. The corrected benchmark-valid snapshot excludes those 45; **that rank 1 is not a held-out ASM claim**. The corrected snapshot has 304 methods/8,718 rankable rows. Its fixed-seed historical `jope-dime-regressor` entry is rank 31/Elo 1687.0, 1 win, 42/45 coverage; TabFM Ensemble is rank 1/Elo 1934.9. It remains historical public-development evidence.

The published view includes 45 test-selected `jope-dime` rows; the corrected benchmark-valid view removes them. The historical `jope-dime` rank-one Elo of 2259.8 belongs to the former view. It is a best-known test-selected result, not a held-out claim. An eligibility change alters the overlap graph and the fitted ladder, so these ratings must not be combined into a single current score.

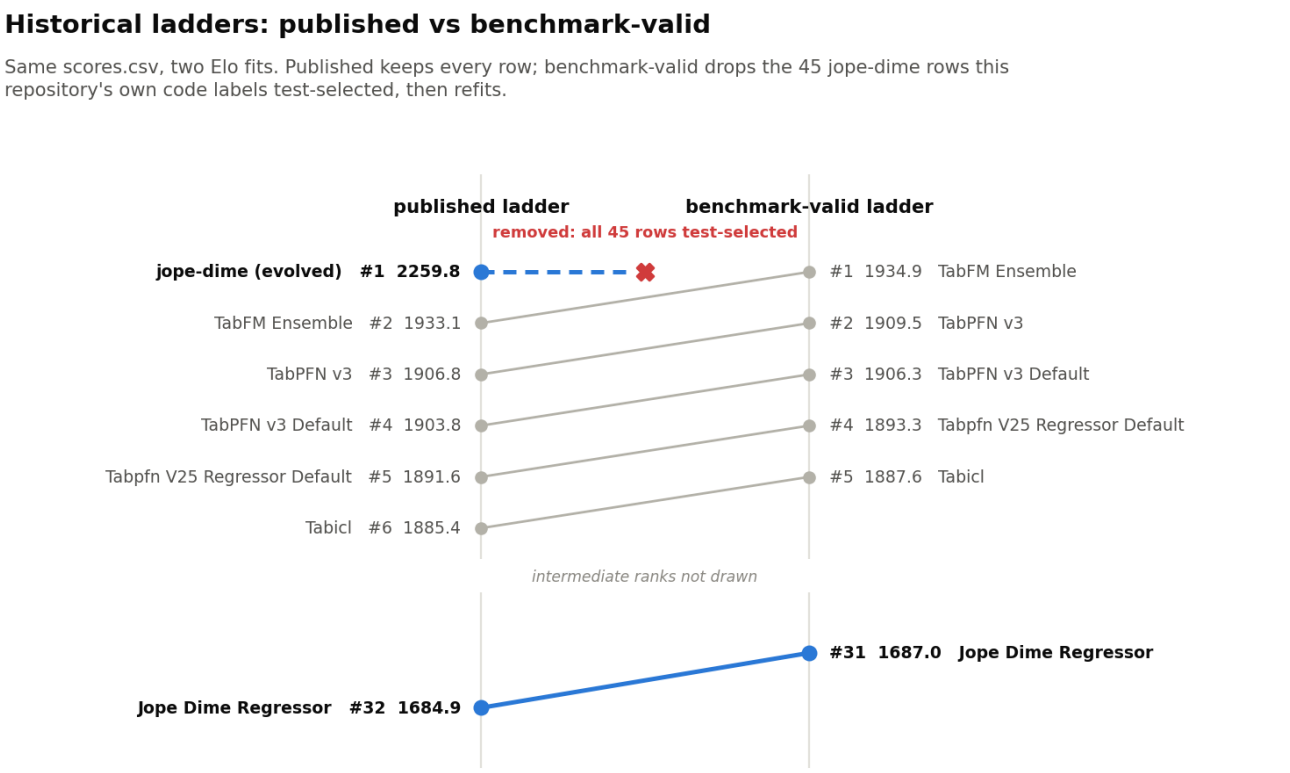


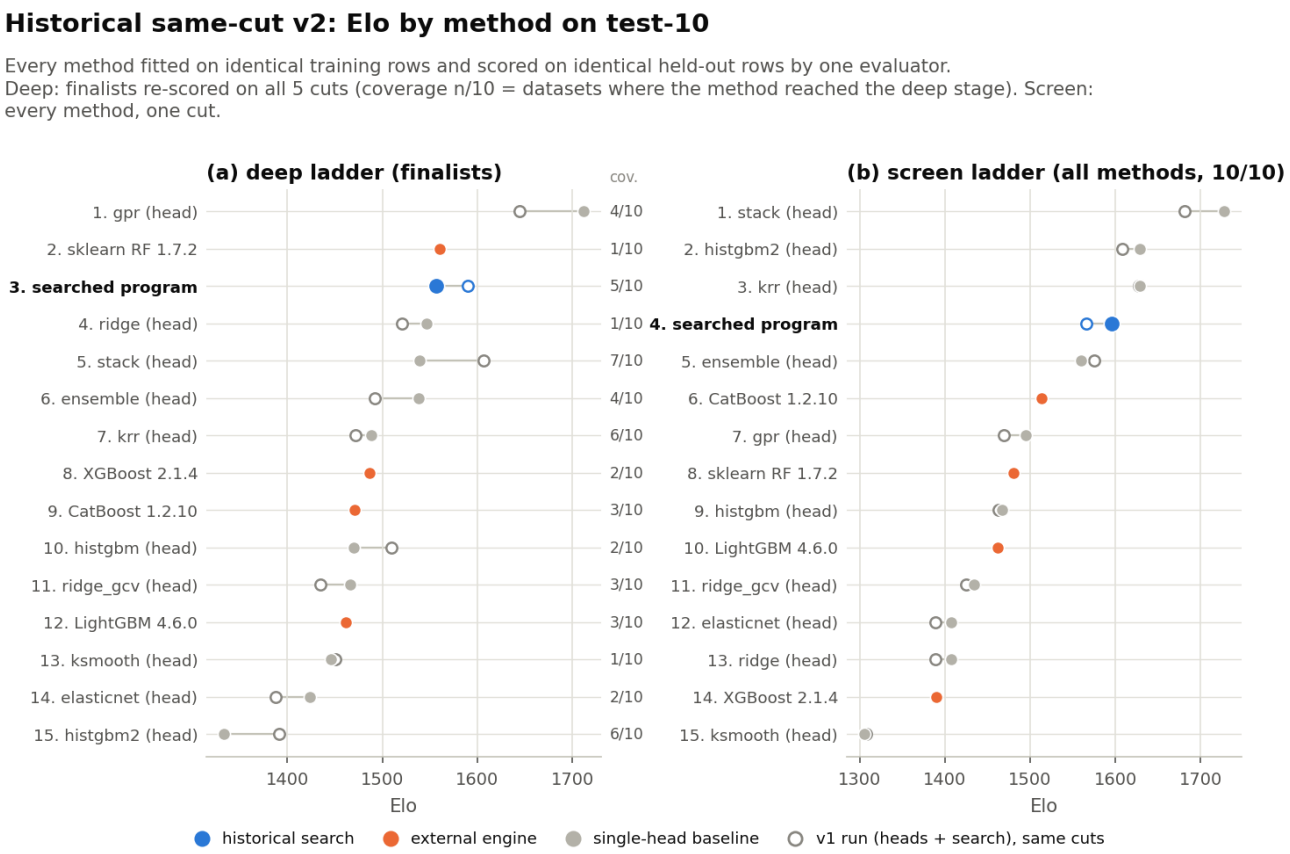
Figure C.4 Historical published and benchmark-valid ladders with their distinct eligibility

C.2.4 Historical same-cut screening and deepening

C.2.4.1 Same-cut v1 and v2 snapshots with screen and deep Elo

Same-cut v1 covers native heads plus resampled search; v2 adds XGBoost 2.1.4, LightGBM 4.6.0, CatBoost 1.2.10 and scikit-learn 1.7.2 random forest under Python 3.12.3. Both use the favorable historical test-10 regression selection, seed 20260907, five folds, top-five deepening and 32×8 search. Deep rows include only screened finalists; coverage must be printed with rankings. Their timing fields are then-recorded wall times under uncontrolled host load, not the present isolated-worker timing protocol.

Same-cut v2 compares all fifteen methods during screening on ten regression datasets. Only the five finalists per dataset are deepened across all five cuts, so the deep-stage denominator differs by method. Hollow markers preserve the earlier v1 snapshot; they are a separate run, not confidence intervals. The historical searched program is distinct from the current qualified neural proposal path.



Source: assets/same-cut/v2/leaderboard-jope-same-cut-v1-{deep,screen}.csv with their elo-*.json snapshots (hash-checked); hollow markers: assets/same-cut/v1 (heads + search only, same cuts). Recorded run, nothing re-run; search budget 32 programs x 8 rounds, re-run on every cut; engines run with library defaults, seed 0, one thread.
Caveat: FAVOURABLE SELECTION. test-10 was chosen where the historical engine ranked first on the published ladder (docs/datasets.md); nothing here generalises to a broader panel. The deep ladder ranks finalists only, so coverage differs by method. As of 2026-09-23.

Figure C.5 Historical same-cut screen and deep Elo with coverage

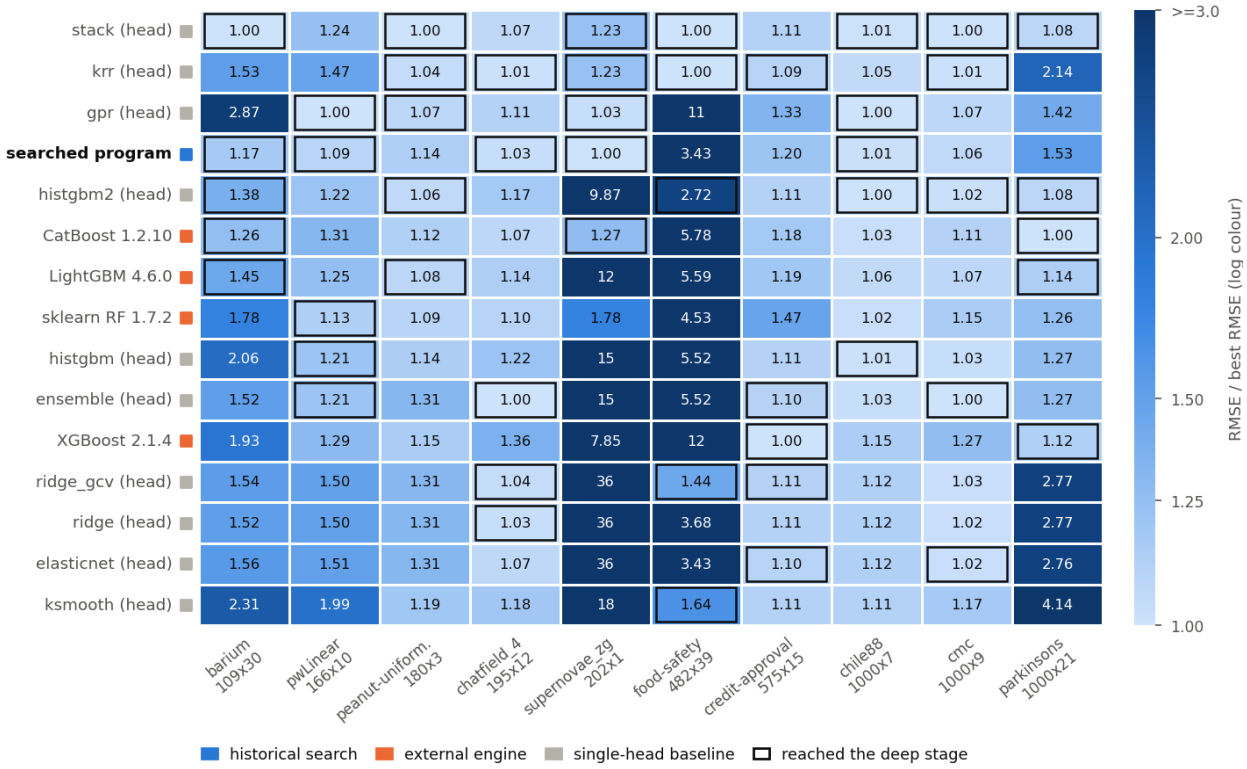
The ten historical datasets were a favorable selection for the earlier engine. Their ranking cannot establish performance on a broader corpus or substitute for the frozen panel (asm-private-30-v1). External engine versions printed in these plots are the historical run versions, not a claim about currently installed adapter versions.

C.2.4.2 Historical relative loss and per-dataset rank

Each heatmap cell is a method's screen RMSE divided by the best screen RMSE on the same dataset. A ratio of one is best; the color saturates at three while displayed values are rounded. This normalization supports within-dataset visual comparison without averaging incompatible raw target units. Outlines show which cells reached the deep stage.

Historical same-cut v2: relative error per dataset (screen cut)

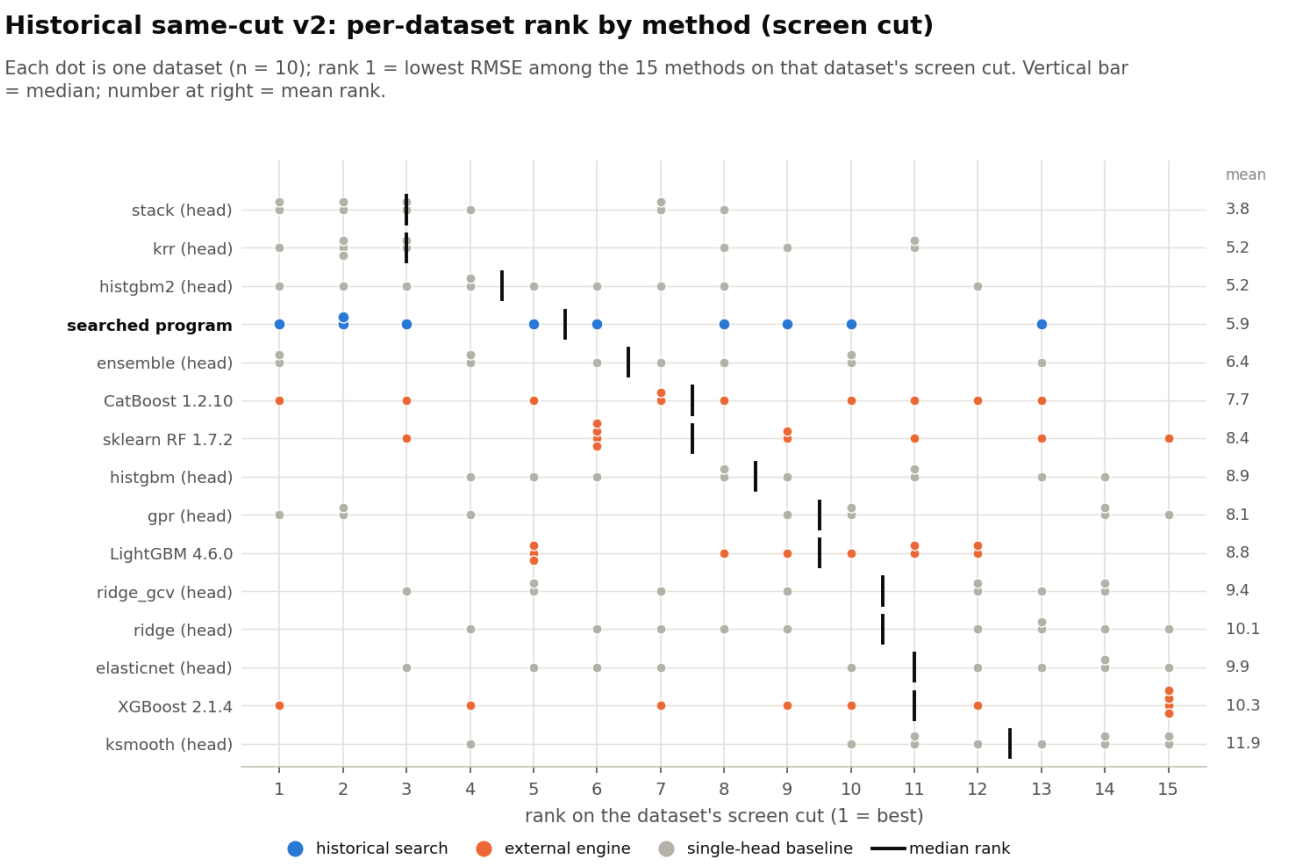
Cell = method RMSE / best RMSE on that dataset (1.00 = best). Rows sorted by median ratio; columns are datasets (training rows x features), sorted by rows.
Outlined cells reached the deep stage (top 5 on the screen cut). Colour saturates at 3.0; labels are rounded ratios.



Source: assets/same-cut/v2/reports/*.bench-report.json (screen metrics per method; the screen cut is fold 0 of fold_of('<dataset_hash>:20260907', row, 5)), extracted to figures/data/fig04_samecut_v2_per_dataset.csv. Recorded run, nothing re-run.
Caveat: FAVOURABLE SELECTION (test-10, docs/datasets.md); one cut per cell, so a single split's noise is included; the deep ladder (fig03) is the pooled 5-cut view. As of 2026-09-23.

Figure C.6 Historical same-cut relative RMSE heatmap

The rank-distribution plot shows all ten per-dataset screening ranks for each method, with mean and median. It reveals variability hidden by a single aggregate ladder. A rank difference conveys ordering, not the magnitude of a loss improvement or a confidence interval.

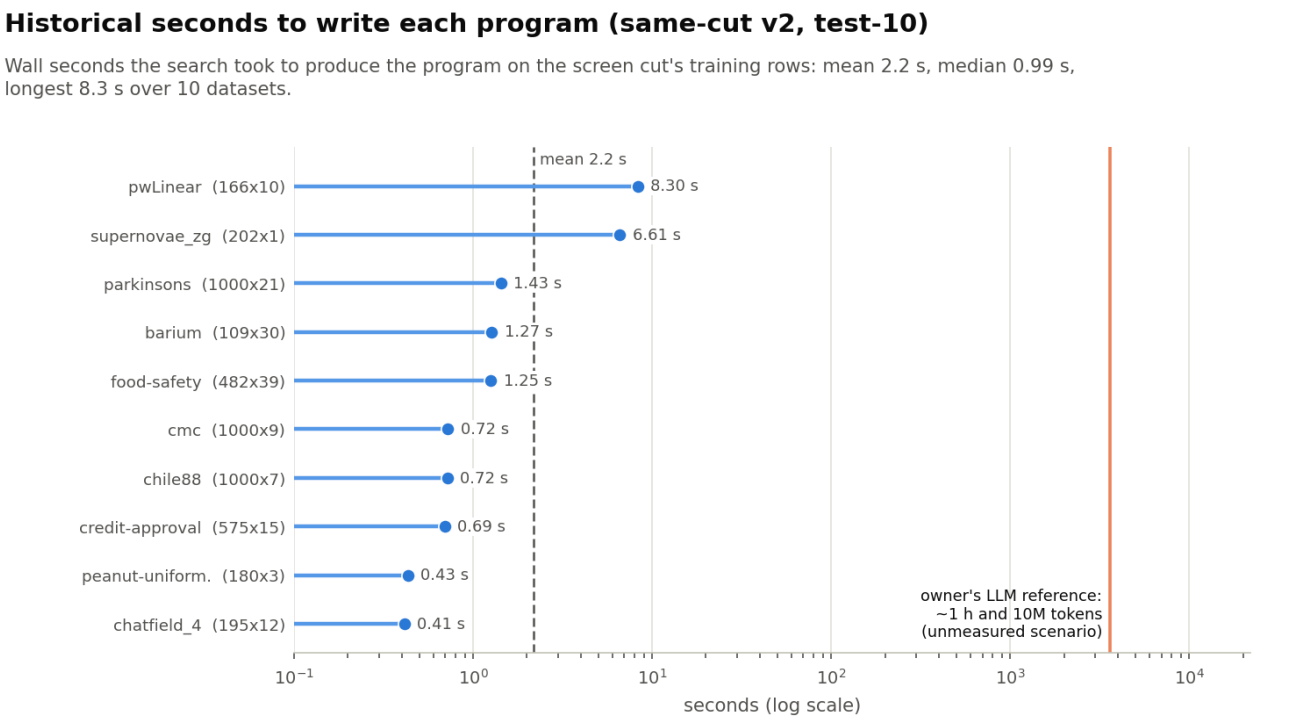


Source: assets/same-cut/v2/reports/*.bench-report.json field screen_rank (ranked by the screen score on one shared cut; on a shared cut R2 and RMSE order identically), extracted to figures/data/fig04_samecut_v2_per_dataset.csv. Recorded run, nothing re-run.
Caveat: FAVOURABLE SELECTION (test-10, docs/datasets.md); one cut per dataset; ranks ignore effect sizes (see fig04 for the ratios). As of 2026-09-23.

Figure C.7 Historical same-cut per-dataset rank distribution

C.2.5 Historical production time and the supplied reference scenario

The retained same-cut reports record mean program-production wall time 2.1837062093 seconds, median 0.9870389355 seconds and maximum 8.299870126 seconds across ten datasets. These are recorded timings under the host's contemporary load, not isolated controlled latency measurements and not the current paired ASM proof's elapsed time.



Source: assets/same-cut/v2/reports/*.bench-report.json, method afm.lite-race.v1.resampled field produce_seconds (budget 32 programs x 8 rounds, 16 workers on a 128-core host), extracted to figures/data/fig06_samecut_v2_search_seconds.csv. Timings are recorded wall time under whatever else the host ran, not controlled.
Reference line: the owner's stated scenario for a large language model writing a program for the same task, about 1 hour and 10,000,000 tokens (deploy/ci/readme-benchmarks.py:73-77). This is a contextual duration reference, not a measured comparison. Workload, hardware, tokens and currency were not matched; no speedup is established. FAVOURABLE SELECTION (test-10). As of 2026-09-23.

Figure C.8 Historical program-production times on the favorable same-cut v2 test-10 selection; the one-hour/ten-million-token reference line is not measured

The one-hour/ten-million-token marker is the user's contextual scenario, drawn as a reference line; it is not measured. Workload, hardware, token trace, currency and result quality were not measured under matching conditions. The chart therefore establishes no matched speedup, cost saving, token-efficiency ratio or quality equivalence. The measured current proof and its per-predictor/per-recipe accounting remain separately reported.

APPENDIX D

Dated data estate, recovery evidence and binary formats

IN THIS APPENDIX

D.1 Configured product stores and local roots 78
D.1.1 Access method, store defaults and path conventions 78
D.1.2 Product S3 buckets 78
D.1.3 Product local roots and the local dataset cache 79
D.2 Additional archive, replica and custody metadata 80
D.2.1 Additional and replica buckets 80
D.2.2 Learning-store prefixes 80
D.2.3 Versioning and Object Lock 81
D.3 Additional host data roots 81
D.3.1 Root inventory and totals 81
D.3.2 Filesystem capacity and dataset-shaped directories 82
D.4 Recorded copy and retention gaps 82
D.4.1 Material without an established remote copy 82
D.4.2 Recorded custody risks and disk capacity 83
D.5 Recovery evidence: what is preserved, and what was lost 83
D.5.1 Lineages A and B and the staging store 83
D.5.2 Lineage-A selections, archives, lost records and exposed score tiers 84
D.6 Retained corpus sizes 84
D.6.1 Six retained corpora 84
D.6.2 Snapshot 421 and snapshot 97 accounting 85
D.7 Binary format: the exact packed teaching record 85
D.7.1 Record layout 85
D.7.2 Packing semantics, legal sets and label policies 86
D.7.3 Verified loading into seal-protected shared memory 86
D.7.4 What the accepted-corpus packing audit measured 87
D.7.4.1 Occupancy, padding and index size 87
D.7.4.2 Compression, the proposed 180-byte layout and label precision 87

This appendix draws mainly on four dated metadata captures made on 2026-09-23; §D.4 Recorded copy and retention gaps also uses the DATA-01 author's copy-risk reconciliation, whose sizes partly come from a 2026-09-20 census. None of the four captures read a dataset payload, sealed label or model weight body. Compare figures within one capture; the captures differ in time and scope.

Capture	Time (UTC)	Scope	Where
Product local roots	16:29:27	Six named product roots; `lstat` metadata plus two cached dataset manifests	§D.1.3 Product local roots and the local dataset cache
Product S3 buckets	16:39:11	Three product buckets; current objects only	§D.1.2 Product S3 buckets

Capture	Time (UTC)	Scope	Where
DATA-01 bucket extension	configuration 17:30:13–17:30:19; learning-store prefix listings 17:30:32–17:33:50; bucket listings 17:34:15–17:37:10; assembled 17:37:54	Configuration of nine buckets, targeted `veox-jopedime-learning-use2` prefixes, and current-object listings of seven additional or reconciled buckets	§D.2 Additional archive, replica and custody metadata
DATA-01 host roots	scans started 17:30:58–17:33:14; report 18:54:01	Twelve named host roots; `find -xdev` metadata	§D.3 Additional host data roots

Both DATA-01 captures reach this manual only as an aggregate subset; §D.2.1 Additional and replica buckets records its release status.

D.1 Configured product stores and local roots

D.1.1 Access method, store defaults and path conventions

AWS metadata access used profile `veox-jepson-prod` and verified account `714044927654` with STS. No credentials are included in the receipts. Current-object listings exclude object versions and deleted objects. ETags are retained as concurrency guards for metadata downloads; they are not assumed to be SHA-256 content hashes. Catalog/collection/bundle manifest downloads use the exact listed ETag and verify size; their bytes receive local SHA-256 pins. No S3 dataset blobs, sealed tables, or model weight bodies were fetched.

`jope-datasets` defaults to `s3://veox-jopedime-datasets-use2`; `jope-models` defaults to `s3://veox-jopedime-models-use2`. The known learning store is `s3://veox-jopedime-learning-use2`; the current `jope-learning` mirror resolver uses `--store`, then `JOPE_LEARNING_URL`, and otherwise no remote mirror. It must not be described as an automatic remote default. No JOPE path/store overrides were set in this audit process.

There are two source-level path conventions to name explicitly: dataset cache resolution defaults to `${XDG_CACHE_HOME:-~/.cache}/jope`, while general `jope_paths::jope_home()` resolves a marked checkout. General mutable state defaults to `${XDG_STATE_HOME:-~/.local/state}/jope`. Use an explicit root or the owning command's resolver; do not assume the shared variable name makes the defaults identical.

D.1.2 Product S3 buckets

S3 bucket / prefix	Current listed objects	Current listed logical bytes
`veox-jopedime-datasets-use2`	15,442	3,234,922,515
↳ `blobs/`	11,577	3,224,104,137
↳ `catalog/`	2	5,363,942
↳ `collections/`	4	94,356
↳ `datasets/`	3,859	5,360,080
`veox-jopedime-learning-use2`	645,281	430,645,459,168
↳ `staging/`	645,281	430,645,459,168
`veox-jopedime-models-use2`	967	3,929,105,273
↳ `blobs/`	21	1,461,921,570

S3 bucket / prefix	Current listed objects	Current listed logical bytes
↳ bundles/	3	32,045
↳ staging/	943	2,467,151,658

The original three bucket listings completed with zero payload-body reads. The learning bucket contains current staging/lineage-a/, staging/lineage-b/ and staging/mirror-weights/ prefixes; the later DATA-01 extension below supplies measured and explicitly derived prefix counts. Current listed learning bytes are 430,645,459,168. A separate CloudWatch daily statistic at 2026-09-23T00:00:00Z reports 645,281 objects and 430,732,116,688 StandardStorage bytes. It is a dated metric with a different scope/capture time; no exact reconciliation with the current listing is claimed. Paginated S3 listing is an observation over an interval, not an atomic bucket snapshot.

D.1.3 Product local roots and the local dataset cache

Local root	Present	Regular files	Logical bytes	Unique-inode logical bytes
/home/ubuntu/.cache/jope	True	170	36,754,881	35,585,244
/home/ubuntu/.local/state/jope	False	0	0	0
/home/ubuntu/JopeDime/datasets/test	True	40	1,096,413	1,096,413
/home/ubuntu/JopeDime/datasets/by-id	False	0	0	0
/home/ubuntu/JopeDime/models	True	5	33,687	33,687
/home/ubuntu/JopeDime/state	False	0	0	0

The local inventory uses lstat, does not follow symlinks, and counts hardlinked payload bytes once in the unique-inode column. It covers the named product roots, not every archive, research checkout or build directory on the host. The standard local state roots were absent. That does not imply the host has no learning evidence: campaign and recovery receipts live in explicit custody paths outside those defaults.

The local dataset cache has two cataloged regression datasets, 0014b9dad5725a1e (2,000 rows, 16 features) and 006bca140f93e411 (2,000 rows, 32 features), each with 1,000 declared training and 1,000 declared test rows. Their payloads were not reopened for this inventory. The committed historical datasets/test/ collection is 10 datasets/40 files/1,096,413 B; it is distinct from the frozen thirty-dataset cache.

D.2 Additional archive, replica and custody metadata

D.2.1 Additional and replica buckets

A separate DATA-01 metadata capture at 2026-09-23T17:37:54Z extends the earlier three-store observation. It used the same verified AWS account and read current-object listings and bucket configuration only. The frozen panel and its row counts are unchanged. The author released the packet at 19:30:23 UTC (manifest `4bebee5a…`, 148 entries), but its handoff records a failed sealed-name gate and `cleared_for_handoff: false`. The only gate hit is in one raw file. A clarification recorded at 21:05:48 UTC (`RELEASE-CLARIFICATION.md`, `380f3402…`, covered by manifest `SHA256SUMS.release-2`, `a4b0c436…`) limits the release to eight pinned files: seven aggregate files and the DOCFIX-01 patch, which is reserved for a separate follow-up documentation PR. Codex applied that patch, with bounded corrections, in documentation-correction PR86 (head `0079ceddb`), which merged at 2026-09-24T00:04:12Z as main `2b375ef11`. Everything under `raw/` and `scratch/`, and the full handoff, stays quarantined in custody. This edition imports only Codex's two-file aggregate projection (stopped subset `558548be…`), which the clarification confirms lies inside the released scope: `data-estate-extension-summary.json` (`e4dbe075…`) and its source/scope pointer `data-estate-extension-release.json` (`366fa000…`). This is an aggregate subset, not a claim that the whole packet passed its release gates.

Additional or reconciled S3 bucket	Current objects	Current logical bytes
`veox-jopedime-archive-use2`	180,547	33,078,118,944
`veox-jopedime-learning-usw2`	0	0
`veox-jopedime-models-usw2`	967	3,929,105,273
`veox-jopedime-models-use2`	967	3,929,105,273
`jopedime-714044927654-us-east-2-dev`	186,305	118,116,270,003
`jopedime-714044927654-us-east-2-prod`	186,304	117,847,288,926
`jopedime-714044927654-us-east-2-sealed`	8,679	246,060,447,894

The models buckets agree in total and reported prefix counts and bytes; this does not prove key-by-key or payload identity. The learning replica is empty at this capture. Its source's four enabled replication filters are `hwm/`, `waves/`, `checkpoints/serving/` and the prefix `training-head`; none covers the current `staging/` objects. This is an observed configuration/coverage gap, not evidence that a matching replication rule failed. A `HeadBucket` probe for `veox-jopedime-datasets-usw2` returned HTTP 404 in this account/context; no dataset replica listing was obtained.

D.2.2 Learning-store prefixes

Learning-store prefix	Objects	Logical bytes	Basis
`staging/lineage-b/accepted60000-restored/`	225,207	110,918,425,513	measured
`staging/lineage-a/encodings/`	419,644	149,840,032,105	derived

Learning-store prefix	Objects	Logical bytes	Basis
staging/lineage-a/preserved-archives-20260908-F-D-E/	34	141,775,927,725	measured
staging/lineage-a/genebank/	5	1,809,949,972	measured
staging/mirror-weights/	391	26,301,123,853	measured

The encodings row is a residual: the earlier full-bucket total minus four later measured sibling prefixes. It was not independently re-listed in full, and the sum closes by construction across capture times. Keep this distinction when comparing inventories. The accepted-60000 restored prefix's 225,207 objects and 110,918,425,513 B also match the retained local upload inventory by file count and size; that comparison is not a fresh content restore test.

D.2.3 Versioning and Object Lock

All nine inspected buckets report versioning enabled. The six `veox-jopedime-*` buckets report Object Lock enabled with no default retention; the separate dev/prod/sealed family returned no Object Lock configuration. Neither result establishes retention for each object or preservation of every prior version. This listing excludes old versions and delete markers. Sealed storage contributes aggregate counts and bytes only; no sealed object body was fetched. Object/file counts are not counts of unique datasets, admitted teachers or recoverable checkpoints.

D.3 Additional host data roots

D.3.1 Root inventory and totals

The DATA-01 report generated at 2026-09-23T18:54:01Z summarizes metadata scans starting at 17:30–17:33 UTC across twelve named roots with `find -xdev` metadata only. Its following apparent-byte totals count each regular-file path, including hardlinks. They are not additive with the earlier product-root table or the S3 copies, and do not measure memory needed to train.

Local root	Regular-file paths	Apparent bytes	Allocated bytes, including directories	Scan
/home/ubuntu/remote_super/quant	6,127,909	740,214,173,579	755,552,710,656	completed
/home/ubuntu/jope-flywheel	465,739	161,547,128,345	162,954,956,800	completed
/home/ubuntu/preserved-archives	23	141,775,926,746	141,776,224,256	completed
/home/ubuntu/jope-preservation	390,889	250,487,278,093	250,592,124,928	completed
/home/ubuntu/jope-evidence	236,213	123,899,684,744	125,110,591,488	completed
/home/ubuntu/jope-migration-evidence	234,158	33,597,305,065	35,018,276,864	partial: 15 unreadable directories

Local root	Regular-file paths	Apparent bytes	Allocated bytes, including directories	Scan
/home/ubuntu/jope-dime-mirror	201,251	23,067,601,488	23,633,518,592	completed
/home/ubuntu/jope-rescue-20260921	70,442	12,508,351,289	12,767,522,816	completed
/home/ubuntu/golden42	37,612	1,749,309,751	1,849,475,072	completed
/home/ubuntu/sotarena	22,602	705,696,476	776,736,768	completed
/home/ubuntu/hyperion/weights	4	253,108,234	253,132,800	completed
/home/ubuntu/.cache/jope	170	36,754,881	37,158,912	completed

Across these roots the scan observed 7,787,012 regular-file paths, 1,489,842,318,691 apparent bytes and 1,510,322,429,952 allocated bytes. Removing the recorded within-root and cross-root hardlink overcount gives 1,472,779,971,117 apparent bytes; content-identical copies on different inodes remain counted. Fifteen permission-denied directories under `jope-migration-evidence` are missing from these totals. Several roots were changing during the scan, including the custody directory holding the inventory itself.

D.3.2 Filesystem capacity and dataset-shaped directories

All twelve roots resolve to the same filesystem. Its contemporaneous readback was 7,936,183,619,584 B total, 6,815,282,892,800 B used and 1,040,879,616,000 B available (969.395 GiB); these are dated observations, not current capacity guarantees. Multiple local paths on that filesystem do not provide independent failure-domain copies. Count/size matches with upload receipts identify recovery candidates; they do not authorize deletion or replace exact-content restore verification.

The `remote_super/quant` scan found 2,322 top-level 16-hex-ID directories with all three `train.csv`, `test.csv` and `meta.json` markers. The corrected `synth*` classification contains 25,197 `synth_<16hex>` dataset directories with those direct markers, plus two containers named `synthetic` and `synthetic_packed`. The dataset-directory class contains 86,923 files and 21,494,046,579 B; it includes more than the three payload markers per dataset. These are name/shape observations, not proof of unique, valid or train-admitted datasets. No rows, feature distributions or label bodies were opened. A pathname containing `sealed` is only a naming statistic and cannot identify every sealed/test-label payload.

D.4 Recorded copy and retention gaps

D.4.1 Material without an established remote copy

The author reconciled the surveyed product/custody listings, upload receipts and local indexes. The following sizes describe material for which that reconciliation did not establish a remote copy. They do not prove absence from every bucket, machine or archive. Counts overlap other tables and risk classes; never sum them as unique at-risk bytes.

Material	Recorded files	Logical bytes	Evidence basis
T3 synthetic-regression train/meta/test payload	75,591	21,471,473,310	September 20 census; 25,197 dataset groups
T3 synthetic-binary payload	305,284	135,798,140,876	September 20 census
Synthetic source data	101,687	126,856,964,874	September 20 census
Binary generator script	1	102,068	September 23 file metadata
Node-archive mirror classes without a recorded S3 match	394,460	75,989,854,135	September 23 sizes joined to retained SHA-256 indexes
Quant model tree	650,015	78,566,865,846	September 23 file metadata; includes potentially replaceable cache
Migration evidence outside the uploaded node-high-value subtree	53,638	22,715,493,488	September 23 metadata; 15 unreadable directories excluded

Payload sizes above include aggregate withheld/test-file sizes from historical metadata; this audit did not open those bodies. Documentation records the T3 regression generator as lost; this is a retained provenance claim, not a fresh exhaustive search. The binary generator has no second copy established in the surveyed records. The 21,471,473,310 B payload figure is the older three-file census, whereas the 21,494,046,579 B current directory class also includes other files: these are different scopes. Model, research, journal, evidence and other retained-root risks are summarized by all 19 source risk IDs in the companion JSON; missing counts remain null. Seven retained roots were not rescanned.

D.4.2 Recorded custody risks and disk capacity

The empty surveyed learning replica and uncovered `staging/` prefix leave cross-region protection unestablished for the current learning objects. Counts matching between model buckets do not establish a successful content restore. Bucket Object Lock without default retention does not prove any object's retention or legal hold; custody buckets have no Object Lock configuration. The retained custody status also records a packer stop for exhausted disk reserve. At the dated local readback, `df` reported 87% used; capacity must be rechecked before admitting later workloads. These findings call for verified preservation and owner decisions about replication/retention. They authorize neither deletion nor new cloud writes, and they do not change the original training or panel budgets. Proposed rules: keep a single learning-store writer until an atomic server-side pointer update is qualified; backup, replication and retention operations remain separately authorized.

D.5 Recovery evidence: what is preserved, and what was lost

D.5.1 Lineages A and B and the staging store

Two historical lineages must remain separate. Lineage A contains July–September flywheel material, old winner/genebank state and preserved archives. Lineage B contains the native corpus/training chain used by the retained step-60000 checkpoint. `ADR-0005` and `docs/learning.md` describe their intended migration and their observed losses; those dated records do not authorize silently promoting old labels under the current ASM rules.

The learning S3 store is a staging estate: at the 2026-09-23 listings all 645,281 objects in `veox-jopedime-learning-use2` were under `staging/`. Targeted metadata listings found `staging/lineage-a/`, `staging/lineage-b/accepted60000-restored/` and `staging/mirror-weights/`. §D.1.2 Product S3 buckets and §D.2.2 Learning-store prefixes record the bucket census and the per-prefix counts. Object upload time is not the time a target was discovered or consumed. Five small metadata objects were fetched with an ETag precondition and hashed: the encoding caveats, frozen slice listing, archive catalog, genebank deprecation note and accepted60000 materialization intent. Their exact keys, byte lengths, ETags, upload times and fetch hashes are in `metadata/fetched-metadata.json` in custody.

D.5.2 Lineage-A selections, archives, lost records and exposed score tiers

The frozen July 25 `slice500` listing chose the first 500 bytewise-sorted names from 25,031 directories, excluding 186 incoming entries and hidden packed storage. That is a reproducible selection of candidate datasets, not proof of 500 admitted teachers. The retained encoding caveat says the historical binary target encoder used a regression-flavored checkpoint: binary values were in range, but that did not establish classification training qualification. The same note describes historical target-vector fusion and encoder pins. It must not be recast as the current verified classification-context contract.

Lineage A preserves archives for snapshots `004754860`, `008903292`, `009658180` and `009915070`. The archives' compressed byte sizes and checksums identify preservation containers, not in-memory training size. This audit did not extract them. The historical records declare the strict-test-exact database, pair registry, later immutable checkpoint descendants, acceptance pointers, sealed-consumption receipts and task-Elo registry lost. A directory number or an old selected pointer cannot recreate that missing evidence. The old genebank has an improvement-event schema with task, dataset, score tier, metric, evaluator, source and ingestion time, but lacks the complete cut/vocabulary/protocol identity required by the current HWM. Its 22,798 improvement-event rows are not 22,798 currently admissible teachers; its timestamps include backfill ingestion.

The exact-current HWM identity and current admission requirements are explained in §7.2.2 The admission guard, label exposure and HWM identity and §9.3 What “better and better” must mean. The historical `test_exact` score tier is optimization-exposed development evidence when used to choose programs. Renaming the tier does not restore independence.

D.6 Retained corpus sizes

D.6.1 Six retained corpora

The following table comes from six directly read local index/provenance files. Snapshot names are identifiers, not a reconstructed wall-clock axis. Several publications have no independently recorded timestamp in their index. These are retained historical corpora, not six steps in one proved monotonic update sequence. Counts must not be summed: corpora overlap.

Corpus	Records	Contexts	Dataset/ family counts	Groups	Record bytes	Audit-label bytes
`qualification-v1`	9,955	3,500	35 / 29	530	5,296,060	2,466,332
`qualification-v2`	85,498	3,500	35 / 29	834	45,484,936	25,540,156

Corpus	Records	Contexts	Dataset/ family counts	Groups	Record bytes	Audit-label bytes
`qualification-v3`	422,840	7,000	35 / 29	1,320	224,950,880	131,176,452
`cohort64-qualification-v1`	779,158	8,799	64 / 64	1,620	414,512,056	243,431,656
`dataset-inclusion-20260909`	5,538,508	8,904	82 / 82	6,815	2,946,486,256	1,753,858,051
Accepted60000 closure, `snapshot-000097`	10,214,332	8,894	80 / 80	7,877	5,434,024,624	3,227,289,461

All six use 532-byte packed records. The first five indexes use `parallel.teaching_corpus.v1`; the accepted closure uses `parallel.teaching_shards.v1`. Their indexes alone range from 1,807,418 to 98,330,379 bytes. The accepted sharded index is 60,589,288 bytes. Large JSON metadata remains a real part of startup work even when target records are binary. Full paths, byte counts, schemas and SHA-256 values are in `corpus-snapshots.csv` and `local-corpus-metadata.json`.

D.6.2 Snapshot 421 and snapshot 97 accounting

A separate September 11 audit captured snapshot `000421` with 8,899,150 records and 24,313,972 observations seen; its original index hash is `74d643be…`. That is a retained audit observation, not a seventh newly payload-verified corpus. The later accepted `000097` belongs to a different recovery run: 97 is not chronologically earlier than 421. The apparent count difference cannot be called a count of newly discovered unique teachers without joining actual program/context identities and admission events.

For accepted snapshot 97, metadata records 34,321,997 observations seen, 109,797 failed programs, 488,028 withheld meta-test observations, 1,338,630 observations without a matching context, 405,322 exact-parameter tokenization rejections and 112,903 out-of-vocabulary parameter rejections. These diagnostics describe the packer's accounting categories; do not assume they form a disjoint partition that can simply be subtracted from the observation total.

D.7 Binary format: the exact packed teaching record

D.7.1 Record layout

The Rust writer and reader agree on the following fixed-width layout. Offsets are byte offsets; all multibyte numbers are little-endian.

Offset	Bytes	Stored field	Meaning
0	256	`128 × u16` token IDs	Exact representable grammar sequence, with padded slots

Offset	Bytes	Stored field	Meaning
256	256	`128 × u16` legal-set IDs	In `teaching_shards.v1`, group-local index into that shard's own `legal_sets` for each sequence position
512	2	`u16` length	Actual token sequence length
514	2	`u16` zero	Reserved space
516	4	`u32` context ID	Index into the shared context metadata/vector table
520	4	`f32` value	Score-derived training value
524	4	`f32` percentile	Ranking supervision under the declared label policy
528	4	`f32` log-runtime	Natural logarithm of one plus runtime
Total	**532**		

D.7.2 Packing semantics, legal sets and label policies

This is binary packing, not general-purpose compression. Every record reserves 128 token slots and 128 legal-state slots. A short program still occupies 532 bytes on disk. Context vectors are shared by context identity. In the accepted sharded format, each group stores its own legal-set table in `shard.json` and its index entry; legal IDs are local to that table. Loading remaps them into one global table (63 distinct sets in the external accepted-corpus audit), shared in host memory and on the GPU. The older monolithic `teaching_corpus.v1` format instead has a top-level legal table. Do not interpret a raw shard ID as a global ID. `groups.bin` stores two little-endian u64 values per group—start record and count—so a group entry is 16 bytes. The JSON index binds context, vocabulary, record hashes and offsets; each immutable shard also has a manifest and an audit-label sidecar.

Original floating-point measurements and exact program provenance are retained separately from the float32 training targets. Historical `reference_v1` orders original f64 RMSEs, uses a median-scaled score value, and assigns ordinal percentiles. The stored value is `1 / (1 + RMSE / max(upper_middle_RMSE, 1e-8))`; the percentile is `1 - i/(n-1)` in the ordered group. Here `upper_middle_RMSE` is the sorted original RMSE at zero-based index `floor(n/2)`, including for an even-sized group; it is not an average of the two middle scores. Its tie behavior is not the later midrank policy. `frozen_z_v1` preserves the first admitted value/percentile for an existing program while new programs receive their initial labels. `strict_midrank_v4` introduces exact-original-precision tie handling and the strict admission contract. These names are different target semantics, not interchangeable storage optimizations. Repacking cannot silently relabel historical targets or snap unsupported program parameters to convenient vocabulary tokens.

D.7.3 Verified loading into seal-protected shared memory

The training reader's verified path copies and hashes the actual record bytes into seal-protected shared memory (a `memfd` carrying write, grow, shrink and seal-lock seals), or obtains a verified shared file carrying those seals, then uses a read-only memory mapping. These Linux file seals, unrelated to sealed evaluation data, prevent later writes, growth and truncation. This protects a running trainer

from source-file mutation and mmap truncation faults. It is **not** a promise of zero-copy disk startup: verification and the initial sealed-memory copy have costs. Index, record and context identities remain content-based; a filesystem timestamp is not a substitute for them.

D.7.4 What the accepted-corpus packing audit measured

D.7.4.1 Occupancy, padding and index size

The external verifier reports scanning all 10,214,332 records in 7,877 accepted-corpus shards, checking recorded payload/sidecar hashes and layout invariants. Its corpus identity is the same `1c840597…` bound in §5.4.1 Record split and the meta-test closure question. This manual independently verifies its selected metadata-file hashes and recomputes the following arithmetic; it does not claim a second scan of the original record or label bodies.

Quantity	Result	Evidence boundary
Occupied token slots	205,657,518 of 1,307,434,496	Sum of the external length histogram
Mean / maximum sequence length	20.1342 / 54 tokens	Accepted snapshot only; not a future grammar bound
Padding fraction	84.2701%	`1 - occupied_slots / (records × 128)`
Pretty-printed `legal_sets` values	37,691,003 bytes, 62.2074% of the 60,589,288-byte index	Raw stored JSON spans; compact serialization is a different 8,335,358 bytes
Original record arrays	5,434,024,624 bytes	10,214,332 × 532
Proposed 180-byte record arrays	1,838,579,760 bytes	Layout calculation; excludes indexes, contexts and runtime overhead
Proposed binary index	1,906,735 bytes	Estimated table components; no qualified reader or writer

D.7.4.2 Compression, the proposed 180-byte layout and label precision

The verifier also reports an actual external `zstd -3 -T1` size experiment: original records compressed separately by group occupy **292,918,135 bytes**, audit-label group frames **331,375,483 bytes**, and one compressed index **5,371,779 bytes**. These named components total **629,665,397 bytes**, versus **8,721,903,373 bytes** uncompressed. This subtotal excludes shard manifests, `groups.bin`, contexts, vocabulary, checkpoint and other provenance; it is not a complete deployable-store size or a host/GPU memory measurement. Experimental 180-byte group records compressed to 259,362,502 bytes in that external experiment. No loader-throughput, optimizer or fitted-search benefit was measured by this metadata audit.

Status: proposed. The owner's packing order follows phase profiling: first a verified Python packed-reader bridge, then immutable shard reuse with bounded staging, then compression, and a narrower layout only if measurements justify it. The 180-byte layout is such a narrower layout: `56 × u16` tokens + `56 × u8` local legal-set IDs + three float32 labels: 112 + 56 + 12 bytes. It would reconstruct length and context from the token prefix and authenticated group metadata. The observed maximum length rounds to 56, and the current loss path already trims each batch to a multiple of eight. Those facts motivate the proposal; they **do not prove bit-identical training**. A production codec must preserve the existing 128-token grammar limit through a versioned extension or verified fallback layout; the observed 54-token maximum does not authorize a smaller grammar cap or truncation. It must reject inconsistent widths and legal-ID overflow, preserve group/sampler order, RNG sequence and all

identity/provenance, reconstruct identical legal masks and every optimizer input, and pass record round trips plus fixed-batch token/mask/context/label, loss, gradient, update and reload/continuation parity. Changing physical bytes also changes the current raw-index corpus digest; compatibility needs an explicit verified migration, not an inherited old hash.

Keep the numerical policy separate from byte layout. The external all-pairs calculation reports that float16 creates new value ties in **0.377392%** of all within-group record pairs and **0.289905%** of teacher-versus-other pairs. These have different denominators from adjacent-score pairs. Quantizing labels therefore changes ranking supervision. Preserve float32 training labels and contexts, and original-precision audit observations; compression of their bytes does not require reducing their precision.

APPENDIX E

Full task catalogue, source/schema index, receipts and glossary

IN THIS APPENDIX

E.1 Primary acceptance items by stage 91
E.1.1 Stage 0: coordination, documentation and cross-cutting work 91
E.1.1.1 LIFE-01 — coordination 91
E.1.1.2 DOC-01 — manual 92
E.1.1.3 DOC-02 — manual examples 92
E.1.1.4 DOC-03 — manual figures 93
E.1.1.5 DOC-04 — manual quality 94
E.1.1.6 DOC-05 — real training and target history 94
E.1.1.7 DOC-06 — versions, KPIs and continuation runbook 94
E.1.1.8 INT-01 — serial integration 95
E.1.1.9 INT-02 — integration qualification 95
E.1.1.10 IP-01 — public privacy boundaries 96
E.1.1.11 MAINT-01 — broad diagnostic qualification 96
E.1.2 Stage 1: correct benchmark integration 97
E.1.2.1 BENCH-01 — frozen benchmark foundation 97
E.1.2.2 BENCH-02 — external comparators 97
E.1.3 Stage 2: shared execution and accounting 98
E.1.3.1 EXEC-01 — bounded workers 98
E.1.3.2 EXEC-02 — fault regression tests 98
E.1.3.3 EXEC-03 — receipt replay regression tests 99
E.1.3.4 EXEC-04 — cutoff semantics regression 99
E.1.3.5 EXEC-05 — accounting and unknown outcomes 100
E.1.4 Stage 3: neural consumption and training 100
E.1.4.1 NEURAL-01 — context qualification 100
E.1.4.2 TEACH-03 — real teacher collection 101
E.1.4.3 TEACH-04 — real corpus materialization 102
E.1.4.4 TRAIN-01 — actual trainer tests 102
E.1.4.5 TRAIN-02 — native supervised training 103
E.1.4.6 TRAIN-03 — bounded real GPU training 103
E.1.4.7 TRAIN-04 — checkpoint continuation and export 104
E.1.4.8 NEURAL-02 — trained-checkpoint consumer readiness 104
E.1.4.9 NEURAL-03 — live newly trained consumption 105
E.1.4.10 PACK-01 — packed training consumer and incremental loading 105
E.1.4.11 LOOP-01 — repeatable measured learning flywheel 106
E.1.5 Stage 4: portable fitted inference 106
E.1.5.1 INFER-01 — fitted state and privacy 106
E.1.5.2 INFER-02 — exact selected-winner portability 107
E.1.5.3 FUTURE-01 — CUDA inference and serving promotion 107
E.1.6 Stage 5: installation and measured demonstration 108
E.1.6.1 INSTALL-01 — private installation 108
E.1.6.2 INSTALL-02 — final input/runtime preparation 108
E.1.6.3 DEMO-01 — measured warm fitted demonstration 109
E.1.6.4 PANEL-01 — unchanged thirty-dataset campaign 109
E.1.6.5 REPORT-01 — complete measured report 110
E.1.7 Stage 6: eligible optimization and reuse 111
E.1.7.1 TEACH-01 — HWM comparison identity 111
E.1.7.2 TEACH-02 — label exposure readers 111
E.1.7.3 MOTIF-01 — eligible reusable fragments 112
E.1.7.4 MOTIF-02 — reuse validation 112
E.1.7.5 MOTIF-03 — new operators 112
E.1.7.6 FOCUS-01 — sustained single-dataset improvement 113
E.2 Coordinated future backlog: 24 proposals 113
E.2.1 Documentation, onboarding and repository hygiene 113
E.2.1.1 Proposal HANDOFF-01 113
E.2.1.2 Proposal HYG-01 114
E.2.1.3 Proposal HELLO-01 114
E.2.1.4 Proposal B6-01 114

E.2.1.5 Proposal DOCFIX-01 114
E.2.1.6 Proposal LAYOUT-01 114
E.2.2 Privacy and naming 114
E.2.2.1 Proposal IP-Q1 114
E.2.2.2 Proposal IP-Q2 115
E.2.2.3 Proposal IP-Q3 115
E.2.2.4 Proposal IP-Q4 115
E.2.2.5 Proposal IP-Q5 115
E.2.2.6 Proposal NAME-01 115
E.2.3 Data, teachers and training 115
E.2.3.1 Proposal DATA-01 115
E.2.3.2 Proposal DATA-02 116
E.2.3.3 Proposal DATA-03 116
E.2.3.4 Proposal TEACH-05 116
E.2.3.5 Proposal TRAIN-05 116
E.2.3.6 Proposal HWM-01 116
E.2.4 Evaluation, export, reuse and cost 117
E.2.4.1 Proposal SEALED-01 117
E.2.4.2 Proposal EMIT-01 117
E.2.4.3 Proposal INFER-03 117
E.2.4.4 Proposal EMIT-03 117
E.2.4.5 Proposal MOTIF-00 117
E.2.4.6 Proposal COST-01 117
E.3 Planned execution and review ownership 118
E.4 Source index 120
E.4.1 Current implementation references 120
E.4.2 Pending source references 122
E.5 Machine-readable companions 124
E.5.1 Companion files 124
E.5.2 Lossless evidence packaging 125
E.5.3 Training-history and continuation companions 125
E.5.3.1 Companion inventory 125
E.5.3.2 Reusing the evidence 126
E.6 Receipts 127
E.6.1 Important local receipts 127
E.6.2 The PR83 lifecycle record 128
E.7 Reproduction boundaries 128
E.8 Names and glossary 129

E.1 Primary acceptance items by stage

E.1.1 Stage 0: coordination, documentation and cross-cutting work

E.1.1.1 LIFE-01 — coordination

Priority: P0. **Current evidence:** reviewed merge, exact-tree remote main and successful main-guard. **Status:** complete with retained readback.

Action: Retain the verified lifecycle receipt; future source changes repeat exact-head CI and eligible independent native review.

Dependencies: None.

Acceptance commands:

```
/home/ubuntu/.local/bin/gh-role writer pr view 83 --json mergedAt,mergeCommit,headRefOid,state
/home/ubuntu/.local/bin/gh-role writer run view 35889731910 --json
headSha,attempt,status,conclusion,jobs
```

Required evidence: PR83 merged 2026-09-23T16:34:47Z; remote main a75e543d2166017cb0139e0a251d51d9a0c98e08 has tree 176b75bc36d76c7ce4eaeb7234677ec013468889, the same tree as reviewed head 68f56404c3e1. Main-guard 35889731910 attempt 1 succeeded; job 107278860508 finished 16:40:11Z. Clean canonical main verified before docs/asm-maintainer-manual branch claim.

Resources: Read-only/static custody work; no fitting, models, GPU or campaign admission.

Source/evidence references: `AGENTS.md:Pull requests`, §1.8.1 Implementation snapshot and this manual's own identity, §13.4 Coverage and reviewed lifecycle, `neural-integration/lifecycle-68f56404c3e199e06413bfd645d3fe0803d7b96f/remote-main-verification.json`.

Qualification limit: This task is complete; subsequent implementation remains pending. Keep the measured source identity 68f56404c3e1 distinct from the rebase-merge commit identity a75e543d2166.

E.1.1.2 DOC-01 — manual

Priority: P1. **Current evidence:** assembled thirteen operating chapters and five evidence appendices with source-audited training and target histories. **Status:** canonical private manual assembled.

Action: Maintain the dated evidence inventory and current source/pending distinctions.

Dependencies: LIFE-01.

Acceptance commands:

```
python3 docs/internal/asm-manual/tools/build_manual.py
```

Required evidence: Input manifest pins every chapter, figure dataset, source receipt and render tool; duplicates resolved; current/pending/historical/unknown labels retained.

Resources: Read-only/static custody work; no fitting, models, GPU or campaign admission.

Source/evidence references: `architecture-training-evidence.json`, `product-language-source-pins.json`, `dataset-inventory.json`, `performance-evidence.json.gz`.

Qualification limit: Root integrated the book and renderer. Claude historical plot inputs were copied under stable before/after hashes; no stopped external renderer handoff is required.

E.1.1.3 DOC-02 — manual examples

Priority: P1. **Current evidence:** fourteen actual commands passed; five backend predictions bit-identical. **Status:** complete at named recipe/refit scope.

Action: Retain the completed fourteen-command receipt and rerun after meaningful recipe/runtime changes; qualify saved fitted winners separately.

Dependencies: DOC-01.

Acceptance commands:

```
python3 "$EXAMPLES/validate-outputs.py" --recipe-output "$EXAMPLE_RUN/recipe"
"$J" pipeline verify-receipt --receipt "$EXAMPLE_RUN/pipeline/receipt.json" --json
```

Required evidence: examples/commands.json instantiated with actual paths and pinned source/tool versions; five outputs with 8 finite values and complete-bit comparison; pipeline receipt verification. Stopped audit SHA256 363fbe62a706b44030205da78411243ba206ad2366d546a277babe966b802c91.

Resources: Completed separate 300-second allocation: one CPU, 4 GiB, zero swap/GPU, network denial declared; 30-second numerical and 60-second compiler caps. Actual receipt retained; independent live property readback limitation is disclosed. No Cargo or frozen-panel campaign; no deadline reset.

Source/evidence references: `examples/commands.json`, `examples/validate-outputs.py`, §10.4.1 Example A: a tiny private recipe, fit and predictions, §10.4.2 Example B: actual legacy pipeline/search shape.

Qualification limit: Actual acceptance: data/example-acceptance/audit.json. Five recipe refits each predict eight synthetic rows; separate legacy pmlb229 pipeline has four successful measurements. No fitted-export, ASM demo, training or full-panel qualification. Resource-readback race retained.

E.1.1.4 DOC-03 — manual figures

Priority: P1. **Current evidence:** twenty-one local figure artifacts; original metrics, complete recovered per-step series and generation points retained. **Status:** figures rendered; local PDF inspection completed; publication pending.

Action: Retain historical/current labels, complete source CSVs and final PDF visual checks.

Dependencies: DOC-01.

Acceptance commands:

```
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 python3 docs/internal/asm-manual/tools/
make_inventory_figures.py
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 python3 docs/internal/asm-manual/tools/make_figures.py
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 python3 docs/internal/asm-manual/tools/
make_training_figures.py
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 python3 docs/internal/asm-manual/tools/
make_target_figures.py
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 python3 docs/internal/asm-manual/tools/
make_legacy_training_figure.py
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 python3 docs/internal/asm-manual/tools/
make_per_step_figure.py
```

Required evidence: Plot source CSV/JSON, tool source, plot hash, precise captions, missing-stat flags and historical-vs-current labels; no invented sealed statistics or causal speedup.

Resources: Read-only/static custody work; no fitting, models, GPU or campaign admission.

Source/evidence references: §A.1.2 Frozen dataset coverage, §C.2.2 Ranking figures: interpretation contract, `dataset-inventory.json`, `performance-evidence.json.gz`.

Qualification limit: Historical Elo belongs to its original ladder/pool/protocol; the current one-dataset proof is not a new Elo ladder. Source and figure checks are retained; authored architecture SVG is a repository source artifact.

E.1.1.5 DOC-04 — manual quality

Priority: P1. **Current evidence:** edition 2 rendered in two volumes; local content, navigation and visual checks completed. **Status:** rendered; final publication gates pending.

Action: Render docs/*.pdf, extract PDF text, inspect all dense tables/code/figures, verify local links and filenames, and produce a readback checklist.

Dependencies: DOC-01, DOC-02, DOC-03.

Acceptance commands:

```
pdftotext -layout "$MANUAL_PDF" "$MANUAL_TEXT"
```

Required evidence: PDF SHA/page count; extracted-text receipt; visual inspections of title/contents, widest tables, commands, plots and final checklist; no clipping/omitted rows; Markdown counterparts present.

Resources: Read-only/static custody work; no fitting, models, GPU or campaign admission.

Source/evidence references: §13.8.5 Updating the book.

Qualification limit: A successful renderer exit is insufficient: the PDF must remain readable and complete.

E.1.1.6 DOC-05 — real training and target history

Priority: P1. **Status:** Original training and target values verified; final composed-document inspection required.

Action: Publish exact observed optimizer/corpus/encoding/target history, component-loss curves, checkpoint lineage, source hashes and explicit comparability/missing-history limits.

Dependencies: DOC-01.

Required evidence: Audited original JSON-pointer-to-CSV equality; every curve bound to model/corpus/monitor/objective; original failed/inconclusive observations retained.

Commands:

```
python3 docs/internal/asm-manual/tools/make_training_figures.py
python3 docs/internal/asm-manual/tools/make_target_figures.py
python3 docs/internal/asm-manual/tools/make_legacy_training_figure.py
python3 docs/internal/asm-manual/tools/make_per_step_figure.py
```

The recovered series preserves 63,000 rows and 378,000 exact metric tokens. Keep all 21 corpus blocks, 20 intervening boundaries and the legacy internal-ladder limitations explicit.

Resources: No new fits, training or model loads; plotting retained metrics only.

Sources: §B.1 Dated lineage and evidence coverage, §9.4 What the historical training and target evidence establishes.

E.1.1.7 DOC-06 — versions, KPIs and continuation runbook

Priority: P1. **Status:** Current and pending interfaces source-audited; independent chapter review complete.

Action: Publish the product/model/encoding/vocabulary/corpus/protocol version tuple, north-star and KPI definitions, real continuation evidence, and exact current-versus-pending commands and code.

Dependencies: DOC-01.

Required evidence: Source-bound command/flag/API audit; held native grant path disclosed; warm-start distinct from optimizer resume; full input closure and 64 GiB budget compatibility stated.

Commands: No complete executable front door is claimed; use the linked chapter's current/pending interface map.

Resources: Documentation and metadata validation only; commands do not admit a new training window.

Sources: §1.8 What version are we running?, §14.2.1 The continuation interfaces that exist today, §9.3 What "better and better" must mean, §6.5 Focused single-dataset search.

E.1.1.8 INT-01 — serial integration

Priority: P0. **Current evidence:** two exact custody replays + guarded hashes. **Status:** stopped unmerged packets.

Action: Refresh guards only after verified main; integrate training64 then product28 then warm/winner add-ons without replacing newer source corrections.

Dependencies: LIFE-01.

Acceptance commands:

```
python3 "$TRAINING_PACKET/replay.py" --expect-head "$SOURCE_SHA" --manifest-sha256
"$TRAINING_MANIFEST_SHA" --output "$TRAINING_PACKET/replay-NEW"
python3 "$PRODUCT_PACKET/replay.py" --expect-head "$SOURCE_SHA" --manifest-sha256
"$PRODUCT_MANIFEST_SHA" --output "$PRODUCT_PACKET/replay-NEW"
```

Required evidence: Exact-source guard receipts; unchanged original packet hashes/skips/resolutions; new applied diff; root scorer digest/cutoff/terminal persistence and current ledger hooks preserved.

Resources: Read-only/static custody work; no fitting, models, GPU or campaign admission.

Source/evidence references: `agent-training-integration/refresh-68f56404/HANDOFF.json`, `agent-product-final-68f56404/HANDOFF.json`, `agent-delivery-refresh-68f56404/HANDOFF.json`.

Qualification limit: NEW is an unused custody directory name. Add-on replay invocation must follow its own pinned helper. Never force stale guards or apply constituent accounting/coverage twice.

E.1.1.9 INT-02 — integration qualification

Priority: P0. **Current evidence:** static syntax/feature checks; no integrated compilation yet. **Status:** pending.

Action: Compile/type-check the actual combined source, format, run meaningful suites and strict Clippy, regenerate coverage/inventories through owners, then obtain exact-head CI/native approval.

Dependencies: INT-01.

Acceptance commands:

```
CARGO_INCREMENTAL=0 CARGO_PROFILE_DEV_DEBUG=line-tables-only cargo test --locked -p jope-bench --
features asm
```

```
CARGO_INCREMENTAL=0 CARGO_PROFILE_DEV_DEBUG=line-tables-only cargo test --locked -p jope-cli --
features asm
CARGO_INCREMENTAL=0 CARGO_PROFILE_DEV_DEBUG=line-tables-only cargo test --locked -p jope-pipeline
--features qualified-context
cargo fmt --all --check
python3 deploy/ci/coverage.py discover
python3 deploy/ci/coverage.py validate
python3 deploy/ci/test-training-proof.py
TMPDIR="$ACCEPTANCE/tmp" JOPE_BUILD_JOBS=4 bash deploy/ci/test-inventory.sh --update
```

Required evidence: Collection/exit logs, actual tests passed/failed/ignored/deselected, refreshed inventory diff, feature/dependency graph, build executable hashes, current-attemptCI and independent exact-head approval.

Resources: Existing exclusive CPU CI allocation; one reused lane target, jobs4 unless lane policy sets otherwise; no concurrent numerical campaign.

Source/evidence references: `deploy/ci/coverage.json`, `deploy/ci/training-proof.py (pending)`, `deploy/ci/fast-lane.sh`, `deploy/ci/full-lane.sh`.

Qualification limit: Custody syntax tests are not this gate. Required 18 Python / 27 Rust / 10 policy names are proof minima, not total coverage. New code needs actual test collection.

E.1.1.10 IP-01 — public privacy boundaries

Priority: P1. **Current evidence:** allowlists/canaries prepared; historicallintblindspots. **Status:** residual D-IP4 and future PR text work.

Action: Resolve the approved public vocabulary/naming scope and test actual CLI help, receipts and NOTICE. Preserve the explicitly requested internal manual and the strict public inference allowlist.

Dependencies: INT-01.

Acceptance commands:

```
bash deploy/ci/ip-scrub.sh
```

Required evidence: Actual shipped-file inventory and help/receipt/NOTICE canaries; private provenance absent; allowlist negative cases. Keep the internal requested manual distinct from public bundles.

Resources: Existing exclusive CPU CI allocation; one reused lane target, jobs4 unless lane policy sets otherwise; no concurrent numerical campaign.

Source/evidence references: `AGENT_CHAT.md:D-IP4`, `deploy/ci/ip-scrub.sh`, `crates/feat-math/src/advanced_v3/fitted_portable.rs (pending)`.

Qualification limit: Current lint success does not prove unscanned surfaces clean. The user explicitly requested the private GP explanation, so ordinary creation of this manual needs no additional approval.

E.1.1.11 MAINT-01 — broad diagnostic qualification

Priority: P2. **Current evidence:** historicalfull-default failures inventoried. **Status:** not closed by fast CI.

Action: Reproduce and repair recorded full-default, scratch, policy and neural families through their owning full lanes. Retain unavailable capabilities and resource limits.

Dependencies: LIFE-01.

Acceptance commands:

```
bash deploy/ci/full-lane.sh full-default
bash deploy/ci/full-lane.sh full-scratch
bash deploy/ci/full-lane.sh full-policy
bash deploy/ci/full-lane.sh full-neural
```

Required evidence: Source-bound full-suite collection and failures, reproductions and corrected receipts. Keep fast-CI and full-diagnostic totals separate; zero tests or skips cannot count as passes.

Resources: Existing exclusive CPU CI allocation; one reused lane target, jobs4 unless lane policy sets otherwise; no concurrent numerical campaign.

Source/evidence references: `deploy/ci/full-lane.sh`, §13.4 Coverage and reviewed lifecycle, `AGENT_CHAT.md:full-default failures`.

Qualification limit: Check lane environment requirements before invocation. The owning qualification supplies allocation and budget; this manual does not admit them.

E.1.2 Stage 1: correct benchmark integration

E.1.2.1 BENCH-01 — frozen benchmark foundation

Priority: P1. **Current evidence:** current source + local/exact-head tests. **Status:** implemented; retain during integration.

Action: Reverify unchanged manifest and all prepared inputs, actual row/group assignments, original category/missing/label semantics, loss directions, full class probability ordering and task method inventory.

Dependencies: INT-02.

Acceptance commands:

```
"$J" benchmark verify-manifest --manifest "$M"
"$J" benchmark verify-inputs --manifest "$M" --datasets "$D"
"$J" benchmark methods
```

Required evidence: Original panel SHA ba7dace4…, 30 datasets, 10 per task, exact input-byte verification, native 17/5/4 recipes and 4 external families; alias/configuration inventory and unavailable reasons.

Resources: Read-only/static custody work; no fitting, models, GPU or campaign admission.

Source/evidence references: `crates/jope-bench/src/panel/data.rs`, `crates/jope-bench/src/panel/methods.rs`, `crates/jope-bench/src/panel/score.rs`, `benchmarks/panel.json`.

Qualification limit: Verification is metadata/data validation, no new fits. Do not replace a failed panel dataset.

E.1.2.2 BENCH-02 — external comparators

Priority: P1. **Current evidence:** four adapters implemented; earlier runtime inventories. **Status:** refresh at final environment.

Action: Pin all runtime/package/adapter bytes and actual constructors; execute relevant adapter contract tests and preserve unavailable dependencies.

Dependencies: INT-02.

Acceptance commands:

```
python3 deploy/prepare-panel-runtime.py --binary "$J" --native-command benchmark --build-receipt
"$CLI_BUILD_RECEIPT" --source-sha "$SOURCE_SHA" --python "$EXTERNAL_PYTHON" --site-packages
"$EXTERNAL_SITE_PACKAGES" --adapters crates/jope-bench/adapters --output "$ACCEPTANCE/
runtimes.json"
"$EXTERNAL_PYTHON" -m pytest crates/jope-bench/adapters/test_panel.py -q
```

Required evidence: Current 4 engine version/source/package hashes, constructors for 3 tasks, actual contract-test receipt and explicit unavailability inventory.

Resources: Preparation owns sharedCPUallocation; tests use allocatedCPU; no panel/training window begins.

Source/evidence references: `deploy/prepare-panel-runtime.py`, `crates/jope-bench/adapters/_panel.py`, `crates/jope-bench/adapters/test_panel.py`.

Qualification limit: Do not nest an outer allocation lock around a front door already taking it. Environment mutation invalidates old inventory.

E.1.3 Stage 2: shared execution and accounting

E.1.3.1 EXEC-01 — bounded workers

Priority: P1. **Current evidence:** ten real synthetic host cases passed at 68. **Status:** implemented; CI coverage gap retained.

Action: Requalify final worker/controller bytes through ten current host cases and make required-lane coverage of critical async cleanup paths explicit.

Dependencies: INT-02.

Acceptance commands:

```
python3 deploy/ci/qualify-execution.py --test-binary "$PIPELINE_TEST_BINARY" --output
"$ACCEPTANCE/host-execution"
```

Required evidence: Ten nonzero actual cases,20concurrentworkers,limit readbacks, process-group drainage, setup failures/cancel gates, original-window equality and fixture code reachability.

Resources: Existing exclusive CPU CI allocation; one reused lane target, jobs4 unless lane policy sets otherwise; no concurrent numerical campaign.

Source/evidence references: `deploy/ci/qualify-execution.py`, `crates/jope-pipeline/src/execution.rs`.

Qualification limit: Current host qualification is not automatically covered by the ordinary required unit lane; missing privileges fail, not skip.

E.1.3.2 EXEC-02 — fault regression tests

Priority: P1. **Current evidence:** production terminal-persistence fix qualified indirectly. **Status:** test gap: drain-and-fixtures-F2.

Action: Add a real async-reap fixture with pre-gate BudgetExhausted and an undrained group. Prove terminal record survives before cleanup and admission remains blocked until drain.

Dependencies: INT-01.

Acceptance commands:

```
CARGO_INCREMENTAL=0 CARGO_PROFILE_DEV_DEBUG=line-tables-only cargo test --locked -p jope-pipeline
--features qualified-context execution::tests
```

Required evidence: Reproduction fails when persist is moved back under drained branch or admission_blocked handling is removed; corrected actual test passes with cleanup ordering evidence.

Resources: Existing exclusive CPU CI allocation; one reused lane target, jobs4 unless lane policy sets otherwise; no concurrent numerical campaign.

Source/evidence references: `crates/jope-pipeline/src/execution.rs:812`, `AGENT_CHAT.md: 2026-09-23T16:17 delta review`.

Qualification limit: Exact new test name is not present yet; existing module command is the truthful current gate. Ten old host cases do not refute this coverage gap.

E.1.3.3 EXEC-03 — receipt replay regression tests

Priority: P1. **Current evidence:** shared per-fold helper fixed and real 68 recovered report passed. **Status:** test gap: resume-fix-F2.

Action: Add PoolBackend-shaped candidate receipt roundtrip asserting live Feedback equals recovered FitAdapter Feedback, including digest and partial completed-fold evidence.

Dependencies: INT-01.

Acceptance commands:

```
CARGO_INCREMENTAL=0 CARGO_PROFILE_DEV_DEBUG=line-tables-only cargo test --locked -p jope-bench --
features asm
CARGO_INCREMENTAL=0 CARGO_PROFILE_DEV_DEBUG=line-tables-only cargo test --locked -p jope-pipeline
--features qualified-context neural_search::tests
```

Required evidence: Regression reproduces the d0f5748b7db1 fitness/replay mismatch; exercises Some(shared fold digest), changed completed outputs refuse, never-admitted normalization cannot gain evidence.

Resources: Existing exclusive CPU CI allocation; one reused lane target, jobs4 unless lane policy sets otherwise; no concurrent numerical campaign.

Source/evidence references: `crates/jope-pipeline/src/neural_search.rs:1735`, `crates/jope-bench/src/panel/optimize_fit.rs`, `AGENT_CHAT.md:resume-fix-F2`.

Qualification limit: The two original FixtureBackend reproductions use absent digests and do not cover the same_folds path.

E.1.3.4 EXEC-04 — cutoff semantics regression

Priority: P1. **Current evidence:** production cutoff ordering corrected; existing test too weak. **Status:** test gap: regressions-F2.

Action: Strengthen cancellation-at-cutoff test to assert original_deadline stop reason and persisted receipt, not only is_ok.

Dependencies: INT-01.

Acceptance commands:

```
CARGO_INCREMENTAL=0 CARGO_PROFILE_DEV_DEBUG=line-tables-only cargo test --locked -p jope-pipeline
--features qualified-context cancellation_at_cutoff_preserves_observed_generation_on_resume
```

Required evidence: Test fails if cancellation check moves before deadline check; successful resume emits no new fit and leaves observed bytes unchanged.

Resources: Existing exclusive CPU CI allocation; one reused lane target, jobs4 unless lane policy sets otherwise; no concurrent numerical campaign.

Source/evidence references: `crates/jope-pipeline/src/neural_search.rs:1380`, `AGENT_CHAT.md:regressions-F2`.

E.1.3.5 EXEC-05 — accounting and unknown outcomes

Priority: P1. **Current evidence:** additive accounting union source + four authored Rust cases. **Status:** pending execution.

Action: Execute started/finished-union accounting tests; retain started-only unknowns separately from terminal refusals and qualify report replay without child retries.

Dependencies: INT-02.

Acceptance commands:

```
CARGO_INCREMENTAL=0 CARGO_PROFILE_DEV_DEBUG=line-tables-only cargo test --locked -p jope-bench --
features asm asm_campaign
```

Required evidence: Actual four new accounting tests collected; started-only attempts counted unknown, no inferredzero, exports counted once inASMworker totals, recovered reporting no launch.

Resources: Existing exclusive CPU CI allocation; one reused lane target, jobs4 unless lane policy sets otherwise; no concurrent numerical campaign.

Source/evidence references: `agent-accounting-union/training-bridge/HANDOFF.json`, `crates/jope-bench/src/panel/asm_campaign.rs (pending)`.

E.1.4 Stage 3: neural consumption and training

E.1.4.1 NEURAL-01 — context qualification

Priority: P0. **Current evidence:** eight fixed native cases passed at 68. **Status:** requires final-source binding.

Action: Build exact context/policy/proof workers; qualify unchanged native references for final source and preserve eight fixed-case plus actual dataset process evidence.

Dependencies: INT-02.

Acceptance commands:

```
"$CONTEXT_PROOF" prepare --sources "$REFERENCE_SOURCES" --sources-sha256 "$REFERENCE_SOURCES_SHA"
--encoder-files "$ENCODER_FILES" --encoder-files-sha256 "$ENCODER_FILES_SHA" --worker
"$CONTEXT_WORKER" --worker-sha256 "$CONTEXT_WORKER_SHA" --source-sha40 "$SOURCE_SHA" --resources
"$CONTEXT_RESOURCES" --resources-sha256 "$CONTEXT_RESOURCES_SHA" --output "$ACCEPTANCE/context-
```

```
proof"
"$CONTEXT_PROOF" run --plan "$CONTEXT_PLAN" --plan-sha256 "$CONTEXT_PLAN_SHA"
```

Required evidence: All eight fixed cases pass the unchanged component/context maximum absolute difference of 0.0003; source, assets and program identities match. Preserve actual bounded execution receipts and the qualified construction token, original labels and row order.

Resources: Separate explicit resource manifest and approved allowance required before admission;20 CPUs/workers maximum,64 GiB maximum,zero swap,one thread,30 seconds per fit,120 seconds per arm maximum. No new allowance granted by this checklist.

Source/evidence references: `crates/jope-policy/src/context_producer/admission.rs:4`, `crates/jope-pipeline/src/cpu_context_bridge.rs`.

Qualification limit: Metadata/path equality or checksum alone is not numerical qualification; no changed tolerance or checkpoint substitution.

E.1.4.2 TEACH-03 — real teacher collection

Priority: P0. **Current evidence:** zero eligible current teachers; two prospective evidenced families. **Status:** blocked on fresh admitted evidence.

Action: Preregister teaching before two separately bounded proof plans for machineCPU and geyser. Obtain actual finite OOF improvement and improvements on four of five folds, or retain the rejection.

Dependencies: BENCH-01, BENCH-02, NEURAL-01, TEACH-01, TEACH-02.

Acceptance commands:

```
"$J" optimize proof-plan --base-plan "$BASE_PLAN" --base-plan-sha256 "$BASE_ID" --configuration
"$TEACHING_CONFIGURATION" --configuration-sha256 "$TEACHING_CONFIGURATION_SHA" --dataset
"$FAMILY_DATASET" --resources "$TEACHING_RESOURCES" --resources-sha256 "$TEACHING_RESOURCES_SHA"
--output "$FAMILY_PLAN"
"$J" optimize run --plan "$FAMILY_PLAN" --plan-sha256 "$FAMILY_PLAN_ID" --manifest "$M" --datasets
"$D" --journal "$FAMILY_JOURNAL"
"$J" optimize report --plan "$FAMILY_PLAN" --plan-sha256 "$FAMILY_PLAN_ID" --manifest "$M" --
datasets "$D" --journal "$FAMILY_JOURNAL"
```

Required evidence: Before-fit freeze, vocabulary and source pins; two independent families and original windows; nine context proofs per dataset; complete five-fold predictions and trivial baselines. Apply the 1e-12 absolute plus 1e-6 relative gain margin and four-of-five rule. Retain no-signal refusals.

Resources: Separate explicit resource manifest and approved allowance required before admission;20 CPUs/workers maximum,64 GiB maximum,zero swap,one thread,30 seconds per fit,120 seconds per arm maximum. No new allowance granted by this checklist.

Source/evidence references: `agent-training-readiness/readiness-v3.json`, §7.4 Present teacher readiness and the smallest real path, `crates/jope-bench/src/panel/teaching.rs (pending)`.

Qualification limit: Execute separately for 230_machine_cpu and 712_chscase_geyser1; the prospective choice predates losses. Both are frozen-panel regression datasets, so if either is used to manufacture teachers, later evaluations on it must be labelled exposed/adaptive (§9.7.1 Freezing the experiment and choosing families; OD-8 in §14.1.4 Proposed owner decisions and external dependencies). Old operational receipts from sources 55e9ed688faf, 3fd7d5b2ebb6 and 68f56404c3e1 cannot be retrofitted. The older queue suggested 300 seconds per family without admitting that allowance.

E.1.4.3 TEACH-04 — real corpus materialization

Priority: P0. **Current evidence:** native exporter prepared. **Status:** blocked on two real families.

Action: Build the request from every eligible requested comparison and its original native-context proof; freeze a family-disjoint corpus and the exact vocabulary.

Dependencies: TEACH-03.

Acceptance commands:

```
"$J" train prepare-inputs --learning-state "$LEARNING_STATE" --request "$CORPUS_REQUEST" --
request-sha256 "$CORPUS_REQUEST_SHA" --output "$CORPUS_OUTPUT"
```

Required evidence: At least two real families; every eligible requested comparison included; seed 7 partition derived from source families; exact 1536 + 512 context values. No invented context, parameter-bin snapping, label remapping or omission of difficult examples.

Resources: Read-only/static custody work; no fitting, models, GPU or campaign admission.

Source/evidence references: `crates/jope-cli/src/train/corpus.rs (pending)`, §8.4.2 Build the real corpus from admitted evidence.

Qualification limit: Only if exactly two families qualify does the recorded machineCPU-development/geyser-training assignment apply. More eligible families must all remain admitted.

E.1.4.4 TRAIN-01 — actual trainer tests

Priority: P0. **Current evidence:** seven pure input tests passed; actual model cases deselected. **Status:** residual training-F3.

Action: Execute actual optimizer, reload and continuation tests across regression, binary and multiclass, plus task, vocabulary and runtime compatibility negatives.

Dependencies: INT-02.

Acceptance commands:

```
python/.venv/bin/python -m pytest python/tests/test_policy_training_contract.py python/tests/
test_training_campaign.py python/tests/test_training_corpus_contract.py python/tests/
test_experimental_proposal.py python/tests/test_experimental_runtime.py -q --
junitxml="$ACCEPTANCE/training-fixtures.xml"
python3 deploy/ci/test-training-proof.py
```

Required evidence: Collected test IDs include all three parameters of test_actual_trained_classifier_reload_sampling_uses_task_head. Prove positive updates, changed parameters, task reward routing, reload and continuation; reject zero updates and missing architecture. The proof gate must see actual execution.

Resources: Existing exclusive CPU CI allocation; one reused lane target, jobs4 unless lane policy sets otherwise; no concurrent numerical campaign.

Source/evidence references: `python/tests/test_experimental_proposal.py (pending)`, `python/src/jope/policy/train_v3.py (pending)`, `deploy/ci/training-proof.py (pending)`.

Qualification limit: Required-name inventories and deselected tests do not count as execution. A synthetic CPU checkpoint remains fixture evidence.

E.1.4.5 TRAIN-02 — native supervised training

Priority: P0. **Current evidence:** supervisor/cgroup orchestration prepared. **Status:** host proof pending.

Action: Exercise the native training front door with bounded synthetic normal and cancellation cases: subgroup entry, release, gate refusal, cleanup and original-window resume.

Dependencies: TRAIN-01.

Acceptance commands:

```
"$J" train plan --state "$FIXTURE_TRAIN_STATE" --python "$TRAINING_PYTHON" --synthetic-fixture --
task all-admitted --device cpu --source-sha "$SOURCE_SHA" --seed 7 --passes 2
"$J" train run --state "$FIXTURE_TRAIN_STATE"
"$J" train verify --state "$FIXTURE_TRAIN_STATE"
```

Required evidence: Actual normal and interrupted CPU workers; controller-owned subgroup joined before execution; original deadline, device and runtime unchanged; children drained and resumed state compatible. The native wrapper must enforce admission.

Resources: Explicitbounded CPU fixture allocation; same planned 3600 seconds/1200scaps are maxima, not authorization to spend them twice; no GPU.

Source/evidence references: `crates/jope-cli/src/train.rs (pending)`, `python/src/jope/policy/training_campaign.py (pending)`.

Qualification limit: Author the cancellation harness against the actual front door and target only its owned processes. CPU mode requires explicit fixture provenance.

E.1.4.6 TRAIN-03 — bounded real GPU training

Priority: P0. **Current evidence:** implemented follow-on; no GPU campaign started. **Status:** blocked on admitted data and consumption readiness.

Action: Run one all-admitted CUDA campaign with real regression and two passes. Retain synthetic classifier fallback separately and report every unavailable category.

Dependencies: TEACH-04, TRAIN-02, NEURAL-02.

Acceptance commands:

```
"$J" train plan --state "$TRAIN_STATE" --python "$TRAINING_PYTHON" --inputs "$CORPUS_OUTPUT/
inputs.json" --task all-admitted --device cuda:0 --source-sha "$SOURCE_SHA" --learning-state
"$LEARNING_STATE" --seed 7 --passes 2
"$J" train run --state "$TRAIN_STATE"
```

Required evidence: Original global/category windows; one RTX3090 UUID; positive updates, changed parameter hashes and task coverage with data_kind/fallback reasons. Save and reload model plus optimizer; record CPU/GPU cost where measured. Requested CUDA must not silently fall back to CPU.

Resources: One original 3600 seconds total window;≤1200 seconds per category including both passes;one RTX 3090,20CPU,64 GiB,zero swap,1Torch thread;no exclusiveCPU CI overlap. Resume never resets.

Source/evidence references: `python/src/jope/policy/training_campaign.py:780 (pending)`, §8.4.3 Plan, run, resume, verify and export.

Qualification limit: NEURAL-02 gates source and fixture readiness before the GPU hour. It does not depend on a newly trained live export, avoiding a circular prerequisite.

E.1.4.7 TRAIN-04 — checkpoint continuation and export

Priority: P0. **Current evidence:** verification/export source prepared. **Status:** requires actual campaign.

Action: Verify two committed passes, optimizer counters and the immutable receipt chain; export complete architecture, vocabulary, task, source, encoder and runtime bindings.

Dependencies: TRAIN-03.

Acceptance commands:

```
"$J" train resume --state "$TRAIN_STATE"
"$J" train verify --state "$TRAIN_STATE"
"$J" train export --state "$TRAIN_STATE" --output "$TRAIN_EXPORT"
```

Required evidence: Resume only when needed inside the original window. Prove increased optimizer steps, identical parameters after reload, changed parameters relative to the prior checkpoint, matching export/resume proof and completed task coverage. Keep experimental non-serving status.

Resources: One original 3600 seconds total window;≤1200 seconds per category including both passes;one RTX 3090,20CPU,64 GiB,zero swap,1Torch thread;no exclusiveCPU CI overlap. Resume never resets.

Source/evidence references: `python/src/jope/policy/training_campaign.py:1203 (pending)`, `python/src/jope/policy/training_campaign.py:1303 (pending)`.

Qualification limit: Resume is conditional, not an instruction to start a third pass after completing the plan. A new state or plan cannot extend the approved hour.

E.1.4.8 NEURAL-02 — trained-checkpoint consumer readiness

Priority: P0. **Current evidence:** independent source review + purechecks; regression packet appended. **Status:** compile/runtime fixture pending.

Action: Qualify the explicit experimental regression consumer before GPU use. Preserve accepted-baseline hard pins and strict architecture, task, vocabulary, source and encoder compatibility.

Dependencies: INT-02, TRAIN-01.

Acceptance commands:

```
CARGO_INCREMENTAL=0 CARGO_PROFILE_DEV_DEBUG=line-tables-only cargo test --locked -p jope-pipeline
--features qualified-context experimental_regression_requires_matching_task_and_executable_head
python/.venv/bin/python -m pytest python/tests/test_experimental_proposal.py -q
```

Required evidence: Execute all three actual task-model cases. Test accepted-versus-experimental decoding and refusal of incomplete architecture, missing state, changed vocabulary, source or encoder. The operational regression route requires a real admitted export.

Resources: Existing exclusive CPU CI allocation; one reused lane target, jobs4 unless lane policy sets otherwise; no concurrent numerical campaign.

Source/evidence references: `agent-experimental-regression/HANDOFF.json`, `crates/jope-pipeline/src/experimental_producer.rs (pending)`, `python/src/jope/policy/experimental_proposal.py (pending)`.

Qualification limit: The new regression consumer is not serving promotion. Synthetic regression may test mechanics but cannot enter the real operational route.

E.1.4.9 NEURAL-03 — live newly trained consumption

Priority: P0. **Current evidence:** current accepted checkpoint origin/evolution proved onone dataset. **Status:** newly trained consumption pending.

Action: Pin the actual exported checkpoints and runtime/canonicalizer into the final configuration. Prove new-checkpoint proposals reach real generation-zero fits and completed descendants.

Dependencies: TRAIN-04, NEURAL-01.

Acceptance commands:

```
"$J" optimize proof-plan --base-plan "$BASE_PLAN" --base-plan-sha256 "$BASE_ID" --configuration
"$TRAINED_CONFIGURATION" --configuration-sha256 "$TRAINED_CONFIGURATION_SHA" --dataset
230_machine_cpu --resources "$CONSUMPTION_RESOURCES" --resources-sha256
"$CONSUMPTION_RESOURCES_SHA" --output "$CONSUMPTION_PLAN"
"$J" optimize run --plan "$CONSUMPTION_PLAN" --plan-sha256 "$CONSUMPTION_PLAN_ID" --manifest "$M"
--datasets "$D" --journal "$CONSUMPTION_JOURNAL"
"$J" optimize report --plan "$CONSUMPTION_PLAN" --plan-sha256 "$CONSUMPTION_PLAN_ID" --manifest
"$M" --datasets "$D" --journal "$CONSUMPTION_JOURNAL"
```

Required evidence: The exported checkpoint hash appears in actual proposal and fit evidence; 64+16 versus 80 origins remain authenticated. Retain completed descendants where claimed, predictions and fitted states. Regression must be admitted and classifier fixture limits explicit; replay the original window successfully.

Resources: Separate explicit resource manifest and approved allowance required before admission;20 CPUs/workers maximum,64 GiB maximum,zero swap,one thread,30 seconds per fit,120 seconds per arm maximum. No new allowance granted by this checklist.

Source/evidence references: `crates/jope-bench/src/panel/experimental_proposal.rs (pending)`, `crates/jope-bench/src/panel/origin_proof.rs (pending)`.

Qualification limit: The current proof-plan scopes one regression dataset. Classification needs its implemented campaign/configuration route; this command alone cannot prove binary or multiclass live consumption.

E.1.4.10 PACK-01 — packed training consumer and incremental loading

Priority: P2. **Status:** Legacy binary store exists; pending Python category route still materializes JSONL.

Action: Design and qualify a lossless packed reader for the bounded trainer, sharing repeated contexts and immutable shards while excluding sealed-family context vectors; measure host/device loading and fresh-example exposure.

Dependencies: TEACH-04, TRAIN-01.

Required evidence: JSONL-versus-packed token/context/label/sampler equivalence; corruption/schema tests; finite complete labels; startup/RSS/I/O/padding measurements; incremental shard reuse and actual optimizer input counts.

Commands: No complete executable front door is claimed; use the linked chapter’s current/pending interface map.

Resources: Source/fixture planning first; any measured training remains inside a separately declared remaining authorized window.

Sources: §D.7.1 Record layout, `python/src/jope/policy/train_v3.py`, `crates/jope-engine/src/teaching_store.rs`.

E.1.4.11 LOOP-01 — repeatable measured learning flywheel

Priority: P1. **Status:** New complete improved-target-to-improved-search cycle not yet qualified.

Action: Demonstrate the regression chain new eligible target → actual exposure → changed parameters → compatible reload → improved fitted search → another new generation, over linked generations; two passes through one corpus are continuation, not two generations. Measure next-search benefit against both the parent checkpoint and generic search, keep the accepted 512×6 step-60000 model separate from experimental regression lineage v1 (256×2, new initialization), and never promote serving automatically.

Dependencies: NEURAL-03, TEACH-04, TRAIN-04.

Required evidence: Per-cycle target deltas, actual fresh exposures, component losses on fixed monitors, optimizer/parent/checkpoint hashes, full failures/cost and matched fitted-program outcomes; promotion separately decided. Decision: the proposed preregistered regression decision rule in learning.md, a proposal and not an achieved result: family-normalized gain versus the parent on at least 10 independent monitor families per round, with sibling fresh-versus-replay-only branches from one immutable parent. Unfunded or insufficient data yields mechanical proof and descriptive results with the causal verdict inconclusive, not an invented success.

Commands: No complete executable front door is claimed; use the linked chapter's current/pending interface map.

Resources: Original one GPU-hour cap remains binding, with at most 20 minutes per category. Its two controlled rounds may remain unfunded or inconclusive within that existing 20-minute regression GPU cap; this item grants no new allocation. Further campaigns require their own explicit allocation.

Sources: §B.1 Dated lineage and evidence coverage, §14.2.1 The continuation interfaces that exist today, §9.3 What "better and better" must mean.

E.1.5 Stage 4: portable fitted inference

E.1.5.1 INFER-01 — fitted state and privacy

Priority: P0. **Current evidence:** native fitted state exists; product28 source prepared. **Status:** qualification pending.

Action: Execute the lossless private codec, complete fitted preprocessing/state reload, and public allowlist/canary tests before export. Unsupported states remain explicit.

Dependencies: INT-02.

Acceptance commands:

```
CARGO_INCREMENTAL=0 CARGO_PROFILE_DEV_DEBUG=line-tables-only cargo test --locked -p feat-math
fitted_portable
CARGO_INCREMENTAL=0 CARGO_PROFILE_DEV_DEBUG=line-tables-only cargo test --locked -p jope-bench --
features asm inference
bash deploy/ci/ip-scrub.sh
```

Required evidence: Complete prediction parity after reload; missing/unseen category, class order and full probability-matrix cases. Preserve the private codec identity; public bundles must omit recipe, discovery and provenance canaries.

Resources: Existing exclusive CPU CI allocation; one reused lane target, jobs4 unless lane policy sets otherwise; no concurrent numerical campaign.

Source/evidence references: `crates/feat-math/src/advanced_v3/fitted_portable.rs (pending)`, `crates/jope-bench/src/panel/inference.rs (pending)`, §10.3.2 State restrictions beyond the head table.

Qualification limit: The private C++ head gate is broader than the public gate. Python/Rust 14 of 26, C++ 2 of 26 and JavaScript 6 of 26 are static head counts, not runtime qualification rates.

E.1.5.2 INFER-02 — exact selected-winner portability

Priority: P1. **Current evidence:** winner/add-on source + statictests. **Status:** actual export/parity unrun.

Action: Qualify the selected existing fold state in order: Python, Rust, the same C++ emitter under C++17 and C++20, then JavaScript. Preserve unsupported external winners and the complete ranking.

Dependencies: INFER-01, PANEL-01.

Acceptance commands:

```
"$J" benchmark export-winner --plan "$PLAN" --plan-sha256 "$PLAN_ID" --manifest "$M" --datasets
"$D" --journal "$JOURNAL" --dataset "$DATASET_ID" --fold "$FOLD"
"$J" benchmark qualify-export --prepared "$PREPARED_EXPORT" --prepared-sha256
"$PREPARED_EXPORT_ID" --tools "$TOOLS" --tools-sha256 "$TOOLS_SHA"
"$J" benchmark report --plan "$PLAN" --plan-sha256 "$PLAN_ID" --manifest "$M" --datasets "$D" --
journal "$JOURNAL"
```

Required evidence: Exact state, artifact, input, class, toolchain and numerical-policy hashes; actual five-backend outcomes. Python/Rust/C++ use bitwise comparison; JavaScript uses 1e-12 absolute and relative tolerance. Recover interrupted work without retrying its child; read-only reports launch nothing. Unsupported never replaces the winner.

Resources: Same original campaign execution window and worker caps; no new job after t0 + 6900 seconds for official panel. No post-panel allowance is implemented.

Source/evidence references: `crates/jope-bench/src/panel/fitted_export.rs (pending)`, `crates/jope-bench/src/panel/fitted_export_run.rs (pending)`, §10.5 Example C: exact fitted winner export after pending integration.

Qualification limit: Library fixtures precede the panel. Actual selected winners require complete confirmations from PANEL-01 and qualification before the original execution cutoff. No separately bounded post-panel qualification front door exists. One fitted fold does not qualify the other four states.

E.1.5.3 FUTURE-01 — CUDA inference and serving promotion

Priority: P2. **Current evidence:** not qualified in this delivery. **Status:** deferred.

Action: After portable CPU backends, design CUDA fitted-inference qualification and independent experimental-to-serving promotion. Preserve the accepted baseline until promotion passes.

Dependencies: INFER-02, REPORT-01.

Command boundary: no truthful executable acceptance command yet; implement or obtain missing interface first.

Required evidence: New backend contract, fixed numerical policy, exact device/toolchain/state parity and promotion evaluation; reviewed pointer compare-and-swap and rollback where used.

Resources: NoCUDAinference/promotionbudgetadmitted bythisroadmap.

Source/evidence references: `docs/adr/ADR-0005-canonical-learning-loop.md`, §8.4.1 What the pending campaign trains, §8.5.2 Compatibility checks and fail-closed rules.

Qualification limit: Training GPU availability does not prove CUDA inference parity. Export is not serving promotion.

E.1.6 Stage 5: installation and measured demonstration

E.1.6.1 INSTALL-01 — private installation

Priority: P1. **Current evidence:** installer source and 15 metadata tests passed. **Status:** cold/live path pending.

Action: Build the CLI from final source, measure cold build separately, checksum-install the exact Linux binary, and verify actual command help plus dependency/ELF boundaries.

Dependencies: INT-02.

Acceptance commands:

```
python3 deploy/install-private.py source --checkout "$R" --source-sha "$SOURCE_SHA" --output
"$ACCEPTANCE/cli-build" --jobs 4
python3 deploy/install-private.py binary --binary "$PRIVATE_BINARY" --sha256 "$BINARY_SHA" --
source-sha "$SOURCE_SHA" --prefix "$ACCEPTANCE/install"
"$J" doctor
```

Required evidence: Cold build/install durations, cache and already_present flags; Cargo executable messages, features, toolchain, dependency graph and ELF checks. Retain checksum/help readbacks and failure receipts. CLI dependencies exclude the engine, Torch and CUDA.

Resources: Existing exclusive CPU CI allocation; one reused lane target, jobs4 unless lane policy sets otherwise; no concurrent numerical campaign.

Source/evidence references: `deploy/install-private.py (pending)`, §4.3 Build and install.

Qualification limit: An installer receipt is not policy/context-worker build authority. Do not include training credentials or model bodies in the public inference bundle.

E.1.6.2 INSTALL-02 — final input/runtime preparation

Priority: P1. **Current evidence:** prepared panel existing; runtime refresh required. **Status:** pending final source.

Action: Install the pinned final Python wheel without changing CUDA dependencies; build worker inventories and verify prepared datasets before campaign planning.

Dependencies: INSTALL-01, BENCH-01, BENCH-02.

Acceptance commands:

```
"$J" benchmark verify-inputs --manifest "$M" --datasets "$D"
"$J" benchmark plan --manifest "$M" --datasets "$D" --source-sha "$SOURCE_SHA" --runtimes
"$RUNTIMES" --output "$BASE_PLAN"
```

Required evidence: Wheel, source, runtime and dependency inventory; context/policy workers identified from actual Cargo messages; semantic base-plan identity; zero fits and no execution clock started.

Resources: Read-only/static custody work; no fitting, models, GPU or campaign admission.

Source/evidence references: §4.4 Prepare the panel and external runtimes, `agent-acceptance-queue/QUEUE.md`.

Qualification limit: Packaging commands depend on final reviewed artifacts and the actual virtual environment. Preserve versions and file identities; changing the environment invalidates a pinned campaign.

E.1.6.3 DEMO-01 — measured warm fitted demonstration

Priority: P1. **Current evidence:** warmtargetadd-onprepared;historicalfirstfit≈50.4s. **Status:** real trained warm proof pending.

Action: Produce actual trained-neural and generic fitted artifacts, reloaded predictions, baseline/comparison and truthful target_met. Do not clip arm budgets to pass sixty seconds.

Dependencies: NEURAL-03, INSTALL-02.

Acceptance commands:

```
"$J" demo --asm-plan "$WARM_PLAN" --asm-plan-sha256 "$WARM_PLAN_ID" --manifest "$M" --datasets
"$D" --output "$WARM_OUTPUT" --export python --export rust
python3 deploy/ci/qualify-demo.py --binary "$J" --sha256 "$J_SHA" --source-sha "$SOURCE_SHA" --
output "$ACCEPTANCE/generic-three-task" --require-under-60
```

Required evidence: Actual fitted state, predictions and baseline; both full paired allowances; start/end, warm elapsed, execution_complete and target_met. Keep generic synthetic qualification separate from trained-neural proof.

Resources: Separate explicit resource manifest and approved allowance required before admission;20 CPUs/workers maximum,64 GiB maximum,zero swap,one thread,30 seconds per fit,120 seconds per arm maximum. No new allowance granted by this checklist.

Source/evidence references: `crates/jope-bench/src/panel/demo_asm.rs (pending)`, `docs/private-warm-target.md (pending)`, §6.4 Warm fitted demonstration after integration.

Qualification limit: Under sixty seconds is the declared goal, not a promise; no qualifying measurement exists yet. The existing twenty-second-arm example is unqualified. An explicitly admitted longer proof may satisfy execution while missing the target. This checklist grants no implicit 300-second allowance.

E.1.6.4 PANEL-01 — unchanged thirty-dataset campaign

Priority: P1. **Current evidence:** protocol/currentfoundation qualified;full panel unstarted. **Status:** pending final trained configuration.

Action: Freeze the final roster/configuration and run the unchanged thirty datasets once. Screen fold0, freeze the top five, confirm five development folds, and add asm.neural diagnostically when absent. Retain every failure.

Dependencies: BENCH-01, BENCH-02, NEURAL-03, DEMO-01, INFER-01, INSTALL-02.

Acceptance commands:

```
"$J" optimize plan --base-plan "$BASE_PLAN" --base-plan-sha256 "$BASE_ID" --configuration
"$TRAINED_CONFIGURATION" --configuration-sha256 "$TRAINED_CONFIGURATION_SHA" --output
"$PANEL_PLAN"
"$J" benchmark run --plan "$PANEL_PLAN" --plan-sha256 "$PANEL_PLAN_ID" --manifest "$M" --datasets
"$D" --journal "$PANEL_JOURNAL"
"$J" benchmark report --plan "$PANEL_PLAN" --plan-sha256 "$PANEL_PLAN_ID" --manifest "$M" --
datasets "$D" --journal "$PANEL_JOURNAL"
```

Required evidence: All thirty dataset identities, ten per task; every applicable method slot; screen/frozen/confirmation losses; selection-exposure row intersection hashes; no sealed reads; winner backend outcomes and actual win counts.

Resources: Exactly 7200 seconds including 300 seconds reporting reserve;20 CPUs/workers,64 GiB,zero swap,one thread per fit,30 seconds per fit,120 seconds per search arm; sealed labels unopened; existing window never reset.

Source/evidence references: `benchmarks/panel.json`, `crates/jope-bench/src/panel/protocol.rs`, `crates/jope-bench/src/panel/selection_exposure.rs`.

Qualification limit: Current code forces asm.neural into extra confirmation, not asm.generic. Selected five-fold summaries are adaptive development, and ten wins per category are not promised. Qualify exports before the original 6900-second cutoff or preserve unattempted status.

E.1.6.5 REPORT-01 — complete measured report

Priority: P1. **Current evidence:** one-dataset, 832 predictor/164recipe operational audit. **Status:** full campaign report pending.

Action: Publish all thirty losses/gaps, outcomes, time/cost per unique artifact and recipe, coverage and failure inventories. Compare the supplied one-hour/ten-million-token scenario only using supported quantities.

Dependencies: PANEL-01, EXEC-05.

Acceptance commands:

```
"$J" benchmark report --plan "$PANEL_PLAN" --plan-sha256 "$PANEL_PLAN_ID" --manifest "$M" --
datasets "$D" --journal "$PANEL_JOURNAL"
```

Required evidence: Report recovery without fits; union of started/finished records; worker CPU distinguished from overlapping service CPU; known and unknown costs; training, installation, preparation and export accounted separately. Tokens and money stay unknown unless measured; retain win/loss/gap tables.

Resources: Original 300 secondsreportingreserve forpanel; laterreadonlyauditopensnofitbudget.

Source/evidence references: §12.5.2 Worked example: CPU, wall time and per-predictor cost of the one-dataset proof, §1.5.2 Canonical proof record and per-arm facts, `crates/jope-bench/src/panel/asm_campaign.rs (pending)`.

Qualification limit: The one-hour comparison supplies wall-duration context, not a matched speedup. Unique artifact hashes do not establish behavioral uniqueness. Never add aggregate service CPU to worker CPU already included in it.

E.1.7 Stage 6: eligible optimization and reuse

E.1.7.1 TEACH-01 — HWM comparison identity

Priority: P0. **Current evidence:** pending full-identity source + test plan. **Status:** not runtime qualified.

Action: Apply and execute full comparison identity enforcement before feedback; ensure stale evaluator/vocab/preprocessing/protocol/class order cannot collide with an incumbent.

Dependencies: INT-02.

Acceptance commands:

```
CARGO_INCREMENTAL=0 CARGO_PROFILE_DEV_DEBUG=line-tables-only cargo test --locked -p jope-learning
```

Required evidence: Changed identity negative cases, old record migration/refusal behavior, hash-chain integrity and ties/strict improvement semantics.

Resources: Existing exclusive CPU CI allocation; one reused lane target, jobs4 unless lane policy sets otherwise; no concurrent numerical campaign.

Source/evidence references: `crates/jope-learning/src/record.rs (pending)`, `crates/jope-learning/src/hwm.rs (pending)`, `crates/jope-learning/src/eligibility.rs (pending)`.

E.1.7.2 TEACH-02 — label exposure readers

Priority: P0. **Current evidence:** burn writer and effective-role mapping prepared. **Status:** residual eligibility-B6.

Action: Wire durable label-burn checks into every path that may claim an untouched test score, including HWM ingest/freeze and report publication, or explicitly prohibit that claim.

Dependencies: TEACH-01.

Acceptance commands:

```
CARGO_INCREMENTAL=0 CARGO_PROFILE_DEV_DEBUG=line-tables-only cargo test --locked -p jope-learning
CARGO_INCREMENTAL=0 CARGO_PROFILE_DEV_DEBUG=line-tables-only cargo test --locked -p jope-bench --features asm
```

Required evidence: Production caller inventory for require_unburned_test_partition; an integration test proving a burned partition is refused by test-valid readers; attempted sealed teaching refused; adaptive development reporting remains truthful.

Resources: Existing exclusive CPU CI allocation; one reused lane target, jobs4 unless lane policy sets otherwise; no concurrent numerical campaign.

Source/evidence references: `crates/jope-learning/src/eligibility.rs:253 (pending)`, `AGENT_CHAT.md:eligibility-B6`, `crates/jope-bench/src/panel/selection_exposure.rs`.

Qualification limit: The panel at source 68f56404c3e1 has no benchmark-valid producer. Separate historical closure questions remain: indexed meta-test contexts and externally reported exposed high-water marks do not by themselves establish training-label leakage. A burn writer alone is not downstream enforcement.

E.1.7.3 MOTIF-01 — eligible reusable fragments

Priority: P1. **Current evidence:** separate typed library branch exists. **Status:** mining front door incomplete.

Action: Integrate the production eligible-State reader, typed fragment/macro miner and umbrella CLI. Preserve full comparison identity, provenance and private/public separation.

Dependencies: TEACH-01, TEACH-02, TEACH-03.

Acceptance commands:

```
"$J" motifs mine --help
"$J" motifs validate --help
```

Required evidence: Actual mine CLI contract/help; eligible-input identity inventory; deterministic typed fragment output. Exclude rejected, no-signal, unbound and sealed examples.

Resources: Read-only/static custody work; no fitting, models, GPU or campaign admission.

Source/evidence references: §1.7 Pending acceptance, `agent-acceptance-queue/QUEUE.md:motifs`.

Qualification limit: These help commands are unavailable at source 68f56404c3e1 and at main a75e543d2166, which has the same tree; they are acceptance targets. No truthful exact mine invocation exists yet; record it after implementation.

E.1.7.4 MOTIF-02 — reuse validation

Priority: P1. **Current evidence:** validationlibrary/preparation only. **Status:** live reuse pending.

Action: Validate fragments/macros on independent host contexts, lossless roundtrips, deterministic fit compatibility and privacy. Measure reuse benefit under the full comparison identity.

Dependencies: MOTIF-01.

Command boundary: no truthful executable acceptance command yet; implement or obtain missing interface first.

Required evidence: Exact implemented validate command and input table; at least three hosts per fragment where its existing contract requires that count; roundtrip and actual fit receipts, independent context identity and canary absence. Support frequency is separate from measured uplift.

Resources: Explicitnewboundedreusequalification allocation required; noresetofpanelortrainingwindows.

Source/evidence references: `agent-acceptance-queue/QUEUE.md:motifs`, `docs/adr/ADR-0005-canonical-learning-loop.md`.

Qualification limit: Do not invent mine flags. The historical validate argument shape alone does not establish umbrella wiring or production State admission.

E.1.7.5 MOTIF-03 — new operators

Priority: P2. **Current evidence:** useracceptance requirement. **Status:** future conditional work.

Action: Admit new operators only after measured ablation and grammar, checkpoint and backend compatibility tests. Preserve the old vocabulary identity.

Dependencies: MOTIF-02.

Command boundary: no truthful executable acceptance command yet; implement or obtain missing interface first.

Required evidence: Preregistered ablation recipes, budget and contexts; measured marginal gain; task/type/codec/emitter/vocabulary compatibility; new identity and migration policy.

Resources: Separatelypreregistered ablationbudget; noneadmitted here.

Source/evidence references: `user approval plan:Eligible optimization and reuse`, §10.6 Future work and acceptance boundaries.

Qualification limit: Typed reusable macros may stay within existing operators. Fragment frequency alone does not justify adding an operator.

E.1.7.6 FOCUS-01 — sustained single-dataset improvement

Priority: P2. **Status:** Current bounded proof exists; no incumbent-seeded attack front door.

Action: Add an authenticated incumbent-aware multi-wave request using full comparison identity, retained original deadline, bounded workers and durable improvement events. Preserve exploration and all non-improvement outcomes.

Dependencies: TEACH-01, TEACH-02, TEACH-03.

Required evidence: Improving, tied, worse, stale and concurrent incumbents; full OOF/baseline admission; no deadline reset or sealed input; successful no-improvement completion; actual cost per eligible improvement.

Commands: No complete executable front door is claimed; use the linked chapter’s current/pending interface map.

Resources: No indefinite loop or additional allowance implied; whole focused experiment has one predeclared finite budget.

Sources: §6.5 Focused single-dataset search, `crates/jope-bench/src/panel/origin_proof.rs`, `crates/jope-pipeline/src/neural_search.rs`.

E.2 Coordinated future backlog: 24 proposals

These are Claude’s coordinated follow-on proposals, retained alongside their overlap with the primary checklist. They are not 24 additional completed features, new budgets, or modifications to the frozen panel. Proposed `OD-*` decisions have not been silently adopted. No sealed evaluation is authorized by this list. The JSON/property list retains each proposed owner and original acceptance description.

E.2.1 Documentation, onboarding and repository hygiene

E.2.1.1 Proposal HANDOFF-01

Proposed owner: Claude. **Priority:** P1. **Primary overlap:** DOC-01, DOC-05.

Proposed acceptance: track-a-20260922/manual-20260923/release/SHA256SUMS verifies; FINDINGS.json posted; sections 06/07/09/11 and figures/tools/data handed off

Disposition: Track with the overlapping primary item; exact implementation and acceptance remain pending.

E.2.1.2 Proposal HYG-01

Proposed owner: Claude. **Priority:** P1. **Primary overlap:** INT-02.

Proposed acceptance: cargo metadata shows workspace default members; offline_cli cpp20 test passes; jope-bench run defaults to min(20, cores)

Disposition: Track with the overlapping primary item; exact implementation and acceptance remain pending.

E.2.1.3 Proposal HELLO-01

Proposed owner: Claude docs, Codex verbs. **Priority:** P1. **Primary overlap:** DEMO-01.

Proposed acceptance: a quickstart of at most 5 commands routing to jope demo runs top to bottom from a clean checkout; transcript retained

Disposition: Track with the overlapping primary item; exact implementation and acceptance remain pending.

E.2.1.4 Proposal B6-01

Proposed owner: Claude. **Priority:** P2. **Primary overlap:** MAINT-01.

Proposed acceptance: A2 seed diagnostic and retained-evidence asserts; spec_drift repointed (3 ids); sotarena repoint; scratch budget 900/1500 s; neural-env.sh resolves CUDA_HOME; 13 ids added to known-broken-tests.txt

Disposition: Track with the overlapping primary item; exact implementation and acceptance remain pending.

E.2.1.5 Proposal DOCFIX-01

Proposed owner: Claude patch, Codex integrates. **Priority:** P1. **Primary overlap:** DOC-01.

Proposed acceptance: learning.md, architecture.md, quickstart.md and datasets.md corrected; AGENTS bucket table with sizes; catalogs/s3.json generated from DATA-01

Disposition: Track with the overlapping primary item; exact implementation and acceptance remain pending.

E.2.1.6 Proposal LAYOUT-01

Proposed owner: Claude. **Priority:** P2. **Primary overlap:** INT-02.

Proposed acceptance: crate package names equal directories; jope-contracts shared types; fewer duplicate Task enums/fold planners

Disposition: Track with the overlapping primary item; exact implementation and acceptance remain pending.

E.2.2 Privacy and naming

E.2.2.1 Proposal IP-Q1

Proposed owner: Claude. **Priority:** P1. **Primary overlap:** IP-01.

Proposed acceptance: CLI help/doctor/README/NOTICE/docs/Cargo descriptions clean; ip-scrub covers NOTICE and schemas; new lint tests fail on old main and pass after

Disposition: Track with the overlapping primary item; exact implementation and acceptance remain pending.

E.2.2.2 Proposal IP-Q2

Proposed owner: Claude. **Priority:** P1. **Primary overlap:** IP-01.

Proposed acceptance: emitter text 2.3.0 with re-pinned fixtures; manifest key reader switch; codec serde aliases

Disposition: Track with the overlapping primary item; exact implementation and acceptance remain pending.

E.2.2.3 Proposal IP-Q3

Proposed owner: Claude. **Priority:** P2. **Primary overlap:** IP-01.

Proposed acceptance: receipts v3 and PR 83 ids renamed with a dual-name journal loader

Disposition: Track with the overlapping primary item; exact implementation and acceptance remain pending.

E.2.2.4 Proposal IP-Q4

Proposed owner: Claude. **Priority:** P2. **Primary overlap:** IP-01, PANEL-01.

Proposed acceptance: same-cut v3 evidence; afm.* method id renamed

Disposition: Track with the overlapping primary item; exact implementation and acceptance remain pending.

E.2.2.5 Proposal IP-Q5

Proposed owner: Claude. **Priority:** P2. **Primary overlap:** IP-01.

Proposed acceptance: publication boundary: private-path rule for docs/internal/** and docs/asm.md plus a packaging exclusion check

Disposition: Track with the overlapping primary item; exact implementation and acceptance remain pending.

E.2.2.6 Proposal NAME-01

Proposed owner: Claude. **Priority:** P1. **Primary overlap:** IP-01.

Proposed acceptance: no public AFM/ASPFM/Prime/Lime text; lint enforces

Disposition: Track with the overlapping primary item; exact implementation and acceptance remain pending.

E.2.3 Data, teachers and training

E.2.3.1 Proposal DATA-01

Proposed owner: Claude. **Priority:** P1. **Primary overlap:** DOC-01, DOC-05.

Proposed acceptance: metadata-only inventory of archive-use2, usw2 replicas, custody dev/prod (sealed: counts only, after Codex agrees), bucket configuration, learning per-prefix reconciliation, replica parity, local roots, single-copy risk list, disk headroom; JSON in the s3-estate schema; zero body reads

Disposition: Author stopped the inventory at 19:30 UTC; this manual includes only the independently checked aggregate subset and 19 qualified risk summaries. The raw packet retains a failed sealed-name gate and is excluded. DOCFIX-01 remains pending; no object-body or numerical access.

E.2.3.2 Proposal DATA-02

Proposed owner: Claude design, Codex prepares. **Priority:** P1. **Primary overlap:** TEACH-04.

Proposed acceptance: frozen teaching manifest: seeded, selected before outcomes, families disjoint from all 30 panel families

Disposition: Track with the overlapping primary item; exact implementation and acceptance remain pending.

E.2.3.3 Proposal DATA-03

Proposed owner: Owner, Claude. **Priority:** P2. **Primary overlap:** TEACH-03.

Proposed acceptance: licence-known multiclass corpus with family registry and frozen splits

Disposition: Track with the overlapping primary item; exact implementation and acceptance remain pending.

E.2.3.4 Proposal TEACH-05

Proposed owner: Codex runs, Claude audits. **Priority:** P1. **Primary overlap:** TEACH-03.

Proposed acceptance: per-wave admission and refusal receipts; every admitted family counted

Disposition: Track with the overlapping primary item; exact implementation and acceptance remain pending.

E.2.3.5 Proposal TRAIN-05

Proposed owner: Codex. **Priority:** P2. **Primary overlap:** TRAIN-03, LOOP-01.

Proposed acceptance: second GPU campaign; consumption reproven

Disposition: Second campaign is outside the existing one-GPU-hour allowance; no additional budget or deadline is granted.

E.2.3.6 Proposal HWM-01

Proposed owner: Codex implements, Claude designs and tests. **Priority:** P2. **Primary overlap:** FOCUS-01.

Proposed acceptance: incumbent-seeded single-dataset attack; replacement only when the candidate beats the incumbent on at least 4 of 5 folds by the margin; negative tests for stale identity, tie, test-high/OOF-flat and burned partition

Disposition: FOCUS-01 covers the user’s focused-search need. Extra incumbent four-of-five-fold rule is a proposed policy, separate from required baseline admission.

E.2.4 Evaluation, export, reuse and cost

E.2.4.1 Proposal SEALED-01

Proposed owner: Codex runs, owner authorizes. **Priority:** P1. **Primary overlap:** REPORT-01.

Proposed acceptance: preregistered one-shot sealed evaluation; burn receipts; no reselection

Disposition: Outside current execution scope: frozen sealed evaluation remains untouched. This proposal does not authorize opening it.

E.2.4.2 Proposal EMIT-01

Proposed owner: Claude. **Priority:** P1. **Primary overlap:** INFER-02.

Proposed acceptance: fitted C++17/20 and JS coverage for winning families with library parity fixtures

Disposition: Track with the overlapping primary item; exact implementation and acceptance remain pending.

E.2.4.3 Proposal INFER-03

Proposed owner: Codex. **Priority:** P2. **Primary overlap:** INFER-02.

Proposed acceptance: post-panel export qualification in its own admitted window

Disposition: Track with the overlapping primary item; exact implementation and acceptance remain pending.

E.2.4.4 Proposal EMIT-03

Proposed owner: Claude. **Priority:** P2. **Primary overlap:** INFER-01.

Proposed acceptance: language-neutral DSL semantics spec, golden corpus, numeric-policy conformance harness

Disposition: Track with the overlapping primary item; exact implementation and acceptance remain pending.

E.2.4.5 Proposal MOTIF-00

Proposed owner: Claude. **Priority:** P1. **Primary overlap:** MOTIF-01.

Proposed acceptance: jope motifs mine|validate --help exit 0; deterministic fixture output; --locked build and tests

Disposition: Track with the overlapping primary item; exact implementation and acceptance remain pending.

E.2.4.6 Proposal COST-01

Proposed owner: Owner constants, Claude table. **Priority:** P1. **Primary overlap:** REPORT-01.

Proposed acceptance: labelled scenario table separate from the measured table; no quality-parity claim

Disposition: Track with the overlapping primary item; exact implementation and acceptance remain pending.

E.3 Planned execution and review ownership

Codex owns canonical integration and numerical admission. Claude contributes source and evidence work in the serial queue. A contributor does not become the independent native reviewer of that same change; the final review is assigned to a separate eligible process. The table below is the current assignment; the original ownership proposal is retained unchanged as history in `data/coordinated-roadmap-proposal.json`.

Item	Execution	Claude contribution / review scope
LIFE-01	Completed PR83 lifecycle	Independent native review retained in lifecycle evidence.
DOC-01	Codex canonical integration	Contribution/source findings only; exact-head native review uses a separate non-contributor Claude process.
DOC-02	Codex (done 16:46Z)	none
DOC-03	Codex	Fresh non-contributing independent native reviewer
DOC-04	Codex	Fresh non-contributing independent native reviewer
INT-01	Codex	pre-review of each refreshed replay; native review at exact head
INT-02	Codex	pre-review; native review
BENCH-01	Codex	read-only receipt audit
BENCH-02	Codex	read-only receipt audit
EXEC-01	Codex	review
EXEC-02	Claude writes the tests; Codex runs them under the CPU lock	custody patch (drain-and-fixtures-F2: async reap, pre-gate BudgetExhausted, undrained group)
EXEC-03	Claude writes the tests; Codex runs them	custody patch (resume-fix-F2: PoolBackend receipt through verify_receipt, Some(shared fold digest))
EXEC-04	Claude writes the tests; Codex runs them	custody patch (regressions-F2: assert original_deadline stop reason and persisted search receipt)
EXEC-05	Codex	review
NEURAL-01	Codex	receipt audit
TEACH-01	Codex	review
TEACH-02	Codex	optional custody patch: production caller wiring for require_unburned_test_partition plus an integration test

Item	Execution	Claude contribution / review scope
TEACH-03	Codex	audit that freeze/vocabulary pins precede fits; independently recompute per-fold gains and the 4-of-5 rule
TEACH-04	Codex	audit that every eligible comparison is included
TRAIN-01	Codex	review
TRAIN-02	Codex	review
TRAIN-03	Codex	review
TRAIN-04	Codex	review
NEURAL-02	Codex	receipt audit
NEURAL-03	Codex	audit that the new checkpoint hash appears in proposal and fit evidence
INFER-01	Codex	review of emitter/parity parts
INFER-02	Codex	review of emitter/parity parts
INSTALL-01	Codex	new-user walkthrough of quickstart and doctor output
INSTALL-02	Codex	new-user walkthrough
DEMO-01	Codex	wins/cost table spec; independent recomputation
PANEL-01	Codex	independent read-only recomputation from the journal
REPORT-01	Codex	independent wins/losses and cost recomputation; every loss published
MOTIF-01	Claude code, Codex live run	via MOTIF-00
MOTIF-02	Claude design, Codex windows	preregistered ablation design
MOTIF-03	Claude design, Codex windows	two-stage promotion design
IP-01	Claude	IP-Q1..Q5
MAINT-01	Claude (B6-01); Codex runs full lanes	author
FUTURE-01	deferred	CUDA design after EMIT-01
DOC-05	Codex integration	Independent source/data cross-check
DOC-06	Codex integration	Independent source/data cross-check
PACK-01	Codex integration	Independent source/data cross-check
FOCUS-01	Codex integration	Independent source/data cross-check
LOOP-01	Codex integration	Independent source/data cross-check

E.4 Source index

E.4.1 Current implementation references

These links bind the reviewed implementation commit `68f56404c3e1`, not a moving branch. PR83 merged it on 2026-09-23 as main `a75e543d2166`, which has the same source tree (`176b75bc36d7`), so each pinned file is also the file at that main commit. Hashes below are SHA256 of file bytes; Git commit/tree identifiers use Git's separate object identity. A link needs access to the private repository.

Current source	SHA256
benchmarks/panel.json	ba7dace4cd535687002cc95538dbc7f7aaf77ec973177c7159b3c6d22471eee0
benchmarks/prepare.py	7600de95f0769309d814d80c6723d65f0588124b3e0d1a9875d8155fcc4a91c9
crates/feat-math/src/advanced_v3/fitted_cpp.rs	c20d7e9e7d530102435bdfd07a7c5275de041742a38d681842b87147e6c82fef
crates/feat-math/src/emit_cpp.rs	531e4d006caac9c41f3934df8f0f84d1f62749a9380ab4b67d1ec21180e65157
crates/feat-math/src/emit_js.rs	2e18c3a0fa85ff4fb0013d610be9404079956d9331c9988486b828bc7ac805cf
crates/feat-math/src/emit_py.rs	138a8c2a279da1b9df7b3c155606ac7e18c145982400ca7ace40f120cf386ca6
crates/feat-math/src/emit_rs.rs	92d1277a90886073ba5b3aa19aa9fb461f97ad3290c2c4a91a1ccaeec7c1d405
crates/feat-math/src/transpile.rs	374d7c1e54c9199b932963f564306deec8ffcc1f0ac7add5554fdcee04e0892a
crates/jope-bench/src/panel/campaign.rs	95e162538f7601f40bd2e7019639a6622259e516fa38cd1f92e83a0d4117b20c
crates/jope-bench/src/panel/campaign_cli.rs	419262eab7516ba0ede8432a5d1f162fd4d64047681764e55edab9e438bb2ad1
crates/jope-bench/src/panel/cli.rs	f1f9879d0b2de5473f8fcd9b3722e5c41147d5e33b26adac0948d0f4541a7b21
crates/jope-bench/src/panel/methods.rs	170b3fcd3fadc140c5d3006e34372a8042d6c8939395a3f633085b884e1cdaa6
crates/jope-bench/src/panel/optimize_cli.rs	17af561a039fd4832dd1bd23c226bbde5065b09b5273eb6e0b1353899042bd80
crates/jope-bench/src/panel/receipt_timing.rs	0a479d36f84c9c22381310e9dd83ca4cf4c48594a7541c666587c0226f4a7367
crates/jope-bench/src/panel/score.rs	10093ed25a81940c7ab985843e742efc49990f3d361c92764d6d6dcae07e81e9
crates/jope-bench/src/panel/selection_exposure.rs	a591787e6eaa95bb2e11d4c46776707076016b06785c4fad1dbc71d1a4d9fc8e
crates/jope-cli/src/kernel.rs	d8f26b2eb49652d61600c699155e7b69664bd0e868e1fc76229da012e6e547c8

Current source	SHA256
crates/jope-cli/src/main.rs	4b2990e5294d0668a8d5f7a6c1e809d91fbb93be1418fc3fb54f291c88251265
crates/jope-cli/src/schema.rs	307ce6a4fa3c22bef7c15e7272bd834100ba78715eba7b1c0547a2e1b6b04d1d
crates/jope-paths/src/lib.rs	abc95bfa0e258642e0101b2f82c07706492dfe1d6c8a6fd99043a475893aa120
crates/jope-pipeline/src/cli.rs	1bd10526c2d58c233f5bb1e9bd575bb5c7afa82691b2238ad2f6993f88bd6ae8
crates/jope-pipeline/src/cpu_context_bridge.rs	48dbe30216bf42520df668ca5b3d4a4138d21ae2731ba07c349fe6ec232729ff
crates/jope-pipeline/src/dataset.rs	3f34436e6d0bbb747c846d63dfe3bcbc4e1ce4e34863caa37d7dce0c7183e7ba
crates/jope-pipeline/src/emit.rs	bb351af0947653caf9bfb5d276efd8b2645599503a101f1100b9fc20bb07edb9
crates/jope-pipeline/src/execution.rs	e38f68bbb765918f421e69d82fa697d822d60a53eb10c51e30cc9f1ec8066a76
crates/jope-pipeline/src/experiment.rs	cec0f36073f73de8c09f508512c18e0098bb3f4028af08081d1b6a8639ab2e1e
crates/jope-pipeline/src/front_door.rs	71dc0402ff413f020a25983a5ba98386324c16af03363f5deae4fafceafd856a
crates/jope-pipeline/src/neural_search.rs	a8cd55c9c4f7b5a1060eab5979deb76db24484360389f2072f6b906c4bae0390
crates/jope-pipeline/src/search_context.rs	e423e9676a75426cdb6fdbbaf908e80c4bc1c89c5d93007aedcdfc0189cd382e
crates/jope-pipeline/src/supervisor.rs	a61265692d61580fe174c44a5f1cc2971e6512b803d91bc480d38c90a73a16b6
crates/jope-policy/src/config.rs	a740ca64edde353c9624f5fdc3b63878c1200ee95c159a71bca1556abc28ed37
crates/jope-policy/src/context_producer.rs	f218d0539e273df4f53263c8ca4f06639d68456ab2057a195bf3253e27aa32d2
crates/jope-policy/src/context_producer/admission.rs	2e98364dfe3994a21ccc2d6115adb7d7396c3de3a9babf15bcae473d4d59c441
crates/jope-policy/src/context_producer/input.rs	0fc3df5ed519439dd47fd27022e4be11594924a46fff4c926901cb514f5e4609
crates/jope-policy/src/producer.rs	bfa9ecfe92bc6224d0409e2a66048a9858958a4833102cd57c0201fc4b083895
crates/jope-policy/src/bin/jope-policy-producer.rs	73a4678ac5fa362396f943ba02e8bc68e1fd5d00d4610d4de5ba597463fb3e6b
crates/jope-policy/src/context_producer/worker.rs	c106275c81e706c3eef8ff4644aa74efaf0d8a3e26eb97eb36b9222d052d82c4
crates/jope-policy/src/seeds.rs	434d8a0cde2e34c57291ec3657bf92707de6fbaf99f53e0fbe6f1b4aee969c15

Current source	SHA256
crates/jope-policy/src/lib.rs	2b0da78dcd569de31fdd0922c90e56e19e404ba9f1e184a1eb61cf634ac733a8
crates/jope-policy/src/model.rs	7d0f464fe4bd51ec4b2afa2257678565f959760c74bc347a7d1cebff3efe1e25
crates/jope-policy/src/ranking.rs	92c31cc55f3943aea2246bc0191325fbd625d2019fd49ce33c54f505b751843c
deploy/ci/e2e-smoke.sh	6becfc97830d87907bfb594aae247a117fb23162382d77bc75fbd770b4668354
deploy/ci/emitters.sh	beb99faf01c60b4f39b0179eab46696bcd5cc10658b8cdd690962bbd35254e6b
deploy/ci/emitters/compare-bits.py	eaee5cc96d3d3161c85fefc02d636874adb4d399c340863534d83db5e8fe6249
deploy/ci/emitters/driver.cpp	899ad86994c93ac37312a66acde6900948948b22bfae1dd0be1516c7e99ca416
deploy/ci/emitters/driver.js	754b98bc8d694ec482e23826412e7dcbde76717e7d2f72d18a9adeb76ad12f75
deploy/prepare-panel-runtime.py	f9a0d1d9ca97dfd7664a39a8df6b1e364e738fdfaaa13a0e327402c073ee46f5
python/src/jope/policy/model_v3.py	11ab74d12bda8a2fb817b073a685d128eff06479f769e4301488339424314ffe

Current interface/semantics: `crates/jope-cli/src/main.rs`, `crates/jope-cli/src/kernel.rs`, `crates/jope-pipeline/src/cli.rs`, `crates/jope-pipeline/src/front_door.rs`, `crates/jope-paths/src/lib.rs`, `crates/jope-bench/src/panel/{cli,campaign_cli,campaign,optimize_cli,methods,score,receipt_timing}.rs`, `crates/jope-pipeline/src/{experiment,execution,supervisor,neural_search}.rs`, `deploy/prepare-panel-runtime.py`, `benchmarks/{panel.json,prepare.py}` and `AGENTS.md`.

E.4.2 Pending source references

These are stopped custody patch views. They are not installed capabilities and are not linked as if they existed at the reviewed Git commit. Their patch identities and acceptance gates are named in §8.4.4 The pending bounded extension path: sequence and limits, §4.1.2 Command surfaces and the stopped integration order and the integration card §E.1.1.8 INT-01 — serial integration.

Pending path	SHA256 of inspected view
agent-delivery-refresh-68f56404/replay-001/source-text/crates/jope-bench/src/panel/demo_asm.rs	dd1a287c4ff3c1447a20e81d2c2990913923cd48c667f3be600f5d6b3c972f62
agent-delivery-refresh-68f56404/replay-001/source-text/crates/jope-bench/src/panel/fitted_export.rs	2f6e75d70e6d89bbfeac7db7ec67c585e6d7fc5c6287b134851db931ef5ffb30
agent-delivery-refresh-68f56404/replay-001/source-text/crates/jope-bench/src/panel/fitted_export_run.rs	3b007f44198aaa7b96fbca4936ed7ad551feea086f6018f1ff0458ddee6c7295
agent-delivery-refresh-68f56404/replay-001/source-text/docs/private-fitted-qualification.md	abf27b0dd7fabc0ee4a1f095d37d2ee4f77c5ffdfd2988699990521089f61ef8
agent-delivery-refresh-68f56404/replay-001/source-text/docs/private-warm-target.md	4234468c7f9b5348f7a264f815d10d6f62baf1f4b5397c6cf1fed644a12d97a8

Pending path	SHA256 of inspected view
agent-product-final-68f56404/replay-001/source-text/crates/feat-math/src/advanced_v3/fitted_portable.rs	f2f8998251f2e7e97900563a7032b7d8b8c80bf43e080b7b622b2e195f75737f
agent-product-final-68f56404/replay-001/source-text/crates/feat-math/src/advanced_v3/fitted_portable/scalar.rs	34823e72c9576c7025d6bbb31a44203f397e1d571642d516a63b5ef328413970
agent-product-final-68f56404/replay-001/source-text/crates/jope-bench/src/panel/demo.rs	00751a3094b42c6f742cffed5f7804803977f47d9f2837b161421e884f7efdfc
agent-product-final-68f56404/replay-001/source-text/crates/jope-bench/src/panel/demo_asm.rs	74c2c6164015bc65f53d021c77f78d9a9ae7238d6c68bb007da424328707064d
agent-product-final-68f56404/replay-001/source-text/crates/jope-bench/src/panel/inference.rs	c4929b553dd6893a48b836c2c3838b62b9d96fbd2c5f76634e235eaeed98e626
agent-product-final-68f56404/replay-001/source-text/deploy/install-private.py	3e4e16d5d7e224624c970aca7217952339c8e679b12752aa8317452c27120e9c
agent-product-final-68f56404/replay-001/source-text/docs/private-install-demo.md	05d728bba5fdc78c1568650ef8179d71ce19cf5c8860ad23317c8cea53dfebb6
crates/jope-bench/src/panel/experimental_proposal.rs	d03c765ac08d1ecc039a82f3dbea0c40bc5158baac042e8df348f0c91a90a018
crates/jope-bench/src/panel/optimize_cli.rs	487a299045d9a24604511cdecd8d4fa0b64ff055e4557339ce34120858549eca
crates/jope-bench/src/panel/origin_proof.rs	229d26685d60646887e5ec30ec4734aa78dfdb2f37788ffbf30f878ce593b675
crates/jope-cli/src/train.rs	486235575fc972306a6294233d94f8a37179945931e74bfb298e342752c634a2
crates/jope-cli/src/train/corpus.rs	a239fd8633848178214270339 63d8c0c5ebc54642130b087a9848aa51f2d93c5
crates/jope-learning/src/eligibility.rs	839e1de7b5f2c869817564ed39f3c30843208f7a00a1a277d27fe50857ce1e8d
crates/jope-learning/src/eligibility/inner.rs	b5fb468fcc554ec7da9227244d435d84593910c254678fb3f4174b5d7a4e9215
crates/jope-learning/src/record.rs	4360d96a4a6c0a27f5ba78fbcab4f5be10a3a800f56acee6379ef805b3815d61
crates/jope-pipeline/src/experimental_producer.rs	9a15e7077651ded2adf364a26a579720ae21aa92f0362c0122d132e34162411c
crates/jope-pipeline/src/search_plan.rs	fe8e402357b9b61359b03760401da9a568b8a1e7092b1222947fc30d2c8fb817
crates/jope-policy/src/context_producer/classification.rs	465af30abcdc7bcc02c046d92a71e0b7a22be910320e22158ef16b216adc4fa6
metadata-receipts/veox-jopedime-models-use2--bundles--hyperion-v7--manifest.json	fdaf938fc7f7957c06959f12c130b2bd81cdaef5e5bb2198ba75cf35da1c5233
metadata-receipts/veox-jopedime-models-use2--bundles--jope-prime-gen31-v2--manifest.json	c8d8d13a2f4cb2eeb21eea364c33e14429c84d344e6b50eb38a472173b97b6f2
metadata-receipts/veox-jopedime-models-use2--bundles--policy-step-60000--manifest.json	d8e259b26ef8d580ddb99ae5826c9501bcdc0c32045945eef9010926fd796e9e

Pending path	SHA256 of inspected view
python/src/jope/policy/experimental_proposal.py	f0338f0505a27c9f2f5eac7579e2c506da5c116b3a206f44c52a1cf87cf9c0aa
python/src/jope/policy/training_campaign.py	9d9cfb3257ecf2b6072cab8adc6d830f3df096605a06ff0677e6338c2aad5a2c

Pending interfaces: final product postimages of deploy/install-private.py, docs/private-install-demo.md and panel {demo,demo_asm,inference}.rs; stopped add-on postimages of panel {fitted_export,fitted_export_run}.rs and docs/{private-warm-target,private-fitted-qualification}.md. Exact source and receipt hashes accompany this appendix's custody handoff.

E.5 Machine-readable companions

E.5.1 Companion files

The manual directory, docs/internal/asm-manual/, contains the following locally reproducible evidence. Paths are relative to it, except the property list, which is docs/asm-maintainer-status.plist from the repository root. The original metadata/custody paths inside receipts describe where observations were obtained; use the copies under this directory when available. Historical preparation receipts keep their original pre-execution wording; the current state of the examples is the executed result in data/example-acceptance/result.json (14 commands, ok: true).

Companion	Purpose
data/dataset-inventory.json	Thirty complete development dataset records and 522 feature records
data/catalog-summary.json	Corpus task, row, feature, origin and license metadata totals
data/metadata-receipts/	Catalog JSONL and collection/model metadata; no payload bodies
data/s3-estate.json, data/local-estate.json	Dated current-object and local logical-byte inventories
data/data-estate-extension-summary.json, data/data-estate-extension-release.json	Reconciled aggregate archive/replica/local-root inventories and 19 qualified recovery risks; exact source mappings and raw-packet exclusion boundary
data/performance-evidence.json.gz	Current proof, historical score cells, reports, ladder identities and scope
data/plot-*.csv	Compact plot inputs with snapshot and source pins
data/roadmap.json	Detailed 43-item dependency/acceptance checklist and 24 separately marked coordinated proposals
docs/asm-maintainer-status.plist	XML property-list representation of the same checklist
data/example-acceptance/	Actual fourteen-command result, audit and handoff
examples/	Private synthetic fixture, exact argument packet and complete-output validator
figures/	Reproducible charts and associated historical data inputs

Companion	Purpose
`render-receipt.json`	PDF digest, pages, consumed input hashes and rendering environment

The machine-readable companions are `dataset-inventory.json` (all thirty datasets, 522 feature records, complete available upstream metadata and detailed development statistics), `performance-evidence.json.gz` (measured current origin plus historical score cells, timing and ladders), `catalog-summary.json`, `local-estate.json`, `s3-estate.json`, and `metadata-receipts/`. The catalog JSONL retains every available per-dataset catalog field for all 3,859 corpus datasets. No claim is made to enumerate every mathematically possible statistic.

E.5.2 Lossless evidence packaging

The full performance inventory and original catalog JSONL are stored as `data/performance-evidence.json.gz` and `data/metadata-receipts/veox-jopedime-datasets-use2--catalog--v1--catalog.jsonl.gz` to respect the repository's 5 MiB file limit. Gzip preserves every original byte; no rows, fields, numeric precision or metadata were removed. `data/evidence-packaging.json` records both compressed and original SHA256 values, sizes and preservation paths. Historical handoffs and the initial assembly manifest intentionally retain their original uncompressed paths and hashes; the packaging receipt supplies the current location mapping.

Read either companion directly with Python's standard library:

```
import gzip
import json
from pathlib import Path

root = Path("docs/internal/asm-manual/data")
with gzip.open(root / "performance-evidence.json.gz", "rt") as stream:
    performance = json.load(stream)
with gzip.open(root / "metadata-receipts/veox-jopedime-datasets-use2--catalog--v1--
catalog.jsonl.gz", "rt") as stream:
    catalog = [json.loads(line) for line in stream if line.strip()]
assert len(catalog) == 3859
```

The plotting tools consume the separately retained CSVs and summary JSON, so this storage change does not change their inputs or measurements. The renderer hashes the compressed companions as well as the packaging receipt.

E.5.3 Training-history and continuation companions

E.5.3.1 Companion inventory

The learning expansion is documented in actual training history, target history and packing, versions, KPIs and continuation, and flywheel/focused search.

Companion	Purpose
`data/training-history/*.csv`	Original loss/diagnostic observations and checkpoint lineage, with source hashes and line/JSON Pointer anchors
`data/training-history/metric-definitions.json`	Exact historical loss and KPI meanings; objective changes and missing source-build binding

Companion	Purpose
`data/training-history/verification.json`	Original-value verification of 3,218 metrics and 78 checkpoint records
`data/training-history/figure-receipt.json`	Input/output hashes for seven historical training figures
`data/legacy-training/`	Four original round-1 and reported self-play summary files, source hashes, recomputed diagnostics and a vector-plot receipt; internal Elo is not benchmark-valid
`data/per-step-history/`	All 63,000 retained recovery steps and 378,000 exact metric tokens in compressed CSV, 21 raw-log hashes, corpus boundaries and explicit 100-step means
`data/per-step-history/independent-review/`	Second complete comparison with original logs, checkpoint identities and all derived block/bin statistics; numerical/source review is separate from final-head CI
`figures/per-step-figure-receipt.json`	Source hashes and recomputation checks for all 630 plotted loss bins and 78 adaptive validation points
`data/packing-routing-audit.json`	Selected external packing metadata, recomputed sizes and strictly scoped paired final-search observations; excludes unjoined campaign rows
`data/original-target-receipts-custody.json`	Durable local archive and verification hashes for all 26,357 original development search receipts; archive stays outside Git, with seeds and off-host backup explicitly excluded
`data/target-history/*.csv`	All 159 repeated historical comparison summaries and all 213 illustrative arm/generation points; exact source identities and limits
`data/target-history/manifest.json`, `data/target-history/verification.json`	Six corpus inventories, binary-format source pins and original target-observation verification
`data/target-history/figure-receipt.json`	Input/output hashes for two historical GP-target figures
`data/extension-guide/`	Current-source version/command/API pins, stopped handoff and static/metadata-only verification
`examples/continue-python-policy.py`	Current Python resume API illustration; metadata-only by default, numerical branch unexecuted and externally supervised
`data/single-dataset-help-receipt.json`	Six actual metadata/help commands, with immutable CLI identity and zero fits

E.5.3.2 Reusing the evidence

The adjacent `data/training-history` CSVs are long-form original measurements. `block-endpoint-metrics.csv` has 1,296 rows; `block-adaptive-validation.csv` 432; `fixed-monitor-metrics.csv` 16; `gen31-epoch-metrics.csv` 462; `a100-checkpoint-lineage.csv` 78; `a100-checkpoint-validation.csv` 312; and `rtx60100-step-metrics.csv` 700. The two original small logs are retained alongside the normalized data. Source hashes and line/JSON Pointer anchors make a plotted point independently traceable. The initial extraction audit covers four CSVs. The shipped `verify_history.py` checks all 3,218 metric `value` cells across the measurement CSVs, plus seven fields (step, model/parameter/optimizer/corpus hashes, world and global batch) and the state-file hash for each of the 78 lineage rows. It does not check every auxiliary numeric column, such as elapsed time, parent/local step, requested fresh fraction or row

counts. Running that helper writes `verification.json` beside the CSVs; use a custody copy when preserving a stopped evidence bundle. Model bodies and sealed target values were not read for this reconstruction.

The additional `data/per-step-history` package preserves the 3,809,970-byte compressed step CSV (SHA `aa431f5a2f5f4d6bfc2e8568ef57bb10b100356750d22eb615dd68cfa1647627`), 21 full source-log hashes, block statistics, explicit 100-step bins and boundary comparisons. `tools/extract_per_step.py` regenerates into a new output directory from the retained raw logs and previously audited checkpoint CSVs; it refuses to overwrite an existing extraction. `tools/make_per_step_figure.py` independently recomputes every plotted bin from the compressed row data before drawing it. Run `python3 docs/internal/asm-manual/tools/make_per_step_figure.py` from the repository root to regenerate this one figure without loading models. The original seven-figure builder remains separate. Both figure receipts bind their exact input hashes.

E.6 Receipts

E.6.1 Important local receipts

The custody root is the single host-local directory `/home/ubuntu/jope-rescue-20260921/asm-integration-20260923/`; this manual records no off-host copy of it. These are host-local receipt locations, not promises that every reader has the same files. The structured documentation evidence identifies the source observations copied into this edition. Weight bodies, source dataset bodies and sealed labels are not embedded in the manual.

Relative custody path	Meaning
`neural-integration/lifecycle-68f56404c3e199e06413bfd645d3fe0803d7b96f/`	Full author/committer inventory, CI, native reviews, merge and main-guard readbacks
`neural-integration/pr83-final-review-68f56404c3e199e06413bfd645d3fe0803d7b96f/`	Independent review prompt, source/proof inspection and native approval receipt
`neural-integration/source-match-68f56404c3e199e06413bfd645d3fe0803d7b96f.json`	All 2,855 Git blobs match the immutable build input inventory
`neural-integration/native-context-68f56404c3e199e06413bfd645d3fe0803d7b96f/`	Eight native encoder qualification cases under unchanged numerical policy
`neural-integration/origin-run-68f56404.json`	Final operational run command, timing and exit status
`neural-integration/origin-report-68f56404.json`	Independent recovered-report command and timing
`agent-final-acceptance/measured-origin-68f56404/audit.json`	Source/report/worker inventory and independently bound supervisor accounting
`neural-integration/final-head-pipeline-bench-68f56404.json`	Local 53 benchmark and 106 pipeline test results
`neural-integration/final-head-host-ten-68f56404.json`	Ten actual privileged host cases and binary identity
`agent-final-acceptance/failed-origin-d0f5748b/audit.json`	Preserved failed prior run; not an accepted current score
`agent-maintainer-manual/`	Section drafts, structured data inventory, example preparation and documentation observations

Treat a receipt's command, source SHA, input hashes, runtime and original deadline as one evidence unit. Copying a number out of that unit can lose its scope. A local path alone proves neither continued availability nor integrity: verify the retained hash and the contents required for the claim.

E.6.2 The PR83 lifecycle record

Gate	Exact observation
PR	`veox-ai/JopeDime#83`, merged
Manual CI dispatch	Run 35887284772, attempt 1, source `68f56404…`, success
PR CI	Run 35887255898, attempt 1, source `68f56404…`, success
Required CI jobs	Eight lanes and aggregate succeeded in both runs
CI test inventory	937 Rust passed, 14 skipped; 221 Python passed per run
First independent native review	5293748183, `neverhuman`, APPROVED, exact full reviewed head, 16:29:01 UTC
Second independent native review	5293788772, same eligible account, APPROVED, same head, 16:32:43 UTC; includes residual disclosures
Author eligibility	All twelve commits associated with `jepsontaylor` as author and committer; reviewer differs
Merge	Guarded rebase with `--match-head-commit` at the reviewed head, completed at 16:34:47 UTC as main `a75e543d2166…` with the reviewed tree `176b75bc36d7…`
Remote-main verification	Fetched main and compared complete Git tree; canonical clean main observed before documentation branch
Main guard	Run 35889731910, attempt 1, success at resulting main; guard job 107278860508

Two review processes overlapped during coordination. Both were independent of the source writer and both approvals are retained; they are not two different human reviewers. One process's supplementary disclosures identify outstanding work rather than silently treating it as fixed. The writer did not select another credential to approve its own source.

The full-main diagnostic run 35889732032 that the PR83 merge queued was cancelled while still queued to preserve the admitted local allocation and let main-guard run. It did not pass or execute. Earlier full-main run 35852022466 failed and remains historical diagnostic evidence requiring repair. Passing fast CI does not replace that outstanding broad qualification.

E.7 Reproduction boundaries

Rendering this manual uses its committed section Markdown, structured data, figure inputs and Python rendering tools. It should not fetch a corpus, load a model, run a fit or contact the sealed store. Plotting historical evidence is distinct from rerunning historical experiments. The included figure data keeps missingness, eligibility and snapshot identities.

The private toy example has its own synthetic input and output receipt. Even a successful five-language recipe/refit comparison cannot qualify a different selected fitted predictor. The pending fitted-inference acceptance must authenticate the retained winner and compare complete saved-state predictions with its original numerical policy.

Operational replay needs more than the document: the exact source/build, immutable workers, model/checkpoint/vocabulary, runtime dependencies, allowed data inputs, original plan and journals, and an available resource allocation. Reproduction that resumes an expired campaign cannot start fresh jobs by assigning a new deadline. A new experiment needs a separately recorded plan; it does not repair the old receipt.

E.8 Names and glossary

ASM is the current product name, settled by the user plan. Historical method IDs such as `afm.lite-race.v1` and `afm.lite-race.v1.resampled` remain exact evidence identifiers in committed same-cut results and the current legacy registry. Use ASM in new prose, but do not rename old identifiers or rewrite frozen reports in place. A method-identity change requires an explicitly versioned new comparison/evidence record. Sources: `crates/jope-bench/src/methods.rs:165`, `:168`; the 2026-09-23 16:07 coordination record.

Term	Meaning in this repository
ASM	Algorithm Synthesis Model: learns to generate and optimize executable algorithms from data
Algorithm / recipe / program	Executable typed procedure and configuration; reusable across fits. The legacy CLI flag for a recipe is `--genome`. A recipe is not its native fitted predictor, an emitted recipe in another language or a fitted inference bundle (§2.6 GP representation: recipes, fitted predictors and the four artifact types)
Method	A comparable benchmark entrant: a registered native recipe (`rust.*`), an external library adapter (`external.*`), an ASM entrant (`asm.neural`, `asm.generic`) or, in historical ladders, a whole system
Candidate	A recipe placed in one search slot
GP	Genetic programming: search over executable programs using variation and measured selection
Arm	One search population in a campaign, with a fixed proposal source and its own declared allowance: the neural arm (policy pool plus exploration slots) or the generic arm (no policy proposals). Historical searches also ran a shuffled-context arm. Unrelated to the Hyperion asset label `arm_b` (the v5 arm-B weights, `hyperion_v5_arm_b.safetensors`)
Fitted predictor	One recipe plus preprocessing and model state learned from exact training inputs
Checkpoint	Serialized neural proposal-model parameters and compatibility metadata; distinct from a fitted tabular predictor
Context	Authenticated numerical dataset/target representation supplied to the proposal policy
Dataset encoder	Learned or specified mapping from permitted dataset structure/rows to context features

Term	Meaning in this repository
Target encoder / target context	Representation of allowed target information; task-specific and restricted to permitted fitting rows. In the regression context it is the 512-float block computed from min–max-rescaled training targets. The 1,024 foundation values in the 1,536-float “dataset” block also use the training targets (§2.7.1.2 Component dimensions and assembly)
Vocabulary	Token identities, grammar and executable restrictions associated with a checkpoint
Proposal policy	The neural model that proposes program tokens for a context; the accepted regression policy is the lineage-B step-60000 checkpoint
Generation zero	Initial candidate slots before evolutionary descendants; 80 per arm in the current ASM search
Seed pool / RNG seed	Seed pool: the programs placed into generation zero of an arm (historically including incumbents and retained teachers). RNG seed: a random-number seed, such as seed 7
Exploration candidate	An admitted generic candidate in the neural arm; not falsely labelled neural-origin
Offspring	Mutation or crossover descendant with authenticated parent identities and actual fit evidence
OOF	Out-of-fold: predict rows using a fit that excluded those rows; adaptive recipe selection can still expose them at another stage
Same-cut cuts (historical)	Hashed cuts built by the historical same-cut benchmark command; not comparable with the frozen panel's cuts (§11.4 Metrics and class order)
Development-hold score	The score of a final refit on a cut's fitting rows, measured on that cut's held fold (report field `development_hold_score`, `crates/jope-bench/src/panel/origin_proof.rs:321`); the one-dataset proof's RMSE uses 34 held rows
Policy-validation families	In the pending teaching partition, the families held out to validate the policy; older text calls them “development” families. Not the panel's development rows
Exposure / adaptive	Exposure: use of a partition's labels for selection, optimization or reporting. Adaptive: measured on data that also helped choose what is measured, so not an independent generalization estimate (§11.2.2 Adaptive development exposure)
Cut identity	Frozen identity of the training/scoring row assignment and evaluator context
Inner folds	Five whole-group folds inside a cut's fitting rows, used for out-of-fold selection and, in the pending eligibility path, for teaching admission (§7.2.1 The preregistered freeze and train-side inner split)

Term	Meaning in this repository
Frozen panel	`asm-private-30-v1` under protocol `asm-panel-v1`: 30 PMLB datasets, ten per task, manifest `ba7dace4…` (§5.1 The frozen thirty-dataset panel: identity, totals, groups and immutability). Older text also says official30 or the unchanged thirty-dataset panel. Training-side validation sets that change between checkpoints are not this panel
Task / category	Task: regression, binary classification or multiclass classification. Training text, `--task all-admitted` and the owner's goal of ten wins per category use “category” in this sense; elsewhere “category” means a categorical feature value
Label	The column a dataset asks a model to predict; one of the senses of “target”
Family	A group of related datasets from one source, such as the UCI computer-hardware source of `230_machine_cpu`. The teaching partition assigns whole families to training or validation. The north star (proposed; owner ratification pending) is measured on previously unseen families (§1.3.1 The proposed north star and matched comparisons)
Development data	Data visible to optimization or recipe/model selection, regardless of a file's name; in the panel, every non-sealed row
Development folds 0–4	The five original folds of the panel's development rows; every group stays within one fold (§5.1 The frozen thirty-dataset panel: identity, totals, groups and immutability)
Cut k	Fit on the development folds other than k and hold out fold k (`dataset.cut(k)`, `crates/jope-bench/src/panel/data.rs:314`). Older text calls cut 0 “fold0” or “development fold zero”. For `230_machine_cpu`, cut 0 has 135 fitting rows and 34 held rows
Sealed evaluation	Separately protected evaluation outside the learning and optimization closure. The “seal-protected shared memory” of the training reader (memfd seals) is unrelated
Burned partition / burn receipt	In the pending training64 patch: a dataset cut originally in the `test` role whose labels were made visible to optimization, so that they now count as development data, and the immutable `jope.label-burn/v1` receipt that records this relabel. Sealed and `meta_test` labels are refused rather than burned. `require_unburned_test_partition` refuses to present a burned partition as untouched; it has no production caller yet (TEACH-02)

Term	Meaning in this repository
HWM	High-water mark: the best retained result for a comparison identity. At `a75e543d2` the current-HWM pointer is keyed only by task, dataset hash and cut (`crates/jope-learning/src/state.rs:13`), so a bound result under a different evaluator, vocabulary or protocol with the same metric can still displace the incumbent (`crates/jope-learning/src/hwm.rs:64`). The pending eligibility/HWM correction (TEACH-01) uses the complete comparison identity (task, dataset, cut, evaluator, split, vocabulary, preprocessing and numerical protocol), so that a result under a different identity cannot displace an unrelated incumbent (§7.2.2 The admission guard, label exposure and HWM identity)
PANEL-WINS (proposed)	**Proposed definition, pending owner ratification; no code at `a75e543d2` computes it.** Per task, the number of the ten frozen datasets on which `asm.neural`'s confirmed five-fold mean development loss is strictly below the minimum over all eligible frozen non-ASM comparators (every `rust.*` and `external.*` recipe). Incomplete, failed or unconfirmed datasets count as not won. `asm.generic` is reported in its own column. Reported as (regression, binary, multiclass); the owner's goal of ten wins per task is (10, 10, 10), a 30/30 sweep (§9.11.2 Win counting and a proposed acceptance record)
Teaching admission	Eligibility decision permitting a measured example into policy training
Motif / macro	Typed reusable program fragment with eligible provenance; not automatically a new primitive operator
Numerical policy	Declared arithmetic, probability, ordering, tolerance and runtime assumptions fixed before scoring
Alias	Alternate spelling of an existing recipe; not another independent method fit. The regression `ridge` head carries the historical alias `logistic` (`crates/feat-math/src/advanced_v3/grammar.rs:1263`), a known misnomer
Front door	The supported CLI entry point for an operation, which enforces its admission rules, identities and budgets
Head	Name the sense: PR head commit, reward or attention head, estimator head (such as the ridge head), or serving/training pointer
Unsupported	A backend or task deliberately refuses a capability outside its admitted contract
Unavailable	A dependency or required runtime is missing; different from a model failure
Failed	An attempted operation ended unsuccessfully; retain its reason and measured/unknown costs
Skipped	A known test or operation did not execute; not passing coverage
Journal	Incremental durable started/finished/observation records for audit and original-window recovery

Term	Meaning in this repository
Receipt	Evidence binding command, source, inputs, runtime, resources, times, status and artifacts
Provenance	The chain of source, data, model, recipe and execution identities supporting an artifact
Custody	Host-local preservation directories outside Git, such as `/home/ubuntu/jope-rescue-20260921/asm-integration-20260923/`, that hold stopped patches, receipts and evidence. “Custody buckets” (also “custody dev/prod”) are something else: the account's separate `jopedime-714044927654-us-east-2-dev`, `-prod` and `-sealed` S3 buckets (§D.2.1 Additional and replica buckets)
Elo	Relative Bradley–Terry rating on the arena's declared method universe and eligible pairwise scores
Warm demo	Demonstration after verified preparation; report separately from cold installation
Promotion	Explicit qualification of an exact artifact/model/backend for its intended serving use; a merge alone is insufficient